

Robust Contracts with Exploration

Chang Liu*

First version: November 30, 2022

This version: January 16, 2026

Abstract

We study a two-period moral hazard problem; there are two agents, with action sets that are unknown to the principal. The principal contracts with each agent sequentially, and seeks to maximize the worst-case discounted sum of payoffs, where the worst case is over the possible action sets. The principal observes the action chosen by the first agent, and then offers a new contract to the second agent based on this knowledge, thus having the opportunity to explore in the first period. We introduce and compare three different notions of dynamic worst-case considerations. Within each notion, we define a suitable rule of updating and characterize the principal's optimal payoff guarantee. We find that linear contracts are robustly optimal not only in static settings, but also in dynamic environments with exploration.

Keywords: Moral hazard, robustness, exploration, linear contracts, maxmin

*School of Economics, UNSW Business School; chang.liu36@unsw.edu.au. I gratefully acknowledge funding from the National Science Foundation under grant DMS-1928930 and from the Alfred P. Sloan Foundation under grant G-2021-16778 during the Fall 2023 semester. I express my deep gratitude to my advisors Shengwu Li, Tomasz Strzalecki, Eric Maskin and Benjamin Golub for their guidance and support throughout the project, and over the years. An earlier version of this paper appeared as the first chapter of my Ph.D. dissertation at Harvard. I would also like to thank Sejal Aggarwal, Ophir Averbuch, Benjamin Brooks, Shani Cohen, Juan Dodyk, Federico Echenique, Yannai Gonczarowski, Olivier Gossner, Jerry Green, Yingni Guo, Bård Harstad, Oliver Hart, Zoë Hitzig, Michihiro Kandori, Jacob Leshno, Jonathan Libgober, John Macke, Erik Madsen, Stephen Morris, Giorgio Saponaro, Brit Sharoni, Cassidy Shubatt, Kathryn Spier, Haoqi Tong, Alexander Wolitzky, and especially Paul Milgrom for valuable comments and discussion. In addition, I thank the audiences at UNSW Sydney, Tsinghua University, Simons Laufer Mathematical Sciences Institute, Monash University, Shanghai University of Finance and Economics, Warsaw School of Economics, Shandong University, HKUST, Peking University, University of Melbourne, CUHK-Shenzhen, University of Queensland, Asian Meeting of the Econometric Society, Stony Brook International Conference on Game Theory, North American Winter Meeting of the Econometric Society, Conference on Mechanism and Institution Design, World Congress of the Game Theory Society (GAMES 2024), Deakin Economic Theory Workshop, and Annual SAET Conference for their insightful feedback.

1 Introduction

Moral hazard models, in which a principal designs a contract to incentivize an agent, have been extensively studied and widely applied. In many canonical moral hazard models, however, optimal contracts require precise knowledge of the environment: the set of all possible actions together with the (stochastic) mappings from actions to outcomes. This aspect raises practical concerns, because in reality the principal’s knowledge is certainly not entirely correct. How should the principal design contracts that have robust guarantees even if some details are incorrect? The emerging area of robust contract design follows the *Wilson Doctrine* (Wilson, 1987), which advocates for realistic approaches that are detail free.

The pioneer work by Carroll (2015) assumes that the principal knows only some of the actions available to the agent, and evaluates contracts based on their worst-case performance, over the unknown actions the agent might take. The results show that, very generally, the optimal contract is linear, which provides new foundations for the common use of linear contracts in practice.

One suspicion, however, about the linear results in Carroll (2015) is how much they hinge on the principal’s inability to explore the unknown, an opportunity that arises naturally in models with multiple interactions.¹ It is not even clear how to model (non-Bayesian) exploration in the robust paradigm. Specifically, if the principal can observe an agent’s chosen action, then she can gain insights into actions that were initially unknown but might be subsequently undertaken. Furthermore, based on the agent’s rationality, she may also exclude certain actions that were not chosen. In such environments, how should the principal design contracts to best utilize exploration opportunities? Specifically, what contracts respond best to new knowledge? Are linear contracts still robustly optimal with exploration?

A suitable class of applications of robust models in contract design involves the principal hiring or consulting specialized agents that surpass her own expertise. This explains the principal’s limited knowledge about all actions available to the agents and her lack of a prior belief regarding the unknown ones. For instance, consider an individual hiring gig workers from online platforms. While long-term contracts are typically not enforceable, she does have the opportunity to interact with a pool of workers. Given that the workers share similar professional training, the individual’s knowledge about the capability of the pool from past experience is valuable for improving future interactions. Within this example, the main theoretical question of this paper is twofold: First, how should the individual structure contracts to best respond to new knowledge gained from exploration? Second, in anticipation of such opportunities, what contracts are

¹One related but distinct criticism of the robust mechanism design literature is that most models are static in construction but assume commitment. We discuss this issue in the literature section. See also Libgober and Mu (2023) for a corresponding perspective in the area of informationally robust mechanism design.

optimal for acquiring new knowledge?

In the baseline model of this paper (Section 2), we study a two-period moral hazard problem. There are two agents, whose action sets are unknown to the principal. The principal contracts with each agent sequentially to provide incentives. She observes the action chosen by the first agent, and then offers a new contract to the second agent based on this knowledge, thus having the opportunity to explore in the first period. The principal and agents are all risk neutral, and payments are constrained by limited liability.

The baseline model assumes that the principal knows only some available actions of the agents, but other unknown actions may also exist, and the principal does not even have a well-defined prior belief about these unknown actions. Faced with this nonquantifiable uncertainty, the principal seeks to maximize her worst-case discounted sum of payoffs, where the worst case is over the possible action sets. Consequently, it is crucial to articulate what actions the principal considers possible in each period, and to determine how the principal’s beliefs about unknown actions are updated across periods.

The main result of this paper is that linear contracts are robustly optimal not just in static settings, but also in dynamic environments with exploration. In order to obtain this conclusion, we introduce and compare three distinct notions of dynamic worst-case considerations: *independent technology*, *advancing technology* and *constant technology*. In the first period, the principal believes that the first agent’s action set could be any set containing the known actions. After the principal offers a contract to the first agent and observes his response, a rule of updating must be specified to determine the actions the principal considers possible in the second period, and these three notions precisely vary based on the principal’s updated beliefs about the subsequent action sets. To better understand the results and analysis, it is helpful to imagine there is an adversarial “nature” that selects the set of actions for the corresponding agent in each period to minimize the principal’s payoff, and the three notions differ in the restrictions imposed on nature’s available moves across periods. Within each notion, we define a suitable rule of updating and characterize the principal’s optimal payoff guarantee, thereby concluding that linear contracts are robustly optimal.

We begin by considering the case of *independent technology*, where the action sets of the two agents are not related; in other words, nature can select the action set for each agent independently. In this case, the choices made by the first agent do not provide the principal with information about what actions the second agent can take. Therefore, the learning aspect is essentially nullified, and the principal’s overall payoff guarantee is maximized by adopting a straightforward approach: offering the optimal static contract identified by [Carroll \(2015\)](#) in both periods. Characterizing the case of independent technology creates a building block that enables us to further analyze the implications of dynamic environments with different levels of interdependence between agents’ actions.

Next, we analyze the first restriction that facilitates meaningful exploration: the case of *advancing technology* (Section 3). In this case, the action set may expand between periods, but cannot shrink. In other words, nature can only introduce new actions across periods, but is not allowed to delete old ones. The main result for the case of advancing technology is that linear contracts are robustly optimal *period-by-period* (Theorem 1). Toward this conclusion, we solve the principal’s dynamic problem via backward induction. After the principal offers some first-period contract and observes the action chosen by the first agent, she learns that this action exists and may be taken again by the second agent. Moreover, this represents the best conjecture the principal can make in the second period, given that nature may introduce new actions that were not present in the first period. Therefore, the principal’s second-period problem simplifies to a single-period problem in Carroll (2015) with respect to the updated knowledge of the set of actions, and thus optimal second-period contracts are linear.

Going back to the first period, when the principal chooses a first-period contract to maximize her overall payoff guarantee, we establish the optimality of a linear first-period contract. The proof of this conclusion boils down to two steps. The first step shows that any nonlinear first-period contract can be improved into another linear contract, thereby (weakly) increasing the overall payoff guarantee (Lemma 1). The second step further shows that the maximum of the principal’s first-period problem exists within the class of linear first-period contracts (Lemma 2). Combining these two steps, we show that, even with the opportunity to use any first-period contract for exploration, no other more complicated form of contracts provides a better payoff guarantee to the principal than linear ones.

Moving on to an alternative notion with more restrictions, the case of *constant technology*, we assume both agents share the same set of actions unknown to the principal (Section 4). In other words, nature can neither introduce new actions across periods nor delete old ones. The main result for the case of constant technology is Theorem 2, which shows that linear contracts are robustly optimal in both periods, although *not period-by-period*. Specifically, the second-period analysis shows that, following *nonlinear* first-period contracts, optimal second-period contracts may also be *nonlinear* in some cases. Nonetheless, upon backward induction to the first period, it is robustly optimal to use linear first-period contracts, thereby ensuring optimal second-period contracts are also linear *on the path*.

The reason for obtaining different results compared to the previous case of advancing technology is a more subtle rule of updating. For simplicity of exposition, we assume the principal only knows one action available to the agents.² After observing the action chosen by the first agent, she believes the action set could be any set that (i) contains the observed action in addition to the initially known action, and (ii) does not

²In Appendix B, we show that analogous results hold if the principal knows a general set of know actions.

contain any action strictly better than the observed action under the first-period contract. We refer to such actions sets as *compatible* (Definition 1). Requirement (i) indicates that the principal learns the existence of the chosen action, and requirement (ii) captures the additional inference she can draw from the rationality of the first agent.

The primary distinction from the previous notion of advancing technology lies in the analysis of the second period. This is not a direct adaptation of the single-period problem in Carroll (2015), precisely because the principal draws additional inferences from the rationality of the first agent, which excludes certain actions. Therefore, the analysis of the second period in the case of constant technology is a significant innovation point of this paper from a technical perspective. We fully characterize the principal's *optimal second-period payoff guarantee*, and identify the contract that attains it in various cases. The analysis reveals four ways the principal may respond to the knowledge gained from observing the chosen action (Lemma 3). Specifically, the principal's optimal guarantee is achieved by offering the best among four contracts: (i) the first-period contract again, (ii) a modified version of the first-period contract with compensation for the second agent, and (iii) & (iv) two linear contracts that correspond to the optimal static contracts in Carroll (2015). As long as the first-period contract is nonlinear, and the observed action is such that one of the first two contracts is optimal, then the optimal guarantee is achieved by nonlinear contracts.

As concluding remarks of the paper, we discuss further results. First, we analyze the situation where the principal knows a set of actions available to the agents in the case of constant technology (Appendix B). We characterize the principal's optimal second-period payoff guarantee in closed form, and identify the contract that attains it in various cases (Lemma 3'). In addition, as long as the set of known actions satisfies a condition called *lower bound on marginal cost* (Definition B.1), linear contracts still outperform nonlinear ones (Theorem 2'). Next, we examine the structure of the optimal linear first-period contract in our dynamic model (Appendix C), and compare it with the optimal static contract identified by Carroll (2015).

Related Literature Foundations for linear incentive contracts have received extensive research attention. The seminal work of Holmström and Milgrom (1987) considers a dynamic framework where output is produced gradually over time, the agent is aware of his own progress, and the principal pays the agent at the end. Although the principal is allowed to use the entire history of output to determine the payment, the optimal contract depends only on the number of realizations of each output level, and is linear in these counts. In a continuous time version of their problem where the agent controls the drift of a multidimensional Brownian motion, the optimal contract can be expressed as a linear function that depends only on

the endpoint.³ However, the stationary structure of their model is critical for this linearity result,⁴ because linear contracts provide the agent with constant incentives to move forward independent of her past performance. In our model, the principal offers multiple contracts during the process, and exploration makes the principal’s problem inherently non-stationary. Therefore, our paper considers a different form of foundation for linear contracts. Furthermore, [Diamond \(1998\)](#) and [Barron, Georgiadis, and Swinkels \(2020\)](#) provide arguments for linear contracts using static Bayesian frameworks.

More recently, pioneered by [Carroll \(2015\)](#), this issue has been investigated by a wave of research using robust models of contract design, which demands contract performance to be robust to limited knowledge of the environment. [Carroll \(2019\)](#) provides a comprehensive review of this approach, as well as an overview of the evolving field of robust mechanism design that adopts many other notions of robustness. Most work in robust contract design, however, analyzes static or one-shot models, which precludes the opportunity for designers to better understand parts of the environment they do not know. While starting with nonquantifiable uncertainty, designers may still be able to gradually gain a better understanding of the environment in which they repeatedly engage through exploration. Our dynamic model provides the principal with the opportunity to explore the unknown, in order to understand how the principal should design contracts that are robustly optimal given this exploration opportunity.

As stated by [Carroll \(2019\)](#), “another challenge is that trying to write dynamic models with non-Bayesian decision makers leads to well-known problems of dynamic inconsistency, except in special cases (e.g., [Epstein and Schneider \(2003\)](#)). This may be one reason why there has been relatively little work to date on robust mechanism design in dynamic settings.” Knowing the difficulty, we carefully specify the principal’s “beliefs” in the second period of our two-period model to follow a recursive structure analogous to [Epstein and Schneider \(2003\)](#), in order to avoid dynamic inconsistency issues.

This paper is relevant to the recent research that examines robust contracting in different organizational environments. Specifically, [Dai and Toikka \(2022\)](#) analyze moral hazard in teams, [Marku, Ocampo, and Tondji \(Forthcoming\)](#) study a common agency model, and [Carroll and Bolte \(2023\)](#) investigate a model with double moral hazard. [Walton and Carroll \(2022\)](#) provide a general framework that goes beyond simple bilateral relationships and allows for rich internal organizational structures. Our model analyzes a simple contracting environment, and aims to capture the main issue in terms of exploration. In particular, due to exploration, the analysis of our dynamic model cannot be directly derived using the conclusions in [Walton](#)

³Following [Holmström and Milgrom \(1987\)](#), [Sung \(1995\)](#) further shows that the optimal contract can still be linear when the agent controls the variance; [Hellwig and Schmidt \(2002\)](#) provide discrete time approximations of the continuous time model.

⁴For example, [Schättler and Sung \(1993\)](#) show that a time-dependent technology makes the optimal contract nonlinear.

and Carroll (2022)).⁵

The revealed preference reasoning in this paper is related to the recent work by [Burkett and Rosenthal \(2023\)](#) and [Antic and Georgiadis \(2023\)](#), who consider a static robust contracting problem with revealed preference data. In [Burkett and Rosenthal \(2023\)](#) and [Antic and Georgiadis \(2023\)](#), the principal’s only knowledge is the agent’s best responses to a finite number of given contracts, and she seeks to maximize her worst-case payoffs over all action sets that can rationalize the data. In the second period of our model, the principal’s additional knowledge is exactly the first agent’s best response to the first-period contract. Therefore, our second-period characterization contains a compensation component similar to their results. However, our model differs in that the principal also initially knows certain available action(s), so the structure of the optimal contracts is not exactly the same.⁶ More importantly, in their settings, the principal’s revealed preference data are exogenously provided, whereas our model places a significant emphasis on endogenizing this aspect through the optimal exploration design in the first period.

From a broader perspective, [Marku, Ocampo, and Tondji \(Forthcoming\)](#) and [Carroll and Bolte \(2023\)](#) are in a similar spirit to our work on how the designers’ robust objectives interact with their policy choices. In [Marku, Ocampo, and Tondji \(Forthcoming\)](#), several principals compete to contract with a common agent. In [Carroll and Bolte \(2023\)](#), the principal faces the choice of supplying input in the process of contracting with an agent. However, the maxmin objective in both studies is applied only once, whereas in our model it needs to be used in each of the two periods. In the area of informationally robust mechanism design, [Libgober and Mu \(2023\)](#) study durable good monopoly without commitment, and introduce the notion of *dynamically-consistent* worst-case information structure.

A number of other recent papers considering static models of robust contracts are related to our work, because the principal is aware of some additional characteristics of the unknown actions in addition to the concern that they may exist. As with [Kambhampati \(2024\)](#), who studies performance evaluation of agents, although we do not place any restrictions on the possible action sets of an individual agent, we assume that the two agents have identical action sets. However, our assumption is for a different reason, in order to make the principal’s observations of chosen actions valuable. In addition, [Antic \(2021\)](#) assumes a lower bound on the productivity of all unknown actions of the principal. Furthermore, in [Dütting, Roughgarden, and Talgam-Cohen \(2020\)](#), the principal only knows the first moment of the distribution over output induced by

⁵We articulate the specific differences between our dynamic model and the general static framework in [Walton and Carroll \(2022\)](#) in Subsection 4.2.

⁶Another reason for similar but not identical results is due to the assumption on the observed actions: [Burkett and Rosenthal \(2023\)](#) and [Antic and Georgiadis \(2023\)](#) assume that the distribution of output (but not the effort cost) associated with the best response is observed. Instead, we assume that both the distribution and the cost are observed, as we believe this is more consistent with the assumption on the principal’s initial knowledge.

each possible action, but not the full distribution.

The rest of the paper is organized as follows. Section 2 lays out the baseline model, and analyzes the case of independent technology. The first main part, Section 3, analyzes the case of advancing technology, and show that linear contracts are robustly optimal *period-by-period*. The second main part, Section 4, then analyzes the case of constant technology and shows that, although optimal second-period contracts may be nonlinear in some cases following nonlinear first-period contracts, linear first-period contracts maximize the overall payoff guarantee, ensuring that optimal second-period contracts remain linear *on the path*. Section 5 concludes. Appendix A contains the proofs of all results in the main text. Appendices B and C present further results.

2 Model

2.1 Notation

We denote by $\Delta(X)$ the set of (Borel) probability measures on a set $X \subseteq \mathbb{R}$, equipped with the weak topology. For $x \in X$, we write δ_x for the degenerate distribution that puts probability one on x .

2.2 Setup

The baseline model is a two-period moral hazard problem, consisting of a principal (she) and two agents (he). The principal contracts with each agent sequentially to provide incentives, and the reservation payoff of the agents is zero. All parties are assumed to be risk neutral. The principal's discount factor is $\beta \in (0, \infty)$.

In each period ($t = 1, 2$), agent t takes a costly action that results in a stochastic output. The realized output y belongs to a set Y of possible output values. Assume Y is a compact subset of $\mathbb{R}$, either finite or infinite, and normalize the lowest possible output to zero: $\min(Y) = 0$.

An *action* of the agents, a , is modeled as a pair $a = (F, c) \in \Delta(Y) \times \mathbb{R}^+$, with the interpretation that if an agent chooses action a , he incurs cost c , and output is drawn $y \sim F$. We equip $\Delta(Y) \times \mathbb{R}^+$ with the natural product topology.

A *technology* is a (nonempty and) compact set of possible actions. Agent t has technology $A_t \subseteq \Delta(Y) \times \mathbb{R}^+$, which only they know but the principal does not. The principal general compact set A_0 of available actions. To ensure that the principal may benefit from contracting with the agents, assume that there exists $(F, c) \in A_0$ such that $\mathbb{E}_F[y] - c > 0$.⁷

⁷Note that it is necessary for the principal to know at least one action that guarantees a strictly positive surplus, because otherwise

To capture the idea of exploration, assume that the principal observes the action chosen by agent 1, and then offers a new contract to agent 2 based on this knowledge. The chosen action itself, however, is not contractible.⁸ Payments to the agents can only depend on the realized output, y .

Assume that the agents have limited liability, so the payment to them can never be strictly negative. A *contract* is a continuous⁹ function $w : Y \rightarrow \mathbb{R}^+$ such that $w(0) = 0$. One foundation for $w(0) = 0$ is *two-sided limited liability*,¹⁰ which also requires that the contracts never pay more than output: $0 \leq w(y) \leq y$ for all values of y . We do not explicitly impose two-sided limited liability, but only view it as a possible explanation for $w(0) = 0$.¹¹

The timing within each period t is summarized as follows:

1. The principal offers a contract w_t .
2. Agent t chooses $a_t = (F_t, c_t) \in A_t$, or quits the relationship (zero payoff for both parties).
3. Output $y_t \sim F_t$ is realized.
4. Payoffs $y_t - w_t(y_t)$ to the principal and $w_t(y_t) - c_t$ to agent t .

The principal's objective is to maximize her worst-case expected discounted sum of payoffs over all possible technologies. Therefore, it is crucial to articulate what actions the principal considers possible in each period, and to determine how the principal's beliefs about unknown actions are updated across periods. Addressing this critical gap in the existing literature, we introduce and compare three distinct notions of dynamic worst-case considerations: (i) *independent technology* $A_1 \perp A_2$, (ii) *advancing technology* $A_1 \subseteq A_2$, and (iii) *constant technology* $A_1 = A_2$.

it is always possible that the agents are not able to produce anything of value.

⁸It is a strong assumption that the chosen action becomes observable to the principal, especially since F represents a distribution. One interpretation is that each period summarizes (the "average" state of) a horizon for which the contract needs to remain fixed, while the agent is repeatedly taking action. During this process, the principal can keep observing him and figure out what action must be taken, in particular what F and c are. However, knowing that the action exists is still not the same as being able to write it into a contract. The action itself may be too complex to be accurately described in contract terms, or its inclusion into the contract may be directly prohibited by law.

⁹The continuity assumption is made only to ensure the existence of best responses of the agents. This assumption becomes vacuous if Y is a finite set, and can also be weakened to upper semicontinuity with additional verifications. See also [Carroll \(2015, footnote 1\)](#), [Walton and Carroll \(2022, footnote 3\)](#), [Carroll and Bolte \(2023, footnote 1\)](#).

¹⁰See also [Burkett and Rosenthal \(2023, Definition 6\)](#).

¹¹Another foundation for $w(0) = 0$ is the standard *free disposal condition*, plus a *lowest support condition* on the agents' possible actions. We say a technology A satisfies the *lowest support condition* if, for all $(F, c) \in A$, the lowest output 0 is in the support of F . Under these two conditions, the principal will only offer contracts with $w(y) \geq w(0)$ for all y , because otherwise the agent may discard output to receive more payments. Given limited liability, it is then without loss of generality to focus on contracts with $w(0) = 0$, since a constant shift does not affect the agent's incentives, but only increases the principal's payoff. That is, if $w(0) > 0$, let $\tilde{w}(y) = w(y) - w(0) \geq 0$ be another valid contract. the agent's chosen action does not change if the principal instead offers $\tilde{w}$, but this increases the principal's payoff by $w(0)$.

In the following sections, we define a suitable rule of updating within each notion and characterize the principal’s optimal payoff guarantee. The conclusion is that linear contracts are robustly optimal in all three notions. To better understand the connections and distinctions among the three notions, it is helpful to imagine there is an adversarial “nature” that selects the technology for the corresponding agent in each period to minimize the principal’s payoff. The three notions differ in the restrictions imposed on the moves available to nature across periods.

2.3 Independent Technology

We begin by considering the case of *independent technology* $A_1 \perp A_2$, where the technology of the two agents A_1 and A_2 are not related; in other words, nature has the flexibility to select the technology for each agent independently. In this case, the choice made by agent 1 does not yield any information for the principal regarding the potential actions agent 2 might take. Therefore, the learning aspect is essentially nullified, and the principal’s overall payoff guarantee is maximized by adopting a straightforward approach: offering the optimal static contract identified by [Carroll \(2015\)](#) in both periods.

We briefly recap the analysis in [Carroll \(2015\)](#), as it lays the foundation for subsequent analyses. It is relatively straightforward to describe the behavior of the agents. In each period t , given contract w and technology A , agent t chooses an action $(F, c) \in A$ to maximize his expected utility, so the best response correspondence is given by

$$BR(w|A) \equiv \arg \max_{(F, c) \in A} \{ \mathbb{E}_F [w(y)] - c \}.$$

The principal’s single-period expected payoff under technology A is denoted by

$$V(w|A) \equiv \max_{(F, c) \in BR(w|A)} \mathbb{E}_F [y - w(y)],$$

where we assume ties are broken in the principal’s favor if the agent is indifferent among several actions.¹²

The principal’s objective is to choose a contract w to maximize her worst-case expected payoff

$$V(w) \equiv \inf_{A \supseteq A_0} V(w|A).$$

¹²This tie-breaking assumption ensures the existence of optimal contracts, and minimizes the departure from standard models. Other tie-breaking rules will lead to essentially the same results, but may introduce technical complications. For example, the principal’s optimal payoff guarantee may be approached, but not achieved, by linear contracts. See also [Carroll \(2015, Section D\)](#), [Dai and Toikka \(2022, footnote 4\)](#), [Carroll and Bolte \(2023\)](#).

The key result of [Carroll \(2015\)](#) is that the principal's optimal single-period payoff guarantee, $\max_w V(w)$, is attained by a linear contract $w(y) = sy$. Specifically, the solution to the principal's static problem can be summarized as follows:

1. Maximize $\sqrt{\mathbb{E}_F[y]} - \sqrt{c}$ over $(F, c) \in A_0$, with solution $a^* = (F^*, c^*)$.
2. Set $s^* = \sqrt{c^*/\mathbb{E}_{F^*}[y]}$ as the share, and offer linear contract $w(y) = s^*y$.

The resulting optimal guarantee is equal to $(\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*})^2$. Consequently, in the case of independent technology, the principal's *overall payoff guarantee* is maximized by offering $w_1(y) = w_2(y) = s^*y$, and is equal to $(1 + \beta)(\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*})^2$.

3 Advancing Technology

The case of independent technology might be overly pessimistic, as it completely prevents the principal from learning about the technology through agent 1's actions. Essentially, with no restriction on nature's moves, the principal is hindered from learning through exploration. In this section, we analyze the first restriction that facilitates meaningful exploration: the case of *advancing technology* $A_1 \subseteq A_2$. Here, the technology may advance between periods, but cannot downgrade. In other words, nature can only introduce new actions across periods, but is not allowed to delete old ones.

The main result for the case of advancing technology is Theorem 1, which shows that linear contracts are robustly optimal *period-by-period*. That is, linear contracts are also optimal in terms of utilizing the exploration opportunity, making them even more robust.

3.1 Rule of Updating and Second Period Analysis

As in the previous case of independent technology, the principal maximizes her worst-case expected discounted sum of payoffs over all possible technologies. In the first period, she believes that agent 1's technology A_1 could be any technology such that $A_1 \supseteq A_0$. Taking into account possible technological advances after the first period, the principal's rule of updating is defined as follows:

$$\begin{aligned} &\text{After the principal offers contract } w_1 \text{ and observes the action } a_1 \text{ chosen by agent 1, she} \\ &\text{believes that agent 2's technology } A_2 \text{ could be any technology such that } A_2 \supseteq A_0 \cup \{a_1\}. \end{aligned} \tag{1}$$

That is, the principal learns that action a_1 exists in A_1 (in addition to the initially known set A_0), and believes that agent 2 may also choose this action again (since $A_1 \subseteq A_2$). Moreover, this represents the best conjecture

the principal can make in the second period, given that nature may introduce new actions that were not present in the first period.

We solve the principal's dynamic problem via backward induction. With the update rule (1), the principal's second-period problem simplifies to a single-period problem in Carroll (2015). Specifically, in the second period, the principal chooses a second-period contract w_2 to maximize her worst-case payoff

$$V_2(w_2|a_1) \equiv \inf_{A_2 \supseteq A_0 \cup \{a_1\}} V(w_2|A_2).$$

Applying Carroll (2015)'s result to the updated knowledge on technology, we conclude that the optimal second-period contract is linear, and the resulting *optimal second-period payoff guarantee* is $V_2^*(a_1) = \Phi(a_1)^2$, where

$$\Phi(a_1) \equiv \max_{a \in A_0 \cup \{a_1\}} \left\{ \sqrt{\mathbb{E}_{F_a}[y]} - \sqrt{c_a} \right\}. \quad (2)$$

Note, here and throughout the analysis below, we denote the output distribution and cost associated with any generic action a by F_a and c_a , respectively.

3.2 First Period Analysis

Going back to the first period, if the principal offers the first-period contract w_1 and agent 1 chooses action $a_1 = (F_1, c_1)$, her *interim payoff guarantee*, defined as her payoff in the first period plus the discounted optimal second-period payoff guarantee, is given by

$$U(w_1|a_1) \equiv \mathbb{E}_{F_1}[y - w_1(y)] + \beta \cdot V_2^*(a_1).$$

Since she believes that agent 1's true technology A_1 could be any technology such that $A_1 \supseteq A_0$, her *overall payoff guarantee*, defined as the worst-case interim payoff guarantee over all possible technologies A_1 , is given by

$$U(w_1) \equiv \inf_{A_1 \supseteq A_0} \left\{ \max_{a_1 \in BR(w_1|A_1)} U(w_1|a_1) \right\},$$

where, once again, we assume ties are broken in her favor.

The principal's first-period problem is to choose a first-period contract w_1 to maximize her overall payoff guarantee $U(w_1)$. We are now ready to state the main result for this section, Theorem 1, which shows the maximum exists and is achieved by a linear contract.

Theorem 1. *In the case of advancing technology, there exists a linear first-period contract w_1 that maximizes the principal's overall payoff guarantee $U(w_1)$.*

Even with the opportunity to use the first-period contract as a means of exploration, no other more complicated form of contracts provides the principal with a better payoff guarantee than linear ones.

The proof of Theorem 1 boils down to two steps. The first step, Lemma 1, shows that any nonlinear first-period contract is outperformed by some linear one. The second step, Lemma 2, further shows that the maximum of the principal's first-period problem exists within the class of linear first-period contracts.

3.2.1 Proof Step 1: Improving Nonlinear Contracts

We start from any arbitrary first-period contract w_1 , and construct another linear contract $\hat{w}_1$ that provides the principal with a weakly higher overall payoff guarantee. Thus, any nonlinear contract can be improved by a linear one.

For any first-period contract w_1 , let $(F_0, c_0) \in A_0$ be agent 1's best response when his technology is A_1 is just the initially known A_0 , and let $\hat{w}_1$ denote the following linear contract:

$$\hat{w}_1(y) = s_1 y \quad \text{with} \quad s_1 = \frac{\mathbb{E}_{F_0}[w_1(y)]}{\mathbb{E}_{F_0}[y]} \geq 0. \quad (3)$$

The procedure of constructing the linear $\hat{w}_1$ is depicted in Figure 1. The solid curve represents first-period

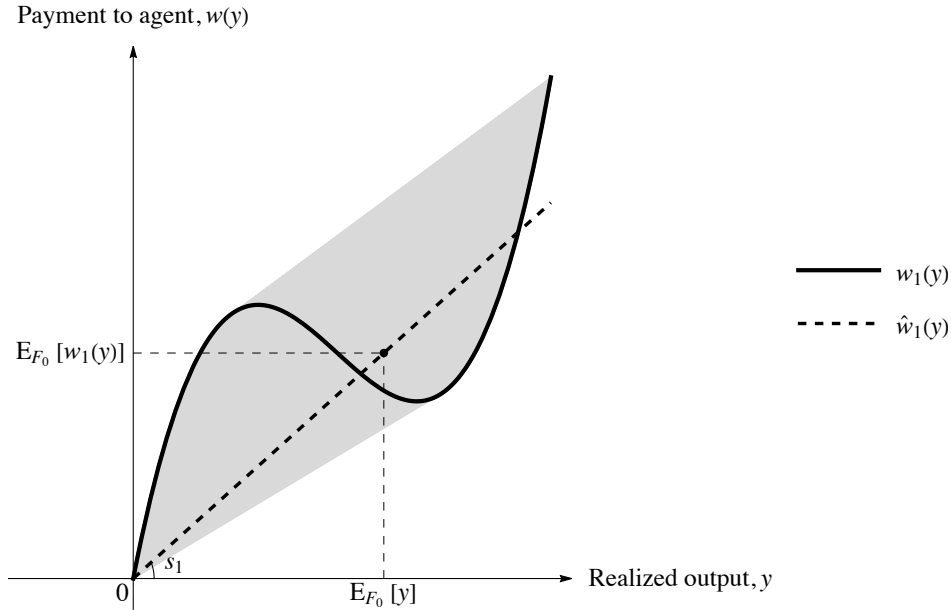


Figure 1: The linear contract $\hat{w}_1$ constructed from w_1 .

contract w_1 , which may be nonlinear and non-monotonic. Consider the point $(\mathbb{E}_{F_0}[y], \mathbb{E}_{F_0}[w_1(y)])$, whose

coordinates are the expected output and the expected payment to agent 1 if he takes action $a_0 = (F_0, c_0)$. This point must lie within the convex hull of the curve w_1 , represented by the shaded area in the figure. The constructed linear contract $\hat{w}_1$ is exactly the dashed line connecting the origin and this point, with a corresponding slope denoted by s_1 .

Note that the linear contract $\hat{w}_1$ is chosen such that if agent 1 takes the action a_0 , his payoff will be exactly equal under $\hat{w}_1$ as under w_1 :

$$\mathbb{E}_{F_0} [\hat{w}_1(y)] - c_0 = s_1 \mathbb{E}_{F_0} [y] - c_0 = \mathbb{E}_{F_0} [w_1(y)] - c_0.$$

We will show that the principal's overall payoff guarantee is at least as high under $\hat{w}_1$ as it is under w_1 ; that is, $U(\hat{w}_1) \geq U(w_1)$.¹³

Lemma 1. *Let w_1 be any first-period contract. The linear contract $\hat{w}_1$ defined by equation (3) satisfies $U(\hat{w}_1) \geq U(w_1)$.*

Proof. All proofs of the results in the main text are in Appendix A. □

Suppose the principal offers the linear first-period contract $\hat{w}_1$, and agent 1 chooses action a_1 from the true technology A_1 . We need to show that the principal's interim payoff guarantee, $U(\hat{w}_1|a_1)$, is at least $U(w_1)$. If there exists another action a'_1 , which may be taken by agent 1 under w_1 and some other technology A'_1 , such that

$$U(\hat{w}_1|a_1) \geq U(w_1|a'_1) \tag{4}$$

holds, then $U(\hat{w}_1|a_1) \geq U(w_1|a'_1) \geq U(w_1)$, and thus the desired conclusion is established. The proof of Lemma 1 explicitly constructs such an alternative action a'_1 for each possible a_1 .

Specifically, the principal's interim payoff guarantee consists of two parts, her payoff in the first period, plus the discounted optimal second-period payoff guarantee. The characterization of the second part in the previous subsection is crucial for the construction of a'_1 , enabling the desired inequality (4) to hold period by period: under $(\hat{w}_1|a_1)$, the principal's payoff in the first period and her guarantee in the second period are both higher than under $(w_1|a'_1)$.

By establishing Lemma 1, we have shown that any nonlinear first-period contract can be improved by a linear one. To finalize the proof of Theorem 1, it suffices to show that, within the class of linear

¹³Unlike the main text of Carroll (2015), which uses linear relations between the principal's and agent's payoffs to characterize the payoff guarantee of any contract, this is an adaptation of the alternative approach suggested by Lucas Maestri in Carroll (2015, Appendix C) to the two-period model.

contracts, the maximum of $U(w_1)$ exists. We will set up a program that characterizes the principal's overall payoff guarantee of an arbitrary linear first-period contract, and prove the existence of maximum through its continuity in the first-period share.

3.2.2 Proof Step 2: Payoff Guarantee of a Linear Contract

To conclude the proof of Theorem 1, we need to establish the following Lemma 2.

Lemma 2. *Within the class of linear first-period contracts, there exists an optimal one for the principal.*

The proof of Lemma 2 requires characterizing the overall payoff guarantee of an arbitrary linear first-period contract, which is the main focus here.

Assume the principal offers a linear first-period contract $w_1(y) = s_1 y$ with $s_1 \in [0, 1]$, and agent 1 chooses $a_1 = (F_1, c_1)$ in response. The principal's optimal second-period payoff guarantee $V_2^*(a_1) = \Phi(a_1)^2$, with Φ defined by equation (2). Thus, her interim payoff guarantee is

$$U(w_1|a_1) = \mathbb{E}_{F_1}[y - w_1(y)] + \beta \cdot V_2^*(a_1) = (1 - s_1) \mathbb{E}_{F_1}[y] + \beta \cdot \Phi(a_1)^2.$$

The worst-case overall payoff guarantee minimizes the above expression over all a_1 that agent 1 may choose under some technology A_1 . Note that agent 1 prefers action a_1 over all known actions $a \in A_0$ if and only if

$$(\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_a}[w_1(y)] - c_a) = (s_1 \mathbb{E}_{F_1}[y] - c_1) - (s_1 \mathbb{E}_{F_a}[y] - c_a) \geq 0, \quad \forall a \in A_0.$$

Moreover, agent 1 obtains at least his reservation payoff of zero, which can also be viewed as his payoff from the null action $(\delta_0, 0)$ that produces zero output at zero cost. Hence, the following program yields a lower bound on the principal's overall payoff guarantee

$$\begin{aligned} \inf_{F_1, c_1} \quad & (1 - s_1) \mathbb{E}_{F_1}[y] + \beta \cdot \Phi(F_1, c_1)^2 \\ \text{s.t.} \quad & (s_1 \mathbb{E}_{F_1}[y] - c_1) - (s_1 \mathbb{E}_{F_a}[y] - c_a) \geq 0, \quad \forall a \in A_0 \cup \{(\delta_0, 0)\}, \end{aligned} \tag{5}$$

because the principal's interim payoff guarantee can never be strictly lower than the infimum given by program (5).

Conversely, for any feasible $a_1 = (F_1, c_1)$ in program (5), agent 1 would take action a_1 in response to w_1 when his technology $A_1 = A_0 \cup \{a_1\}$. The worst case over all such technologies leaves the principal with exactly her interim payoff guarantee, $U(w_1|a_1) = (1 - s_1) \mathbb{E}_{F_1}[y] + \beta \cdot \Phi(a_1)^2$. Thus, if a solution to program

(5) exists (i.e., if infimum may be replaced by minimum), then the principal's payoff guarantee cannot be strictly higher than its minimum value.

The above analysis shows that the worst-case overall payoff guarantee of any linear first-period contract $w_1(y) = s_1 y$ is exactly characterized by program (5). In the proof of Lemma 2 in Appendix A.1, we formally show the existence of minimum in this program, and its continuity in the first-period share s_1 . We first reformulate program (5) as an equivalent maximization problem with continuous objective function and compact feasible region, and then invoke Berge's maximum theorem to prove the required existence and continuity. Since the overall payoff guarantee of a linear first-period contract $w_1(y) = s_1 y$ is continuous in the first-period share s_1 , it achieves a maximum. This maximum is also the optimal guarantee over all linear contracts.

Specifically, under a linear first-period contract w_1 , the expression of $V_2^*(a_1) = \Phi(a_1)^2$ given by equation (2) gets simplified, thus showing that both the objective and the constraint of program (5) depend on the choice variables (F_1, c_1) only through the value of $(\mathbb{E}_{F_1}[y], c_1)$, and are continuous. To complete the proof, we only need to show that the value of $(\mathbb{E}_{F_1}[y], c_1)$ can be restricted to a compact region without affecting the infimum value of program (5), and that region changes in a continuous¹⁴ manner when the first period share s_1 changes.

Combining Lemmas 1 and 2, we prove the main result of this section, Theorem 1, which establishes the optimality of a linear first-period contract.

4 Constant Technology

In the previous section, we have focused on the case of advancing technology ($A_1 \subseteq A_2$) and show that linear contracts are robustly optimal period-by-period in that notion of dynamic worst-case consideration. This section analyzes an alternative notion with more restrictions: the case of *constant technology* $A_1 = A_2 = A$. Here, the two agents have the same action set unknown to the principal. In other words, nature can neither introduce new actions across periods nor delete old ones.

For simplicity of exposition, assume the principal knows only one action $a_0 = (F_0, c_0) \in A$ available to the agents, with $\mathbb{E}_{F_0}[y] - c_0 > 0$. In Appendix B, we show that analogous results hold if the principal knows a general set of known actions A_0 as in the baseline model.

The main result for the case of constant technology is Theorem 2, which shows that linear contracts are robustly optimal in both periods, although *not period-by-period*. Specifically, second period analysis

¹⁴In the language of correspondences, both upper and lower hemicontinuous.

(Subsection 4.2) shows that, following nonlinear first-period contracts, optimal second-period contracts may also be nonlinear in some cases. Nonetheless, upon backward induction to the first period (Subsection 4.3), it is robustly optimal to use linear first-period contracts, so optimal second-period contracts are also linear *on the path*. The reason for obtaining different results compared to the previous case of advancing technology is due to a different and more subtle rule of updating, which we refer to as *compatibility* (Definition 1).

4.1 Rule of Updating: Compatibility

As in the previous two cases, the principal maximizes her worst-case expected discounted sum of payoffs over all possible technologies. In the first period, she only knows the action a_0 , and believes that the true technology A could be any technology such that $A \ni a_0$. After the principal offers contract w_1 and observes the action a_1 chosen by agent 1, a rule of updating needs to be specified to determine the technologies that the principal considers possible. We say those possible technologies *compatible* with (w_1, a_1) , formally defined as follows.¹⁵

Definition 1 (Compatible). *Given w_1 and $a_1 = (F_1, c_1)$, a technology A is compatible with (w_1, a_1) if*

1. $A \supseteq \{a_0, a_1\}$.
2. $\mathbb{E}_F [w_1(y)] - c \leq \mathbb{E}_{F_1} [w_1(y)] - c_1$ for all $(F, c) \in A$.

Roughly speaking, a technology A is compatible with (w_1, a_1) if it contains a_1 (in addition to a_0), and does not contain any action strictly better than a_1 under w_1 . The first requirement in Definition 1 indicates that the principal learns that action a_1 exists (in addition to the initially known a_0), and believes that agent 2 may also take this action again. The second requirement in Definition 1 captures the additional inference she can draw from agent 1's rationality in this case of constant technology: the true technology A cannot contain any action (F, c) that leads to a strictly higher payoff for agent 1, i.e., it is impossible that $\mathbb{E}_F [w_1(y)] - c > \mathbb{E}_{F_1} [w_1(y)] - c_1$.

The principal's dynamic problem is again solved via backward induction. In the second period, since the principal believes that A could be any technology compatible with (w_1, a_1) , her problem is to choose a second-period contract w_2 to maximize her worst-case payoff:

$$V_2(w_2|w_1, a_1) \equiv \inf_{A \text{ compatible with } (w_1, a_1)} V(w_2|A).$$

¹⁵This is an analogue of *consistency* in solution concepts like perfect Bayesian equilibrium.

Note that this is not a direct adaptation of the single-period problem in [Carroll \(2015\)](#) (where A could be any technology containing $\{a_0, a_1\}$), precisely because of her additional inference from agent 1's rationality in the definition of compatibility, which rules out the possibility that certain actions exist in A . In Subsection 4.2, we characterize the principal's optimal second-period payoff guarantee, $\hat{V}_2^*(w_1, a_1)$, showing that this distinction matters. The maximum always exists, as we identify the contract that attains it; however, it may be achieved by a nonlinear w_2 if the corresponding w_1 is nonlinear.

Going back to the first period, if the principal offers first-period contract w_1 and the true technology A is such that agent 1 chooses action $a_1 = (F_1, c_1)$, her interim payoff guarantee is given by

$$\hat{U}(w_1|a_1) \equiv \mathbb{E}_{F_1}[y - w_1(y)] + \beta \cdot \hat{V}_2^*(w_1, a_1).$$

Since she believes that the true technology A could be any technology such that $A \ni a_0$, her overall payoff guarantee is given by

$$\hat{U}(w_1) \equiv \inf_{A \ni a_0} \left\{ \max_{a_1 \in BR(w_1|A)} \hat{U}(w_1|a_1) \right\},$$

where again we assume ties are broken in her favor.

The principal's first-period problem is to choose a first-period contract w_1 to maximize her overall payoff guarantee. In Subsection 4.3, we show the maximum exists and is achieved by a linear contract.

4.2 Second Period Analysis

We begin our analysis with the second period of the dynamic relationship, where the principal has offered some first-period contract w_1 and observed agent 1's selected action a_1 . We fully characterize the principal's optimal second-period payoff guarantee, $\hat{V}_2^*(w_1, a_1)$, and identify the contract that attains it in various cases. The analysis reveals four ways the principal may respond to the knowledge gained from observing a_1 , and in particular shows that if w_1 is nonlinear, then the optimal second-period payoff guarantee may be achieved by a nonlinear w_2 .

The main result for the second period analysis is Lemma 3, which shows that $\hat{V}_2^*(w_1, a_1)$ is achieved by offering the best among four contracts: (i) the first-period contract w_1 again, (ii) a modified w_1 with compensation for agent 2, and (iii) & (iv) two linear contracts that correspond to the optimal static contracts in [Carroll \(2015\)](#). As long as the first-period contract w_1 is nonlinear, and the observed action a_1 is such that one of the first two contracts is optimal, then $\hat{V}_2^*(w_1, a_1)$ is achieved by nonlinear contracts.

Lemma 3 reveals that the analysis in this section is not a direct adaptation of the single-period problem in [Carroll \(2015\)](#), since optimal contracts may not be linear. This difference is precisely due to the second

requirement of compatibility, where the principal draws additional inferences from the rationality of agent 1, excluding certain actions. Note that the analysis is also not covered by the recent work of [Walton and Carroll \(2022\)](#), which establishes a general static framework that allows for rich organizational structures, and identifies two properties of the counterparty's possible responses which jointly imply that a linear contract solves the principal's single-period maxmin problem. Specifically, their *Richness* property requires that the set of possible responses to a given contract be sufficiently and unboundedly broad. The Richness property is violated in the case of constant technology exactly because of the principal's exploration and inference in the first period, since the true technology cannot contain any action that is strictly better for agent 1 than the observed action under the first-period contract.¹⁶

Suppose that in the first period the principal offers contract w_1 and observes agent 1's action $a_1 = (F_1, c_1)$. She learns that the true technology A is compatible with (w_1, a_1) ; that is, it contains a_0 and a_1 , and does not contain any action strictly better than a_1 for agent 1 under w_1 .

In the second period, if she offers the same contract $w_2 = w_1$, then she knows that agent 2 will choose a_1 again because the two agents have the same technology. This exactly repeats her first-period payoff $\mathbb{E}_{F_1}[y - w_1(y)]$ in the second period. Part 1 of Lemma 3 below shows that, in some cases, doing so is already optimal for the principal, which means that an optimal second-period contract may be nonlinear following nonlinear first-period contracts.

Offering the same contract again is only one response of the principal to the knowledge gained by observing a_1 , and there are plenty of other possible responses. For example, if the initially known action a_0 may lead to a higher payoff for the principal (i.e., $\mathbb{E}_{F_0}[y - w_1(y)] > \mathbb{E}_{F_1}[y - w_1(y)]$), then it might be tempting for the principal to try to obtain the payoff $\mathbb{E}_{F_0}[y - w_1(y)]$ instead. However, achieving this payoff requires the principal to use w_1 to induce action a_0 , and this would violate agent 2's incentive constraint. Indeed, in the first period, the chosen action a_1 provides agent 1 with a (weakly) higher payoff compared to the known action a_0 , and this relationship gets transferred to the second period because both agents have the same technology. This gives rise to the following notion of the *incentive gap*.

Definition 2 (Incentive gap). *Given w_1 and $a_1 = (F_1, c_1)$, the incentive gap, $g(w_1, a_1)$, denotes the difference*

¹⁶The other property in [Walton and Carroll \(2022\)](#), *Responsiveness*, indicates that the counterparty's behavior is responsive to the incentive provided by expected payment, and allows comparison of the principal's payoff guarantees from two different contracts. The Responsiveness property is satisfied in our model. As a converse result, [Walton and Carroll \(2022\)](#) also show that Responsiveness is necessary for linearity under a strengthened version of Richness. This result is in parallel with our analysis, since it is Richness that is not satisfied in our model.

in agent 1's payoff between choosing a_1 and a_0 . Formally,

$$g(w_1, a_1) \equiv (\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_0}[w_1(y)] - c_0).$$

If the principal wants to induce action a_0 using a contract “similar to” w_1 , then agent 2 needs to be compensated for not choosing a_1 , and the amount of compensation increases with the incentive gap $g(w_1, a_1)$. Part 2 of Lemma 3 shows that the incentive gap sometimes becomes a real cost. Specifically, if $\mathbb{E}_{F_0}[y - w_1(y)] > g(w_1, a_1)$, then the principal can offer to agent 2 a modified version of w_1 with compensation in order to guarantee that her payoff in the second period is at least $(\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g(w_1, a_1)})^2$. Moreover, the proof of Lemma 3 shows that this is the optimal payoff guarantee using a modified version of w_1 . Note that if the incentive gap is small, this value becomes close to $\mathbb{E}_{F_0}[y - w_1(y)]$, and may be better for the principal than simply offering $w_2 = w_1$ again.

After observing a_1 , the principal learns that the true technology A must contain $\{a_0, a_1\}$. If the principal ignores the second requirement of compatibility (Definition 1) and applies the single-period problem in Carroll (2015), her optimal guarantee would be equal to $(\max\{\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0}, \sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}\})^2$, achieved by offering the better one of the two linear contracts, $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_0/\mathbb{E}_{F_0}[y]}$ or $s_2 = \sqrt{c_1/\mathbb{E}_{F_1}[y]}$. With the additional inference in place, the guarantee from this procedure can only increase. Parts 3 and 4 of Lemma 3 show that, when this payoff guarantee is larger than the previous two cases (w_1 again, or a modified w_1 with compensation), it is optimal for the principal to offer the better of the two linear contracts, and doing so exactly attains this payoff guarantee.

We are now ready to present the main result of this subsection, Lemma 3, which establishes the optimality of the aforementioned contracts. The principal's optimal second-period payoff guarantee is achieved by offering the best among the four contracts described above: w_1 again, modified w_1 with compensation, and the two linear contracts.

Lemma 3. *Suppose the principal offers first-period contract w_1 , and agent 1 chooses $a_1 = (F_1, c_1)$ in response. The principal's optimal second-period payoff guarantee is $\hat{V}_2^*(w_1, a_1) = \hat{\Phi}(w_1, a_1)^2$, where*

$$\hat{\Phi}(w_1, a_1) \equiv \max\left\{\sqrt{\mathbb{E}_{F_1}[y - w_1(y)]}, \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g(w_1, a_1)}, \sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0}, \sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}\right\}$$

(with $\sqrt{x} = -\infty$ for $x < 0$ by convention). (6)

Specifically,

1. If $\sqrt{\mathbb{E}_{F_1}[y - w_1(y)]}$ attains the maximum in equation (6), then the principal's optimal second-period

payoff guarantee is achieved by $w_2 = w_1$.

2. If $\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g(w_1, a_1)}$ attains the maximum in equation (6), then the principal's optimal second-period payoff guarantee is achieved by

$$w_2(y) = w_1(y) + m \cdot (y - w_1(y)) \quad \text{with} \quad m = \sqrt{\frac{g(w_1, a_1)}{\mathbb{E}_{F_0}[y - w_1(y)]}} \in [0, 1]. \quad (7)$$

3. If $\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0}$ attains the maximum in equation (6), then the principal's optimal second-period payoff guarantee is achieved by $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_0 / \mathbb{E}_{F_0}[y]}$.
4. If $\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}$ attains the maximum in equation (6), then the principal's optimal second-period payoff guarantee is achieved by $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_1 / \mathbb{E}_{F_1}[y]}$.

The proof of Lemma 3 mainly consists of two parts. The first part is to prove that, when each element in the quadruple defined by equation (6) attains the maximum, the principal's payoff guarantee in the second period from offering the corresponding contract is exactly as claimed in the statement of Lemma 3. This requires providing lower bounds on the principal's second-period payoffs, and constructing worst-case technologies to show that the bounds are tight. The second part is to show that, under arbitrary second-period contracts, the principal's payoff guarantee is not strictly higher than $\hat{\Phi}(w_1, a_1)^2$. This requires constructing worst-case technologies to show that the payoff guarantee is lower than (the square of) at least one of element in the quadruple.

Note that compared to the case of advancing technology, the principal acquires more knowledge from the observation of a_1 under constant technology. As an implication, her optimal second-period payoff guarantee takes a more complex form that depends directly on the first-period contract w_1 : how you exploit is related to how you explore.

Lemma 3 indicates that, as long as the first-period contract w_1 is nonlinear, and the observed action a_1 is such that one of the first two elements in the quadruple defined by equation (6) attains the maximum, then the principal's optimal second-period guarantee $\hat{V}_2^*(w_1, a_1)$ is achieved by nonlinear contracts. On the other hand, for linear first-period contracts w_1 , the four contracts mentioned in the statement of Lemma 3 are all linear. This shows that optimal way for the principal to respond to the knowledge gained is closely related to the specific approach she chooses to explore in the first period.

4.3 First Period Analysis

In the previous subsection, we have focused on principal's problem in the second period and fully characterized her optimal second-period payoff guarantee. This section analyzes the principal's first-period problem in the dynamic relationship, that is, choosing a first-period contract w_1 to maximize her overall payoff guarantee $\hat{U}(w_1)$.

We first state the main result of this section, Theorem 2, which establishes the optimality of a linear first-period contract.

Theorem 2. *In the case of constant technology, there exists a linear first-period contract w_1 that maximizes the principal's overall payoff guarantee $\hat{U}(w_1)$.*

The principal's optimal overall payoff guarantee is achieved through a linear first-period contract, together with an optimally chosen linear second-period contract.

Similar to Theorem 1, the proof of Theorem 2 takes two steps: (1) improve any nonlinear first-period contract to a linear one; (2) prove that the maximum of the principal's first-period problem exists within the class of linear first-period contracts. Since the principal's optimal second-period payoff guarantee in the previous subsection takes a more complicated form (equation (6)), the proof here is more lengthy, but the main idea remains the same. In particular, the closed-form characterization is very useful. First, it provides a tool to compare the overall payoff guarantee between different first-period contracts, essential for showing that any nonlinear first-period contract can be improved by a linear one. Second, the expression (6) is the maximum of four continuous functions (in the appropriate sense of continuity), and the continuity is key to show existence of an optimal linear contract.

Although Lemma 3 shows that, following nonlinear first-period contracts, optimal second-period contracts may also be nonlinear in some cases, here we demonstrate that the principal's optimal overall payoff guarantee is achieved by a linear first-period contract (along with an optimally chosen linear second-period contract). The principal has the opportunity to explore in the first period, and linear first-period contracts are optimal in terms of utilizing the exploration opportunity, making them even more robust.

5 Conclusion

In this paper, we study a two-period moral hazard problem, where the principal does not know the action sets available to the agents and demands contracts to be robust to this uncertainty; she has the opportunity to explore in the first period and observes the chosen action, and then offers a new contract to the second

agent based on this knowledge. We introduce and compare three different notions of dynamic worst-case considerations. Within each notion, we define a suitable rule of updating and characterize the principal’s optimal payoff guarantee, thereby identifying how the principal should respond to knowledge and design new contracts. The results show that linear contracts are robustly optimal not just in static settings, but also in dynamic environments with exploration.

We consider a contribution of this paper to propose possible ways to extend robust models in mechanism design to allow for multiple interactions and exploration. Despite the presence of nonquantifiable uncertainty, designers can gradually improve their understanding of the environment in which they repeatedly engage, using the appropriate rule of updating. We hope the generalizability of this approach across other models will be further explored in future work.

References

- Antic, Nemanja, (2021). “Contracting with Unknown Technologies.” Working paper.
- Antic, Nemanja and George Georgiadis, (2023). “Robust Contracts: A Revealed Preference Approach.” Working paper.
- Barron, Daniel, George Georgiadis, and Jeroen M. Swinkels, (2020). “Optimal Contracts with a Risk-taking Agent.” *Theoretical Economics* 15 (2):715–761.
- Burkett, Justin and Maxwell Rosenthal, (2023). “Data-Driven Contract Design.” Working paper. Available at SSRN: <https://ssrn.com/abstract=4672184>.
- Carroll, Gabriel, (2015). “Robustness and Linear Contracts.” *American Economic Review* 105 (2):536–63.
- , (2019). “Robustness in Mechanism Design and Contracting.” *Annual Review of Economics* 11 (1):139–166.
- Carroll, Gabriel and Lukas Bolte, (2023). “Robust Contracting under Double Moral Hazard.” *Theoretical Economics* 18 (4):1623–1663.
- Dai, Tianjiao and Juuso Toikka, (2022). “Robust Incentives for Teams.” *Econometrica* 90 (4):1583–1613.
- Diamond, Peter, (1998). “Managerial Incentives: On the Near Linearity of Optimal Compensation.” *Journal of Political Economy* 106 (5):931–957.

- Dütting, Paul, Tim Roughgarden, and Inbal Talgam-Cohen, (2020). “Simple versus Optimal Contracts.” Working paper. Available at arXiv: <https://arxiv.org/abs/1808.03713>.
- Epstein, Larry G. and Martin Schneider, (2003). “Recursive Multiple-Priors.” *Journal of Economic Theory* 113 (1):1–31.
- Hellwig, Martin F. and Klaus M. Schmidt, (2002). “Discrete-Time Approximations of the Holmström-Milgrom Brownian-Motion Model of Intertemporal Incentive Provision.” *Econometrica* :2225–2264.
- Holmström, Bengt and Paul Milgrom, (1987). “Aggregation and Linearity in the Provision of Intertemporal Incentives.” *Econometrica* :303–328.
- Kambhampati, Ashwin, (2024). “Robust Performance Evaluation of Independent and Identical Agents.” Working paper.
- Libgober, Jonathan and Xiaosheng Mu, (2023). “Coasian Dynamics under Informational Robustness.” Working paper. Available at arXiv: <https://arxiv.org/abs/2202.04616>.
- Marku, Keler, Sergio Ocampo, and Jean-Baptiste Tondji, (Forthcoming). “Robust Contracts in Common Agency.” *The RAND Journal of Economics* .
- Schättler, Heinz and Jaeyoung Sung, (1993). “The First-Order Approach to the Continuous-Time Principal-Agent Problem With Exponential Utility.” *Journal of Economic Theory* 61 (2):331–371.
- Sung, Jaeyoung, (1995). “Linearity With Project Selection and Controllable Diffusion Rate in Continuous-Time Principal-Agent Problems.” *The RAND Journal of Economics* :720–743.
- Walton, Daniel and Gabriel Carroll, (2022). “A General Framework for Robust Contracting Models.” *Econometrica* 90 (5):2129–2159.
- Wilson, Robert, (1987). “Game-Theoretic Analyses of Trading Processes.” In *Advances in Economic Theory: Fifth World Congress*, edited by Truman Fassett Bewley. Cambridge University Press, 33–70.

Appendix A Proofs of Results in the Main Text

A.1 Proofs for Section 3

Proof of Lemma 1. Consider an arbitrary action $a_1 = (F_1, c_1)$ agent 1 would take under contract $\hat{w}_1$. We need to show that the principal's interim payoff guarantee, $U(\hat{w}_1|a_1)$, is at least $U(w_1)$. Note that

$$U(\hat{w}_1|a_1) = \mathbb{E}_{F_1}[y - \hat{w}_1(y)] + \beta \cdot V_2^*(a_1),$$

where $V_2^*(a_1) = \Phi(a_1)^2$ with

$$\Phi(a_1) = \max_{a \in A_0 \cup \{a_1\}} \left\{ \sqrt{\mathbb{E}_{F_a}[y]} - \sqrt{c_a} \right\}.$$

It suffices to construct another action a'_1 , which may be taken by agent 1 under w_1 and some other technology, such that $U(w_1|a'_1) \leq U(\hat{w}_1|a_1)$. By assumption, a_0 is agent 1's best response if $A_1 = A_0$, so an action a'_1 may be taken by agent 1 under w_1 if and only if his payoff from choosing a'_1 is higher than from choosing a_0 . Consider the following two cases.

Case 1. $\mathbb{E}_{F_1}[y] \geq \mathbb{E}_{F_0}[y]$.

Let $a'_1 = a_0$. When agent 1 takes action a_0 in response to w_1 , the principal's resulting payoff in the first period is

$$\mathbb{E}_{F_0}[y - w_1(y)] = (1 - s_1) \mathbb{E}_{F_0}[y] \leq (1 - s_1) \mathbb{E}_{F_1}[y] = \mathbb{E}_{F_1}[y - \hat{w}_1(y)],$$

so her payoff in the first period under $(w_1|a_0)$ is weakly lower than under $(\hat{w}_1|a_1)$.

Moreover, the principal's optimal second-period payoff guarantee is $V_2^*(a_0) = \Phi(a_0)^2$ with

$$\Phi(a_0) = \max_{a \in A_0} \left\{ \sqrt{\mathbb{E}_{F_a}[y]} - \sqrt{c_a} \right\}.$$

By definition we have $0 < \Phi(a_0) \leq \Phi(a_1)$, which implies $V_2^*(a_0) \leq V_2^*(a_1)$. The principal's interim payoff guarantee is

$$\begin{aligned} U(w_1|a_0) &= \mathbb{E}_{F_0}[y - w_1(y)] + \beta \cdot V_2^*(a_0) \\ &\leq \mathbb{E}_{F_1}[y - \hat{w}_1(y)] + \beta \cdot V_2^*(a_1) = \hat{U}(\hat{w}_1|a_1), \end{aligned}$$

as desired.

Case 2. $\mathbb{E}_{F_1}[y] < \mathbb{E}_{F_0}[y]$.

Let $\lambda = \mathbb{E}_{F_1}[y]/\mathbb{E}_{F_0}[y] \in [0, 1]$ and let F'_1 be the mixture $\lambda F_0 + (1 - \lambda) \delta_0$. Note that $\mathbb{E}_{F'_1}[y] = \mathbb{E}_{F_1}[y]$. Consider $a'_1 = (F'_1, c_1)$. Note that

$$\mathbb{E}_{F'_1}[w_1(y)] - c_1 = \lambda \mathbb{E}_{F_0}[w_1(y)] - c_1 = \lambda s_1 \mathbb{E}_{F_0}[y] - c_1 = s_1 \mathbb{E}_{F_1}[y] - c_1 = \mathbb{E}_{F_1}[\hat{w}_1(y)] - c_1,$$

and

$$\mathbb{E}_{F_0}[w_1(y)] - c_0 = s_1 \mathbb{E}_{F_0}[y] - c_0 = \mathbb{E}_{F_0}[\hat{w}_1(y)] - c_0.$$

Thus,

$$(\mathbb{E}_{F'_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_0}[w_1(y)] - c_0) = (\mathbb{E}_{F_1}[\hat{w}_1(y)] - c_1) - (\mathbb{E}_{F_0}[\hat{w}_1(y)] - c_0) \geq 0,$$

implying that a'_1 may be chosen by agent 1 in response to w_1 under some technology.

When agent 1 chooses action a'_1 in response, the principal's resulting payoff in the first period is

$$\mathbb{E}_{F'_1} [y - w_1(y)] = \lambda \mathbb{E}_{F_0} [y - w_1(y)] = \lambda (1 - s_1) \mathbb{E}_{F_0} [y] = (1 - s_1) \mathbb{E}_{F_1} [y] = \mathbb{E}_{F_1} [y - \hat{w}_1(y)],$$

so her payoff in the first period under $(w_1|a'_1)$ and under $(\hat{w}_1|a_1)$ is exactly equal.

Moreover, the principal's optimal second-period payoff guarantee is $V_2^*(a'_1) = \Phi(a'_1)^2$ with

$$\Phi(a'_1) = \max_{a \in A_0 \cup \{a'_1\}} \left\{ \sqrt{\mathbb{E}_{F_a} [y]} - \sqrt{c_a} \right\}.$$

From $\mathbb{E}_{F'_1} [y] = \mathbb{E}_{F_1} [y]$, it follows that $\Phi(a'_1) = \Phi(a_1)$, which implies that $V_2^*(a'_1) = V_2^*(a_1)$. The principal's interim payoff guarantee is

$$\begin{aligned} U(w_1|a'_1) &= \mathbb{E}_{F'_1} [y - w_1(y)] + \beta \cdot V_2^*(a'_1) \\ &= \mathbb{E}_{F_1} [y - \hat{w}_1(y)] + \beta \cdot V_2^*(a_1) = U(\hat{w}_1|a_1), \end{aligned}$$

as desired.

This completes the proof. $\square$

Proof of Lemma 2. We first reformulate program (5) as an equivalent maximization problem with continuous objective function and compact feasible region. Slightly abusing notation, we use $U(s_1)$ instead of $U(w_1)$ to denote the infimum value of program (5). Note that both the objective and the constraints of program (5) depend on the choice variables (F_1, c_1) only through the value of $(\mathbb{E}_{F_1} [y], c_1)$. Rewrite $\mathbb{E}_{F_1} [y] = x$ and $c_1 = z$ with $x, z \geq 0$. Plugging into the original program (5), we obtain an equivalent program

$$\begin{aligned} U(s_1) &= \inf_{x, z} (1 - s_1)x + \beta \cdot \phi(x, z)^2 \\ \text{s.t.} \quad &s_1x - z \geq \max_{a \in A_0 \cup \{(\delta_0, 0)\}} \{s_1 \mathbb{E}_{F_a} [y] - c_a\}, \quad x, z \geq 0, \end{aligned} \tag{A.1}$$

where

$$\phi(x, z) \equiv \max \left\{ \sqrt{x} - \sqrt{z}, \max_{a \in A_0} \left\{ \sqrt{\mathbb{E}_{F_a} [y]} - \sqrt{c_a} \right\} \right\}. \tag{A.2}$$

Let $\bar{x} \equiv \max_{a \in A_0} \mathbb{E}_{F_a} [y] > 0$, and $\bar{v} \equiv \max_{a \in A_0} \left\{ \sqrt{\mathbb{E}_{F_a} [y]} - \sqrt{c_a} \right\} > 0$. Suppose

$$(F_0, c_0) \in \arg \max_{a \in A_0 \cup \{(\delta_0, 0)\}} \{s_1 \mathbb{E}_{F_a} [y] - c_a\}.$$

Note that $(x_0, z_0) = (\mathbb{E}_{F_0} [y], c_0)$ is feasible in program (A.1) and leads to objective value

$$(1 - s_1)x_0 + \beta \cdot \phi(x_0, z_0)^2 \leq (1 - s_1)\bar{x} + \beta \cdot \bar{v}^2.$$

If $x \geq \bar{x}$, then

$$(1 - s_1)x + \beta \cdot \phi(x, z)^2 \geq (1 - s_1)\bar{x} + \beta \cdot \bar{v}^2.$$

Therefore, restricting $x \in [0, \bar{x}]$ will not change the infimum of program (A.1). Moreover,

$$s_1 x - z \geq 0 \quad \Rightarrow \quad z \leq s_1 x \leq x,$$

so restricting $(x, z) \in [0, \bar{x}]^2$ will not change the infimum of program (A.1).

Consider the following program

$$\begin{aligned} \Psi^*(s_1) &\equiv \sup_{x, z} \Psi(x, z; s_1) \equiv -\left((1 - s_1)x + \beta \cdot \phi(x, z)^2\right) \\ \text{s.t.} \quad &(x, z) \in \Gamma(s_1), \end{aligned} \tag{A.3}$$

where $\hat{\Phi}$ is defined by equation (A.2), and Γ is defined as follows:

$$\Gamma(s_1) \equiv \left\{ (x, z) \in [0, \bar{x}]^2 : s_1 x - z \geq \max_{a \in A_0 \cup \{(\delta_0, 0)\}} \{s_1 \mathbb{E}_{F_a}[y] - c_a\} \right\}.$$

By definition, $\Psi : [0, \bar{x}]^2 \times [0, 1] \rightarrow \mathbb{R}$ is a continuous function, and $\Gamma : [0, 1] \rightrightarrows [0, \bar{x}]^2$ is a compact-valued and nonempty-valued correspondence. Moreover, the infimum of program (A.1), $U(s_1)$, is given by $-\Psi^*(s_1)$.

Note that for each s_1 , $\Gamma(s_1)$ defines a half plane intersecting a square, and that the half plane shifts linearly in s_1 . Thus, Γ is both upper and lower hemicontinuous. It then follows from Berge's maximum theorem that Ψ^* is continuous, and

$$\Gamma^*(s_1) \equiv \{(x, z) \in \Gamma(s_1) : \Psi(x, z; s_1) = \Psi^*(s_1)\}$$

is upper hemicontinuous with nonempty and compact values. As a consequence, a solution to program (A.3) exists for all s_1 , and the supremum can be replaced by maximum.

It follows that the infimum in program (A.1) and therefore the original program (5) can both be replaced by minimum, and the resulting minimum value $U(s_1) = -\Psi^*(s_1)$ is continuous in s_1 . Hence, $U(s_1)$ achieves a maximum over $[0, 1]$. This maximum is also the optimal guarantee over all linear contracts. $\square$

Proof of Theorem 1. According to Lemma 2, among all linear first-period contracts, there exists an optimal one, call it w_1^* . If w_1 is any other (nonlinear) first-period contract that outperforms w_1^* , then by Lemma 1, there is a linear contract that in turn does at least as well as w_1 . But this contradicts the fact that w_1^* is an optimal linear contract. Therefore, w_1^* is optimal among all first-period contracts. $\square$

A.2 Proofs for Section 4

A.2.1 Proofs for Subsection 4.2

If the principal offers $w_2 = w_1$, agent 2 will choose a_1 again. This just repeats her first-period payoff $\mathbb{E}_{F_1}[y - w_1(y)]$ in the second period.

To prove Lemma 3, we start by establishing three lemmas, Lemmas A.1, A.2, A.3, to prove that the principal's payoff guarantee in the second period from offering the remaining three contracts, (i) $w_2(y) = w_1(y) + m \cdot (y - w_1(y))$ with m defined by equation (7), (ii) $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_0 / \mathbb{E}_{F_0}[y]}$, and (iii) $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_1 / \mathbb{E}_{F_1}[y]}$, is exactly as claimed in the statement of Lemma 3.

Lemma A.1. *If $\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g(w_1, a_1)}$ attains the maximum in equation (6), and the principal offers $w_2(y) = w_1(y) + m \cdot (y - w_1(y))$ with m defined by equation (7), then her payoff guarantee in the second period is exactly $\left(\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g(w_1, a_1)}\right)^2$.*

Proof of Lemma A.1. Let $g_0 \equiv g(w_1, a_1) = (\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_0}[w_1(y)] - c_0)$.

If $\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g_0}$ attains the maximum in equation (6), then it holds that $\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g_0} \geq \sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0} > 0$, which implies that $m \in [0, 1]$.

Suppose the principal offers $w_2(y) = w_1(y) + m \cdot (y - w_1(y))$ with m defined by equation (7). We first show that this guarantees her at least $(\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g_0})^2$.

Let (F_2, c_2) be the action chosen by agent 2. By agent 1's rationality, we have

$$\mathbb{E}_{F_1}[w_1(y)] - c_1 \geq \mathbb{E}_{F_2}[w_1(y)] - c_2.$$

By agent 2's rationality, we have

$$\mathbb{E}_{F_2}[w_2(y)] - c_2 \geq \mathbb{E}_{F_0}[w_2(y)] - c_0.$$

Summing up the two inequalities, we obtain

$$\begin{aligned} m \cdot \mathbb{E}_{F_2}[y - w_1(y)] &= \mathbb{E}_{F_2}[w_2(y) - w_1(y)] \geq (\mathbb{E}_{F_0}[w_2(y)] - c_0) - (\mathbb{E}_{F_1}[w_1(y)] - c_1) \\ &= m \cdot \mathbb{E}_{F_0}[y - w_1(y)] - g_0, \end{aligned}$$

implying that

$$\mathbb{E}_{F_2}[y - w_1(y)] \geq \mathbb{E}_{F_0}[y - w_1(y)] - g_0/m.$$

Therefore, the principal's payoff in the second period is

$$\begin{aligned} \mathbb{E}_{F_2}[y - w_2(y)] &= \mathbb{E}_{F_2}[y - w_1(y)] - m \cdot \mathbb{E}_{F_2}[y - w_1(y)] = (1 - m) \mathbb{E}_{F_2}[y - w_1(y)] \\ &\geq (1 - m) (\mathbb{E}_{F_0}[y - w_1(y)] - g_0/m) = \left(\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g_0} \right)^2, \end{aligned}$$

as desired.

Next we show that her payoff guarantee from $w_2(y) = w_1(y) + m \cdot (y - w_1(y))$ cannot be strictly higher than $(\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g_0})^2$, since this is exactly her payoff when the technology is $A = \{a_0, a_1, (F', c')\}$, with $F' = (1 - m)F_0 + m \cdot \delta_0$ and $c' = c_0 - (m \cdot \mathbb{E}_{F_0}[w_1(y)] + g_0)$.

The proof takes three steps.

Step 1 $\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g_0} \geq \sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0}$ implies $c_0 \geq m \cdot \mathbb{E}_{F_0}[w_1(y)] + g_0$, so c' is indeed nonnegative.

It suffices to show

$$\begin{aligned} &\left(\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} + \sqrt{g_0} \right)^2 \geq m \cdot \mathbb{E}_{F_0}[w_1(y)] + g_0 \\ \Leftrightarrow &\left(\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} \right)^2 \geq m \cdot \mathbb{E}_{F_0}[w_1(y)] - 2\sqrt{g_0} \cdot \left(\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} \right) \\ \Leftrightarrow &\left(\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} \right)^2 \geq m \cdot \left(\mathbb{E}_{F_0}[w_1(y)] - 2\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} \cdot \left(\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} \right) \right). \end{aligned} \tag{A.4}$$

Note that

$$\begin{aligned}
& \mathbb{E}_{F_0} [w_1(y)] - 2\sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \cdot \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \right) \\
&= \mathbb{E}_{F_0} [w_1(y)] - 2\sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \cdot \frac{\mathbb{E}_{F_0} [w_1(y)]}{\sqrt{\mathbb{E}_{F_0} [y]} + \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]}} \\
&= \frac{\mathbb{E}_{F_0} [w_1(y)]}{\sqrt{\mathbb{E}_{F_0} [y]} + \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]}} \cdot \left(\sqrt{\mathbb{E}_{F_0} [y]} + \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} - 2\sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \right) \\
&= \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \right) \cdot \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \right) = \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \right)^2.
\end{aligned}$$

Therefore, inequality (A.4) is equivalent to

$$\left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \right)^2 \geq m \cdot \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \right)^2,$$

which is implied by the assumption that $\sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \geq \sqrt{g_0}$ (or equivalently, $m \leq 1$).

Step 2 $A = \{a_0, a_1, (F', c')\}$ is compatible with (w_1, a_1) . That is, agent 1 chooses a_1 in response to w_1 .

Agent 1's payoff from (F', c') is

$$\begin{aligned}
\mathbb{E}_{F'} [w_1(y)] - c' &= (1 - m) \mathbb{E}_{F_0} [w_1(y)] - c_0 + (m \cdot \mathbb{E}_{F_0} [w_1(y)] + g_0) \\
&= (\mathbb{E}_{F_0} [w_1(y)] - c_0) + g_0 = \mathbb{E}_{F_1} [w_1(y)] - c_1,
\end{aligned}$$

so he would choose $a_1 = (F_1, c_1)$ in response to w_1 .

Note that agent 1 is actually indifferent between (F_1, c_1) and (F', c') , and we will show below that agent 2 is indifferent between (F_0, c_0) and (F', c') . Technically to ensure that agent 1 chooses (F_1, c_1) and agent 2 chooses (F', c') we can set $F' = (1 - m + \varepsilon) F_0 + (m - \varepsilon) \delta_0$ and $c' = c_0 - (m \cdot \mathbb{E}_{F_0} [w_1(y)] + g_0) + \varepsilon \cdot \mathbb{E}_{F_0} [w_1(y) + (m/2) \cdot (y - w_1(y))]$ then let $\varepsilon \downarrow 0$. Many of the following cases of potential indifference shall be treated similarly, and we omit them for brevity.

Step 3 If $A = \{a_0, a_1, (F', c')\}$, then agent 2 chooses (F', c') in response to w_2 , leading to a payoff of $\left(\sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} - \sqrt{g_0} \right)^2$ for the principal.

Agent 2's payoff from (F', c') is

$$\begin{aligned}
\mathbb{E}_{F'} [w_2(y)] - c' &= (1 - m) \mathbb{E}_{F_0} [w_1(y) + m \cdot (y - w_1(y))] - c_0 + (m \cdot \mathbb{E}_{F_0} [w_1(y)] + g_0) \\
&= \mathbb{E}_{F_0} [w_1(y)] + m \cdot \mathbb{E}_{F_0} [y - w_1(y)] - m^2 \cdot \mathbb{E}_{F_0} [y - w_1(y)] - c_0 + g_0 \\
&= \mathbb{E}_{F_0} [w_2(y)] - g_0 - c_0 + g_0 = \mathbb{E}_{F_0} [w_2(y)] - c_0,
\end{aligned}$$

and his payoff from $a_1 = (F_1, c_1)$ is

$$\begin{aligned}
\mathbb{E}_{F_1} [w_2(y)] - c_1 &= \mathbb{E}_{F_1} [w_1(y) + m \cdot (y - w_1(y))] - c_1 \\
&= m \cdot \mathbb{E}_{F_1} [y - w_1(y)] + (\mathbb{E}_{F_0} [w_1(y)] - c_0) + g_0 \\
&\leq m \cdot \left(\sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} - \sqrt{g_0} \right)^2 + (\mathbb{E}_{F_0} [w_1(y)] - c_0) + g_0 \\
&\leq m \cdot \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} \left(\sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} - \sqrt{g_0} \right) + (\mathbb{E}_{F_0} [w_1(y)] - c_0) + g_0 \\
&= m \cdot \mathbb{E}_{F_0} [y - w_1(y)] - g_0 + (\mathbb{E}_{F_0} [w_1(y)] - c_0) + g_0 = \mathbb{E}_{F_0} [w_2(y)] - c_0,
\end{aligned}$$

so he would choose (F', c') in response to w_2 .

This leaves the principal with payoff of

$$\begin{aligned}
\mathbb{E}_{F'} [y - w_2(y)] &= \mathbb{E}_{F'} [y - w_1(y)] - m \cdot \mathbb{E}_{F'} [y - w_1(y)] = (1 - m) \mathbb{E}_{F'} [y - w_1(y)] \\
&= (1 - m)^2 \mathbb{E}_{F_0} [y - w_1(y)] = \left(\sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} - \sqrt{g_0} \right)^2,
\end{aligned}$$

as desired.

This completes the proof. $\square$

Lemma A.2. *If $\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0}$ attains the maximum in equation (6), and the principal offers the linear contract $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_0/\mathbb{E}_{F_0}[y]}$, then her payoff guarantee in the second period is exactly $(\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0})^2$.*

Proof of Lemma A.2. Suppose that $\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0}$ attains the maximum in equation (6), and the principal offers the linear contract $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_0/\mathbb{E}_{F_0}[y]}$. We first show that this guarantees her at least $(\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0})^2$.

Let (F_2, c_2) be the action chosen by agent 2. By agent 2's rationality, we have

$$\mathbb{E}_{F_2} [w_2(y)] - c_2 \geq \mathbb{E}_{F_0} [w_2(y)] - c_0,$$

which further implies that

$$s_2 \mathbb{E}_{F_2} [y] = \mathbb{E}_{F_2} [w_2(y)] \geq \mathbb{E}_{F_2} [w_2(y)] - c_2 \geq \mathbb{E}_{F_0} [w_2(y)] - c_0 = s_2 \mathbb{E}_{F_0} [y] - c_0,$$

and hence

$$\mathbb{E}_{F_2} [y] \geq \mathbb{E}_{F_0} [y] - c_0/s_2.$$

Therefore, the principal's payoff in the second period is

$$\mathbb{E}_{F_2} [y - w_2(y)] = \mathbb{E}_{F_2} [(1 - s_2)y] \geq (1 - s_2)(\mathbb{E}_{F_0} [y] - c_0/s_2) = (\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0})^2,$$

as desired.

Next we show that her payoff guarantee from this linear contract cannot be strictly higher, since $(\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0})^2$ is exactly her payoff when the technology is $A = \{a_0, a_1, (F', 0)\}$, with $F' = \lambda F_0 + (1 - \lambda)\delta_0$ where $\lambda = 1 - \sqrt{c_0/\mathbb{E}_{F_0}[y]} \in [0, 1]$.

The proof takes two steps. Let $g_0 \equiv g(w_1, a_1) = (\mathbb{E}_{F_1} [w_1(y)] - c_1) - (\mathbb{E}_{F_0} [w_1(y)] - c_0)$.

Step 1 $A = \{a_0, a_1, (F', 0)\}$ is compatible with (w_1, a_1) . That is, agent 1 chooses a_1 in response to w_1 .

Agent 1's payoff from $(F', 0)$ is $\mathbb{E}_{F'} [w_1(y)] = \lambda \mathbb{E}_{F_0} [w_1(y)] = \left(1 - \sqrt{c_0/\mathbb{E}_{F_0} [y]}\right) \mathbb{E}_{F_0} [w_1(y)]$, and we have

$$\begin{aligned} \left(1 - \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}}\right) \mathbb{E}_{F_0} [w_1(y)] &\leq \mathbb{E}_{F_1} [w_1(y)] - c_1 \quad \Leftrightarrow \quad \left(1 - \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}}\right) \mathbb{E}_{F_0} [w_1(y)] \leq (\mathbb{E}_{F_0} [w_1(y)] - c_0) + g_0 \\ &\Leftrightarrow \quad \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}} \mathbb{E}_{F_0} [w_1(y)] - c_0 + g_0 \geq 0. \end{aligned}$$

From

$$\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0} \geq \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} - \sqrt{g_0},$$

we obtain

$$\mathbb{E}_{F_0} [w_1(y)] \geq \mathbb{E}_{F_0} [y] - \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0} + \sqrt{g_0}\right)^2,$$

and thus

$$\begin{aligned} \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}} \mathbb{E}_{F_0} [w_1(y)] - c_0 + g_0 &\geq \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}} \cdot \left(\mathbb{E}_{F_0} [y] - \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0} + \sqrt{g_0}\right)^2\right) - c_0 + g_0 \\ &= \left(1 - \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}}\right) \left(\sqrt{c_0} - \sqrt{g_0}\right)^2 \geq 0, \end{aligned}$$

as desired. So we indeed have $\mathbb{E}_{F'} [w_1(y)] \leq \mathbb{E}_{F_1} [w_1(y)] - c_1$, implying that agent 1 would choose $a_1 = (F_1, c_1)$ in response to w_1 .

Step 2 If $A = \{a_0, a_1, (F', 0)\}$, then agent 2 chooses $(F', 0)$ in response to w_2 , leading to a payoff of $\left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0}\right)^2$ for the principal.

Agent 2's payoff from $(F', 0)$ is

$$\begin{aligned} \mathbb{E}_{F'} [w_2(y)] &= \lambda \mathbb{E}_{F_0} [s_2 y] = \left(1 - \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}}\right) \cdot \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}} \cdot \mathbb{E}_{F_0} [y] \\ &= \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0}\right) \sqrt{c_0} = \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}} \cdot \mathbb{E}_{F_0} [y] - c_0 \\ &= s_2 \mathbb{E}_{F_0} [y] - c_0 = \mathbb{E}_{F_0} [w_2(y)] - c_0. \end{aligned}$$

His payoff from $a_1 = (F_1, c_1)$ is $\mathbb{E}_{F_1} [w_2(y)] - c_1 = \sqrt{c_0/\mathbb{E}_{F_0} [y]} \cdot \mathbb{E}_{F_1} [y] - c_1$, and we have

$$\sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}} \cdot \mathbb{E}_{F_1} [y] - c_1 \leq \mathbb{E}_{F_0} [w_2(y)] - c_0 \quad \Leftrightarrow \quad \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}} \cdot \mathbb{E}_{F_1} [y] - c_1 \leq \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0}\right) \sqrt{c_0}.$$

From $\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0} \geq \sqrt{\mathbb{E}_{F_1} [y]} - \sqrt{c_1}$, we obtain $\mathbb{E}_{F_1} [y] \leq \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0} + \sqrt{c_1}\right)^2$, and thus

$$\begin{aligned} &\left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0}\right) \sqrt{c_0} - \left(\sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}} \cdot \mathbb{E}_{F_1} [y] - c_1\right) \\ &\geq \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0}\right) \sqrt{c_0} - \left(\sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}} \cdot \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0} + \sqrt{c_1}\right)^2 - c_1\right) \\ &= \left(1 - \sqrt{\frac{c_0}{\mathbb{E}_{F_0} [y]}}\right) \left(\sqrt{c_0} - \sqrt{c_1}\right)^2 \geq 0, \end{aligned}$$

as desired. So we indeed have $\mathbb{E}_{F_1}[w_2(y)] - c_1 \leq \mathbb{E}_{F_0}[w_2(y)] - c_0 = \mathbb{E}_{F'}[w_2(y)]$, implying that agent 2 would choose $(F', 0)$ in response to w_2 .

This leaves the principal with payoff of

$$\begin{aligned}\mathbb{E}_{F'}[y - w_2(y)] &= \lambda \mathbb{E}_{F_0}[(1 - s_2)y] = \left(1 - \sqrt{\frac{c_0}{\mathbb{E}_{F_0}[y]}}\right) \left(1 - \sqrt{\frac{c_0}{\mathbb{E}_{F_0}[y]}}\right) \cdot \mathbb{E}_{F_0}[y] \\ &= \left(\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0}\right)^2,\end{aligned}$$

as desired.

This completes the proof. $\square$

Lemma A.3. *If $\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}$ attains the maximum in equation (6), and the principal offers the linear contract $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_1/\mathbb{E}_{F_1}[y]}$, then her payoff guarantee in the second period is exactly $\left(\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}\right)^2$.*

Proof of Lemma A.3. If $\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}$ attains the maximum in equation (6), then it holds that $\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1} \geq \sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0} > 0$, which implies that $c_1/\mathbb{E}_{F_1}[y] \in [0, 1]$.

Suppose the principal offers the linear contract $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_1/\mathbb{E}_{F_1}[y]}$. We first show that this guarantees her at least $\left(\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}\right)^2$. Let (F_2, c_2) be the action chosen by agent 2. By agent 2's rationality, we have

$$\mathbb{E}_{F_2}[w_2(y)] - c_2 \geq \mathbb{E}_{F_1}[w_2(y)] - c_1,$$

which further implies that

$$s_2 \mathbb{E}_{F_2}[y] = \mathbb{E}_{F_2}[w_2(y)] \geq \mathbb{E}_{F_2}[w_2(y)] - c_2 \geq \mathbb{E}_{F_1}[w_2(y)] - c_1 = s_2 \mathbb{E}_{F_1}[y] - c_1,$$

and hence

$$\mathbb{E}_{F_2}[y] \geq \mathbb{E}_{F_1}[y] - c_1/s_2.$$

Therefore, the principal's payoff in the second period is

$$\mathbb{E}_{F_2}[y - w_2(y)] = \mathbb{E}_{F_2}[(1 - s_2)y] \geq (1 - s_2)(\mathbb{E}_{F_1}[y] - c_1/s_2) = \left(\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}\right)^2,$$

as desired.

Next we show that her payoff guarantee from this linear contract cannot be strictly higher, since this is exactly her payoff when the technology is $A = \{a_0, a_1, (F', 0)\}$, with $F' = \lambda F_1 + (1 - \lambda)\delta_0$ where $\lambda = 1 - \sqrt{c_1/\mathbb{E}_{F_1}[y]} \in [0, 1]$.

The proof takes two steps.

Step 1 $A = \{a_0, a_1, (F', 0)\}$ is compatible with (w_1, a_1) . That is, agent 1 chooses a_1 in response to w_1 .

Agent 1's payoff from $(F', 0)$ is $\mathbb{E}_{F'}[w_1(y)] = \lambda \mathbb{E}_{F_1}[w_1(y)] = \left(1 - \sqrt{c_1/\mathbb{E}_{F_1}[y]}\right) \mathbb{E}_{F_1}[w_1(y)]$, and we have

$$\begin{aligned}\left(1 - \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}}\right) \mathbb{E}_{F_1}[w_1(y)] &\leq \mathbb{E}_{F_1}[w_1(y)] - c_1 \quad \Leftrightarrow \quad \left(1 - \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}}\right) \mathbb{E}_{F_1}[w_1(y)] \leq \mathbb{E}_{F_1}[w_1(y)] - c_1 \\ &\Leftrightarrow \quad \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}} \mathbb{E}_{F_1}[w_1(y)] - c_1 \geq 0.\end{aligned}$$

From $\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1} \geq \sqrt{\mathbb{E}_{F_1}[y - w_1(y)]}$, we obtain $\mathbb{E}_{F_1}[w_1(y)] \geq \mathbb{E}_{F_1}[y] - (\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1})^2$, and thus

$$\sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}} \mathbb{E}_{F_1}[w_1(y)] - c_1 \geq \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}} \cdot (\mathbb{E}_{F_1}[y] - (\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1})^2) - c_1 = \left(1 - \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}}\right) c_1 \geq 0,$$

as desired. So we indeed have $\mathbb{E}_{F'}[w_1(y)] \leq \mathbb{E}_{F_1}[w_1(y)] - c_1$, implying that agent 1 would choose (F_1, c_1) in response to w_1 .

Step 2 If $A = \{a_0, a_1, (F', 0)\}$, then agent 2 chooses $(F', 0)$ in response to w_2 , leading to a payoff of $(\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1})^2$ for the principal.

Agent 2's payoff from $(F', 0)$ is

$$\begin{aligned} \mathbb{E}_{F'}[w_2(y)] &= \lambda \mathbb{E}_{F_1}[s_2 y] = \left(1 - \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}}\right) \cdot \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}} \cdot \mathbb{E}_{F_1}[y] \\ &= (\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}) \sqrt{c_1} = \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}} \cdot \mathbb{E}_{F_1}[y] - c_1 \\ &= s_2 \mathbb{E}_{F_1}[y] - c_1 = \mathbb{E}_{F_1}[w_2(y)] - c_1. \end{aligned}$$

His payoff from (F_0, c_0) is $\mathbb{E}_{F_0}[w_2(y)] - c_0 = \sqrt{c_1/\mathbb{E}_{F_1}[y]} \cdot \mathbb{E}_{F_0}[y] - c_0$, and we have

$$\sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}} \cdot \mathbb{E}_{F_0}[y] - c_0 \leq \mathbb{E}_{F_1}[w_2(y)] - c_1 \quad \Leftrightarrow \quad \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}} \cdot \mathbb{E}_{F_0}[y] - c_0 \leq (\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}) \sqrt{c_1}.$$

From $\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1} \geq \sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0}$, we obtain $\mathbb{E}_{F_0}[y] \leq (\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1} + \sqrt{c_0})^2$, and thus

$$\begin{aligned} &(\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}) \sqrt{c_1} - \left(\sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}} \cdot \mathbb{E}_{F_0}[y] - c_0\right) \\ &\geq (\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}) \sqrt{c_1} - \left(\sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}} \cdot (\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1} + \sqrt{c_0})^2 - c_0\right) \\ &= \left(1 - \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}}\right) (\sqrt{c_1} - \sqrt{c_0})^2 \geq 0, \end{aligned}$$

as desired. So we indeed have $\mathbb{E}_{F_0}[w_2(y)] - c_0 \leq \mathbb{E}_{F_1}[w_2(y)] - c_1 = \mathbb{E}_{F'}[w_2(y)]$, implying that agent 2 would choose $(F', 0)$ in response to w_2 .

This leaves the principal with payoff of

$$\begin{aligned} \mathbb{E}_{F'}[y - w_2(y)] &= \lambda \mathbb{E}_{F_0}[(1 - s_2)y] = \left(1 - \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}}\right) \left(1 - \sqrt{\frac{c_1}{\mathbb{E}_{F_1}[y]}}\right) \cdot \mathbb{E}_{F_1}[y] \\ &= (\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1})^2, \end{aligned}$$

as desired.

This completes the proof. $\square$

We are now ready to prove Lemma 3.

Proof of Lemma 3. If the principal offers $w_2 = w_1$, this guarantees her payoff in the first-period, which is

equal to $\mathbb{E}_{F_1} [y - w_1(y)]$. Note that her payoff guarantee from $w_2 = w_1$ cannot be strictly higher, since this is exactly her payoff when the technology is $A = \{a_0, a_1\}$, which is compatible with (w_1, a_1) .

Together with Lemmas A.1, A.2 and A.3, we have shown that by offering the best among the four contracts: (i) $w_2 = w_1$, (ii) $w_2(y) = w_1(y) + m \cdot (y - w_1(y))$ with m defined by equation (7), (iii) $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_0/\mathbb{E}_{F_0}[y]}$, and (iv) $w_2(y) = s_2 y$ with $s_2 = \sqrt{c_1/\mathbb{E}_{F_1}[y]}$, the principal's payoff guarantee in the second period is exactly given by $\hat{\Phi}(w_1, a_1)^2$, where $\hat{\Phi}$ is defined by equation (6). The principal's optimal second-period payoff guarantee, $\hat{V}_2^*(w_1, a_1)$, is thus at least $\hat{\Phi}(w_1, a_1)^2$.

Now consider an arbitrary second-period contract w_2 . It suffices to show that the principal's payoff guarantee is not strictly higher than $\hat{\Phi}(w_1, a_1)^2$ under w_2 .

Consider the following two cases.

Case 1. $\mathbb{E}_{F_1} [w_2(y)] - c_1 \geq \mathbb{E}_{F_0} [w_2(y)] - c_0$.

1. If $\mathbb{E}_{F_1} [w_2(y)] \geq \mathbb{E}_{F_1} [w_1(y)]$, consider the second-period contract w_2 when the technology is $A = \{a_0, a_1\}$, which is compatible with (w_1, a_1) . Agent 2 would prefer to take action $a_1 = (F_1, c_1)$. This leaves the principal with a payoff of

$$\mathbb{E}_{F_1} [y - w_2(y)] \leq \mathbb{E}_{F_1} [y - w_1(y)] \leq \hat{\Phi}(w_1, a_1)^2,$$

as desired.

2. If $\mathbb{E}_{F_1} [w_2(y)] < c_1$, consider the second-period contract w_2 when $A = \{a_0, a_1, (\delta_0, 0)\}$, which is compatible with (w_1, a_1) . Agent 2's payoff from $(\delta_0, 0)$ is

$$w_2(0) \geq 0 > \mathbb{E}_{F_1} [w_2(y)] - c_1,$$

so he would prefer to take action $(\delta_0, 0)$. This leaves the principal with a payoff of

$$-w_2(0) \leq 0 \leq \hat{\Phi}(w_1, a_1)^2,$$

as desired.

3. If $c_1 \leq \mathbb{E}_{F_1} [w_2(y)] < \mathbb{E}_{F_1} [w_1(y)]$, let $\lambda = 1 - c_1/\mathbb{E}_{F_1} [w_2(y)] \in [0, 1]$ and let F' be the mixture $\lambda F_1 + (1 - \lambda)\delta_0$. Consider the technology $A = \{a_0, a_1, (F', 0)\}$.

We proceed with two steps.

Step 1 A is compatible with (w_1, a_1) . That is, agent 1 chooses a_1 in response to w_1 .

Agent 1's payoff from $(F', 0)$ is

$$\mathbb{E}_{F'} [w_1(y)] = \lambda \mathbb{E}_{F_1} [w_1(y)] = \mathbb{E}_{F_1} [w_1(y)] - \frac{\mathbb{E}_{F_1} [w_1(y)]}{\mathbb{E}_{F_1} [w_2(y)]} c_1 < \mathbb{E}_{F_1} [w_1(y)] - c_1,$$

so he would prefer to take action $a_1 = (F_1, c_1)$ when $A = \{a_0, a_1, (F', 0)\}$.

Step 2 Agent 2 chooses $(F', 0)$ in response to w_2 , resulting in the principal's payoff no more than $(\sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1})^2$.

Agent 2's payoff from $(F', 0)$ is

$$\begin{aligned}\mathbb{E}_{F'} [w_2(y)] &= \lambda \mathbb{E}_{F_1} [w_2(y)] + (1 - \lambda)w_2(0) \\ &\geq \lambda \mathbb{E}_{F_1} [w_2(y)] = \mathbb{E}_{F_1} [w_2(y)] - c_1,\end{aligned}$$

which is also larger than $\mathbb{E}_{F_0} [w_2(y)] - c_0$ by assumption. So he would prefer to take action $(F', 0)$.

This leaves the principal with a payoff of

$$\begin{aligned}\mathbb{E}_{F'} [y - w_2(y)] &= \lambda \mathbb{E}_{F_1} [y - w_2(y)] + (1 - \lambda)(0 - w_2(0)) \\ &\leq \lambda \mathbb{E}_{F_1} [y - w_2(y)] = \left(1 - \frac{c_1}{\mathbb{E}_{F_1} [w_2(y)]}\right) (\mathbb{E}_{F_1} [y] - \mathbb{E}_{F_1} [w_2(y)]) \\ &\leq \left(\sqrt{\mathbb{E}_{F_1} [y]} - \sqrt{c_1}\right)^2,\end{aligned}\tag{A.5}$$

which is no more than $\hat{\Phi}(w_1, a_1)^2$, as desired. The last inequality (A.5),

$$\begin{aligned}&\left(1 - \frac{c_1}{\mathbb{E}_{F_1} [w_2(y)]}\right) (\mathbb{E}_{F_1} [y] - \mathbb{E}_{F_1} [w_2(y)]) \leq \left(\sqrt{\mathbb{E}_{F_1} [y]} - \sqrt{c_1}\right)^2 \\ \Leftrightarrow &\left(\sqrt{\mathbb{E}_{F_1} [w_2(y)]} - \sqrt{\frac{c_1 \mathbb{E}_{F_1} [y]}{\mathbb{E}_{F_1} [w_2(y)]}}\right)^2 \geq 0,\end{aligned}$$

which always holds.

Case 2. $\mathbb{E}_{F_1} [w_2(y)] - c_1 < \mathbb{E}_{F_0} [w_2(y)] - c_0$.

1. If $\mathbb{E}_{F_0} [w_2(y)] < c_0$, consider the second-period contract w_2 when $A = \{a_0, a_1, (\delta_0, 0)\}$, which is compatible with (w_1, a_1) . Agent 2's payoff from $(\delta_0, 0)$ is

$$w_2(0) \geq 0 > \mathbb{E}_{F_0} [w_2(y)] - c_0,$$

so he would prefer to take action $(\delta_0, 0)$. This leaves the principal with a payoff of

$$-w_2(0) \leq 0 \leq \hat{\Phi}(w_1, a_1)^2,$$

as desired.

2. If $\mathbb{E}_{F_0} [w_2(y)] \geq c_0$, and it holds that

$$\begin{aligned}&\text{either (i) } \mathbb{E}_{F_0} [w_1(y)] \leq \mathbb{E}_{F_1} [w_1(y)] - c_1, \\ &\text{or (ii) } \mathbb{E}_{F_0} [w_2(y)] < \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_1} [w_1(y)] - c_1)} c_0,\end{aligned}\tag{A.6}$$

let $\lambda = 1 - c_0/\mathbb{E}_{F_0} [w_2(y)] \in [0, 1]$ and let F' be the mixture $\lambda F_0 + (1 - \lambda)\delta_0$. Consider the technology $A = \{a_0, a_1, (F', 0)\}$.

We proceed with two steps.

Step 1 A is compatible with (w_1, a_1) . That is, agent 1 chooses a_1 in response to w_1 .

Agent 1's payoff from $(F', 0)$ is

$$\begin{aligned}\mathbb{E}_{F'} [w_1(y)] &= \lambda \mathbb{E}_{F_0} [w_1(y)] = \mathbb{E}_{F_0} [w_1(y)] - \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_2(y)]} c_0 \\ &< \mathbb{E}_{F_1} [w_1(y)] - c_1.\end{aligned}\tag{A.7}$$

Note that inequality (A.7) holds exactly due to the assumptions in (A.6). So agent 1 would prefer to take action $a_1 = (F_1, c_1)$ when $A = \{a_0, a_1, (F', 0)\}$.

Step 2 Agent 2 chooses $(F', 0)$ in response to w_2 , resulting in the principal's payoff no more than $(\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0})^2$.

Agent 2's payoff from $(F', 0)$ is

$$\begin{aligned}\mathbb{E}_{F'} [w_2(y)] &= \lambda \mathbb{E}_{F_0} [w_2(y)] + (1 - \lambda) w_2(0) \\ &\geq \lambda \mathbb{E}_{F_0} [w_2(y)] = \mathbb{E}_{F_0} [w_2(y)] - c_0,\end{aligned}$$

which is also larger than $\mathbb{E}_{F_1} [w_2(y)] - c_1$ by assumption. So he would prefer to take action $(F', 0)$ when $A = \{a_0, a_1, (F', 0)\}$.

This leaves the principal with a payoff of

$$\begin{aligned}\mathbb{E}_{F'} [y - w_2(y)] &= \lambda \mathbb{E}_{F_0} [y - w_2(y)] + (1 - \lambda) (0 - w_2(0)) \\ &\leq \lambda \mathbb{E}_{F_0} [y - w_2(y)] = \left(1 - \frac{c_0}{\mathbb{E}_{F_0} [w_2(y)]}\right) (\mathbb{E}_{F_0} [y] - \mathbb{E}_{F_0} [w_2(y)]) \\ &\leq (\sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0})^2,\end{aligned}\tag{A.8}$$

which is no more than $\hat{\Phi}(w_1, a_1)^2$, as desired. The last inequality (A.8) holds for the same reason as (A.5).

3. If both inequalities in (A.6) are reversed, i.e.,

$$\mathbb{E}_{F_0} [w_1(y)] > \mathbb{E}_{F_1} [w_1(y)] - c_1 \quad \text{and} \quad \mathbb{E}_{F_0} [w_2(y)] \geq \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_1} [w_1(y)] - c_1)} c_0,$$

let

$$\begin{aligned}\lambda &= \frac{(\mathbb{E}_{F_0} [w_2(y)] - c_0) - (\mathbb{E}_{F_1} [w_1(y)] - c_1)}{\mathbb{E}_{F_0} [w_2(y)] - \mathbb{E}_{F_0} [w_1(y)]}, \\ c' &= \frac{\mathbb{E}_{F_0} [w_1(y)] (\mathbb{E}_{F_0} [w_2(y)] - c_0) - \mathbb{E}_{F_0} [w_2(y)] (\mathbb{E}_{F_1} [w_1(y)] - c_1)}{\mathbb{E}_{F_0} [w_2(y)] - \mathbb{E}_{F_0} [w_1(y)]},\end{aligned}$$

and let F' be the mixture $\lambda F_0 + (1 - \lambda) \delta_0$. Consider the technology $A = \{a_0, a_1, (F', c')\}$.

We proceed with three steps.

Step 1 $\lambda \in [0, 1]$ and $c' \geq 0$, so (F', c') is a valid action.

Note that

$$\mathbb{E}_{F_0} [w_2(y)] \geq \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_1} [w_1(y)] - c_1)} c_0 \geq \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_0} [w_1(y)] - c_0)} c_0 = \mathbb{E}_{F_0} [w_1(y)],$$

so the denominator of λ and c' is positive.

Moreover,

$$\begin{aligned} \mathbb{E}_{F_0} [w_2(y)] - c_0 &\geq \frac{\mathbb{E}_{F_1} [w_1(y)] - c_1}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_1} [w_1(y)] - c_1)} c_0 \\ &\geq \frac{\mathbb{E}_{F_1} [w_1(y)] - c_1}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_0} [w_1(y)] - c_0)} c_0 = \mathbb{E}_{F_1} [w_1(y)] - c_1, \end{aligned}$$

so the numerator of λ is positive.

The numerator of c' is positive because

$$\begin{aligned} &\mathbb{E}_{F_0} [w_1(y)] (\mathbb{E}_{F_0} [w_2(y)] - c_0) \geq \mathbb{E}_{F_0} [w_2(y)] (\mathbb{E}_{F_1} [w_1(y)] - c_1) \\ \Leftrightarrow \quad &\mathbb{E}_{F_0} [w_2(y)] \geq \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_1} [w_1(y)] - c_1)} c_0. \end{aligned}$$

Finally,

$$\begin{aligned} &(\mathbb{E}_{F_0} [w_2(y)] - c_0) - (\mathbb{E}_{F_1} [w_1(y)] - c_1) \leq \mathbb{E}_{F_0} [w_2(y)] - \mathbb{E}_{F_0} [w_1(y)] \\ \Leftrightarrow \quad &\mathbb{E}_{F_0} [w_1(y)] - c_0 \leq \mathbb{E}_{F_1} [w_1(y)] - c_1, \end{aligned}$$

so λ is indeed smaller than 1.

Step 2 A is compatible with (w_1, a_1) . That is, agent 1 chooses a_1 in response to w_1 .

Agent 1's payoff from (F', c') is

$$\mathbb{E}_{F'} [w_1(y)] - c' = \lambda \mathbb{E}_{F_0} [w_1(y)] - c' = \mathbb{E}_{F_1} [w_1(y)] - c_1,$$

so he would prefer to take action $a_1 = (F_1, c_1)$ when $A = \{a_0, a_1, (F', c')\}$.

Step 3 Agent 2 chooses (F', c') in response to w_2 , resulting in the principal's payoff no more than $\left(\sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} - \sqrt{g(w_1, a_1)} \right)^2$.

Agent 2's payoff from (F', c') is

$$\begin{aligned} \mathbb{E}_{F'} [w_2(y)] - c' &= \lambda \mathbb{E}_{F_0} [w_2(y)] + (1 - \lambda) w_2(0) - c' \\ &\geq \lambda \mathbb{E}_{F_0} [w_2(y)] - c' = \mathbb{E}_{F_0} [w_2(y)] - c_0, \end{aligned}$$

which is also larger than $\mathbb{E}_{F_1} [w_2(y)] - c_1$ by assumption. So he would prefer to take action (F', c') when $A = \{a_0, a_1, (F', c')\}$.

This leaves the principal with a payoff of

$$\begin{aligned}
\mathbb{E}_{F'}[y - w_2(y)] &= \lambda \mathbb{E}_{F_0}[y - w_2(y)] + (1 - \lambda)(0 - w_2(0)) \\
&\leq \lambda \mathbb{E}_{F_0}[y - w_2(y)] = \frac{(\mathbb{E}_{F_0}[w_2(y)] - c_0) - (\mathbb{E}_{F_1}[w_1(y)] - c_1)}{\mathbb{E}_{F_0}[w_2(y)] - \mathbb{E}_{F_0}[w_1(y)]} (\mathbb{E}_{F_0}[y] - \mathbb{E}_{F_0}[w_2(y)]) \\
&\leq \left(\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g(w_1, a_1)} \right)^2,
\end{aligned} \tag{A.9}$$

which is no more than $\hat{\Phi}(w_1, a_1)^2$, as desired. The last inequality (A.9),

$$\begin{aligned}
&\frac{(\mathbb{E}_{F_0}[w_2(y)] - c_0) - (\mathbb{E}_{F_1}[w_1(y)] - c_1)}{\mathbb{E}_{F_0}[w_2(y)] - \mathbb{E}_{F_0}[w_1(y)]} (\mathbb{E}_{F_0}[y] - \mathbb{E}_{F_0}[w_2(y)]) \leq \left(\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g(w_1, a_1)} \right)^2 \\
\Leftrightarrow &\frac{(\mathbb{E}_{F_0}[w_2(y)] - \mathbb{E}_{F_0}[w_1(y)] - \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} \cdot \sqrt{g(w_1, a_1)})^2}{\mathbb{E}_{F_0}[w_2(y)] - \mathbb{E}_{F_0}[w_1(y)]} \geq 0,
\end{aligned}$$

which always holds. (Recall that $g(w_1, a_1) = (\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_0}[w_1(y)] - c_0) \geq 0$.)

Summing up the above cases, we prove that the principal's payoff guarantee is not strictly higher than $\hat{\Phi}(w_1, a_1)^2$ under any second-period contract w_2 .

This completes the proof. $\square$

A.3 Proofs for Section 4.3

To prove Theorem 2, we start by establishing two lemmas, Lemmas A.4 and A.5. Lemma A.4 shows that any nonlinear contract is outperformed by some linear one, and Lemma A.5 further shows that the maximum of the principal's first-period problem exists within the class of linear first-period contracts.

Lemma A.4. *In the case of constant technology, the linear contract $\hat{w}_1$ defined by equation (3) satisfies $\hat{U}(\hat{w}_1) \geq \hat{U}(w_1)$.*

Proof of Lemma A.4. Consider an arbitrary action $a_1 = (F_1, c_1)$ agent 1 would take under contract $\hat{w}_1$. We need to show that the principal's interim payoff guarantee, $\hat{U}(\hat{w}_1|a_1)$, is at least $\hat{U}(w_1)$. The incentive gap is

$$g(\hat{w}_1, a_1) = (\mathbb{E}_{F_1}[\hat{w}_1(y)] - c_1) - (\mathbb{E}_{F_0}[\hat{w}_1(y)] - c_0) \geq 0,$$

and Lemma 3 shows that the principal's optimal second-period payoff guarantee is $\hat{V}_2^*(\hat{w}_1, a_1) = \hat{\Phi}(\hat{w}_1, a_1)^2$, where

$$\begin{aligned}
\hat{\Phi}(\hat{w}_1, a_1) &= \max \left\{ \sqrt{\mathbb{E}_{F_1}[y - \hat{w}_1(y)]}, \sqrt{\mathbb{E}_{F_0}[y - \hat{w}_1(y)]} - \sqrt{g(\hat{w}_1, a_1)}, \sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0}, \sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1} \right\}, \\
&\quad (\text{with } \sqrt{x} = -\infty \text{ for } x < 0 \text{ by convention}).
\end{aligned} \tag{A.10}$$

The principal's interim payoff guarantee is

$$\hat{U}(\hat{w}_1|a_1) = \mathbb{E}_{F_1}[y - \hat{w}_1(y)] + \beta \cdot \hat{V}_2^*(\hat{w}_1, a_1).$$

It suffices to construct another action a'_1 , which may be taken by agent 1 under w_1 and some other technology, such that $\hat{U}(w_1|a'_1) \leq \hat{U}(\hat{w}_1|a_1)$. Note that an action may be taken by agent 1 if and only if the incentive gap is nonnegative, i.e., $g(w_1, a'_1) \geq 0$.

Case 1. $\mathbb{E}_{F_1}[y] \geq \mathbb{E}_{F_0}[y]$.

Consider $a'_1 = a_0$. The corresponding incentive gap is $g(w_1, a_0) = 0$. When agent 1 takes action a_0 in response, the principal's resulting payoff in the first period is

$$\mathbb{E}_{F_0}[y - w_1(y)] = (1 - s_1) \mathbb{E}_{F_0}[y] \leq (1 - s_1) \mathbb{E}_{F_1}[y] = \mathbb{E}_{F_1}[y - \hat{w}_1(y)],$$

so her payoff in the first period under $(w_1|a_0)$ is weakly lower than under $(\hat{w}_1|a_1)$.

Moreover, it follows from Lemma 3 that the principal's optimal second-period payoff guarantee is $\hat{V}_2^*(w_1, a_0) = \hat{\Phi}(w_1, a_0)^2$, where

$$\hat{\Phi}(w_1, a_0) = \max \left\{ \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]}, \sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0} \right\}.$$

Note that we have shown $\mathbb{E}_{F_0}[y - w_1(y)] \leq \mathbb{E}_{F_1}[y - \hat{w}_1(y)]$, so $\hat{\Phi}(w_1, a_0)$ is also weakly smaller than $\hat{\Phi}(\hat{w}_1, a_1)$ (given by equation (A.10)). This implies that $\hat{V}_2^*(w_1, a_0) \leq \hat{V}_2^*(\hat{w}_1, a_1)$.

Therefore, the principal's interim payoff guarantee is

$$\begin{aligned} \hat{U}(w_1|a_0) &= \mathbb{E}_{F_0}[y - w_1(y)] + \beta \cdot \hat{V}_2^*(w_1, a_0) \\ &\leq \mathbb{E}_{F_1}[y - \hat{w}_1(y)] + \beta \cdot \hat{V}_2^*(\hat{w}_1, a_1) = \hat{U}(\hat{w}_1|a_1), \end{aligned}$$

as desired.

Case 2. $\mathbb{E}_{F_1}[y] < \mathbb{E}_{F_0}[y]$.

Let $\lambda = \mathbb{E}_{F_1}[y]/\mathbb{E}_{F_0}[y] \in [0, 1]$ and let F'_1 be the mixture $\lambda F_0 + (1 - \lambda) \delta_0$. Note that $\mathbb{E}_{F'_1}[y] = \mathbb{E}_{F_1}[y]$. Consider $a'_1 = (F'_1, c_1)$. The corresponding incentive gap is

$$g(w_1, a'_1) = (\mathbb{E}_{F'_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_0}[w_1(y)] - c_0).$$

Note that

$$\mathbb{E}_{F'_1}[w_1(y)] - c_1 = \lambda \mathbb{E}_{F_0}[w_1(y)] - c_1 = \lambda s_1 \mathbb{E}_{F_0}[y] - c_1 = s_1 \mathbb{E}_{F_1}[y] - c_1 = \mathbb{E}_{F_1}[\hat{w}_1(y)] - c_1,$$

and

$$\mathbb{E}_{F_0}[w_1(y)] - c_0 = s_1 \mathbb{E}_{F_0}[y] - c_0 = \mathbb{E}_{F_0}[\hat{w}_1(y)] - c_0.$$

Thus,

$$\begin{aligned} g(w_1, a'_1) &= (\mathbb{E}_{F'_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_0}[w_1(y)] - c_0) \\ &= (\mathbb{E}_{F_1}[\hat{w}_1(y)] - c_1) - (\mathbb{E}_{F_0}[\hat{w}_1(y)] - c_0) \\ &= g(\hat{w}_1, a_1) \geq 0. \end{aligned}$$

When agent 1 takes action a'_1 in response, the principal's resulting payoff in the first period is

$$\mathbb{E}_{F'_1}[y - w_1(y)] = \lambda \mathbb{E}_{F_0}[y - w_1(y)] = \lambda (1 - s_1) \mathbb{E}_{F_0}[y] = (1 - s_1) \mathbb{E}_{F_1}[y] = \mathbb{E}_{F_1}[y - \hat{w}_1(y)],$$

so her payoff in the first period under $(w_1|a'_1)$ and under $(\hat{w}_1|a_1)$ are exactly equal.

Moreover, the quadruple in equation (6) with respect to (w_1, a'_1) ,

$$\left\{ \sqrt{\mathbb{E}_{F'_1} [y - w_1(y)]}, \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} - \sqrt{g(w_1, a'_1)}, \sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0}, \sqrt{\mathbb{E}_{F'_1} [y]} - \sqrt{c_1} \right\},$$

takes the same value as the quadruple in equation (6) with respect to $(\hat{w}_1, a_1)$,

$$\left\{ \sqrt{\mathbb{E}_{F_1} [y - \hat{w}_1(y)]}, \sqrt{\mathbb{E}_{F_0} [y - \hat{w}_1(y)]} - \sqrt{g(\hat{w}_1, a_1)}, \sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0}, \sqrt{\mathbb{E}_{F_1} [y]} - \sqrt{c_1} \right\}.$$

It follows from Lemma 3 that the principal's optimal second-period payoff guarantee also takes the same value: $\hat{V}_2^*(w_1, a'_1) = \hat{V}_2^*(\hat{w}_1, a_1)$.

Therefore, the principal's interim payoff guarantee is

$$\begin{aligned} \hat{U}(w_1|a'_1) &= \mathbb{E}_{F'_1} [y - w_1(y)] + \beta \cdot \hat{V}_2^*(w_1, a'_1) \\ &= \mathbb{E}_{F_1} [y - \hat{w}_1(y)] + \beta \cdot \hat{V}_2^*(\hat{w}_1, a_1) = \hat{U}(\hat{w}_1|a_1), \end{aligned}$$

as desired.

This completes the proof. $\square$

Lemma A.5. *In the case of constant technology, within the class of linear first-period contracts, there exists an optimal one for the principal.*

Proof of Lemma A.5. Assume the principal offers a linear first-period contract $w_1(y) = s_1 y$ with $s_1 \in [0, 1]$. If agent 1's payoff from taking a_0 is strictly negative, i.e., $\mathbb{E}_{F_0} [w_1(y)] - c_0 = s_1 \mathbb{E}_{F_0} [y] - c_0 < 0$, then the principal cannot guarantee any positive payoff in the first period, since it is possible that the action $(\delta_0, 0) \in A$, and the agent would strictly prefer this action to a_0 . Moreover, according to Lemma 3, the principal's optimal second-period payoff guarantee is $\hat{V}_2^*(w_1, (\delta_0, 0)) = (\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0})^2$. This is already strictly worse than offering the alternative contract $s'_1 y$ with $s'_1 = \sqrt{c_0 / \mathbb{E}_{F_0} [y]}$ instead, because doing so guarantees a strictly positive payoff $(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0})^2$ in the first period, and the payoff guarantee in the second period can only get better.

Therefore, when searching for optimal linear contracts, we may concentrate on those with $s_1 \geq c_0 / \mathbb{E}_{F_0} [y]$. For any such linear first-period contract, suppose that agent 1 chooses $a_1 = (F_1, c_1)$ in response. As is shown in Lemma 3, the principal's optimal second-period payoff guarantee is $\hat{\Phi}(w_1, a_1)^2$, with $\hat{\Phi}$ defined by equation (6). Thus, her interim payoff guarantee is

$$\hat{U}(w_1|a_1) = \mathbb{E}_{F_1} [y - w_1(y)] + \beta \cdot \hat{\Phi}(w_1, a_1)^2 = (1 - s_1) \mathbb{E}_{F_1} [y] + \beta \cdot \hat{\Phi}(w_1, a_1)^2.$$

The worst-case overall payoff guarantee minimizes the above expression over all a_1 that agent 1 may choose under some technology. Note that agent 1 prefers action a_1 over the known action a_0 if and only if the incentive gap is nonnegative, i.e., $g(w_1, a_1) \geq 0$, which is equivalent to

$$(\mathbb{E}_{F_1} [w_1(y)] - c_1) - (\mathbb{E}_{F_0} [w_1(y)] - c_0) = (s_1 \mathbb{E}_{F_1} [y] - c_1) - (s_1 \mathbb{E}_{F_0} [y] - c_0) \geq 0.$$

Hence, the following program yields a lower bound on the principal's overall payoff guarantee

$$\begin{aligned} \inf_{F_1, c_1} \quad & (1 - s_1) \mathbb{E}_{F_1} [y] + \beta \cdot \hat{\Phi}(w_1, (F_1, c_1))^2 \\ \text{s.t.} \quad & (s_1 \mathbb{E}_{F_1} [y] - c_1) - (s_1 \mathbb{E}_{F_0} [y] - c_0) \geq 0, \end{aligned} \tag{A.11}$$

because the principal's interim payoff guarantee can never be strictly lower than the infimum given by program (A.11).

Conversely, if $s_1 \geq c_0/\mathbb{E}_{F_0}[y]$, then for any feasible $a_1 = (F_1, c_1)$ in program (A.11), agent 1 would take action a_1 in response to w_1 whenever the technology A is compatible with (w_1, a_1) . The worst case over all such technologies leaves the principal with exactly her interim payoff guarantee, $\hat{U}(w_1|a_1) = (1 - s_1)\mathbb{E}_{F_1}[y] + \beta \cdot \hat{\Phi}(w_1, a_1)^2$. Thus, if a solution to program (A.11) exists, then the principal's payoff guarantee cannot be strictly higher than its minimum value.

The above analysis shows that, for $s_1 \geq c_0/\mathbb{E}_{F_0}[y]$, the worst-case overall payoff guarantee of any linear first-period contract $w_1(y) = s_1 y$ is exactly characterized by program (A.11).

Suppose $s_1 \geq c_0/\mathbb{E}_{F_0}[y]$. We now reformulate program (A.11) as an equivalent maximization problem with continuous objective function and compact feasible region. Slightly abusing notation, we use $\hat{U}(s_1)$ instead of $\hat{U}(w_1)$ to denote the infimum value of program (A.11).

Plug $w_1(y) = s_1 y$ into equation (6) and let $s_0 \equiv \sqrt{c_0/\mathbb{E}_{F_0}[y]}$. We may rewrite $\hat{\Phi}(w_1, a_1)$ as

$$\hat{\Phi}(w_1, a_1) = \max \left\{ \sqrt{(1 - s_1)\mathbb{E}_{F_1}[y]}, \sqrt{(1 - s_1)\mathbb{E}_{F_0}[y]} - \sqrt{g(w_1, a_1)}, (1 - s_0)\sqrt{\mathbb{E}_{F_0}[y]}, \sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1} \right\}.$$

Similarly,

$$g(w_1, a_1) = (s_1\mathbb{E}_{F_1}[y] - c_1) - (s_1 - s_0^2)\mathbb{E}_{F_0}[y] \geq 0.$$

Note that both the objective and the constraints of program (A.11) depend on the choice variables (F_1, c_1) only through the value of $(\mathbb{E}_{F_1}[y], c_1)$. Rewrite $\mathbb{E}_{F_1}[y] = x\mathbb{E}_{F_0}[y]$, $c_1 = z\mathbb{E}_{F_0}[y]$, and let $g(w_1, a_1) = h\mathbb{E}_{F_0}[y]$ with $x, z, h \geq 0$. Plugging into the original program (A.11) and cancelling out $\mathbb{E}_{F_0}[y]$ from both sides of the constraints, we obtain an equivalent program

$$\begin{aligned} \hat{U}(s_1) &= \inf_{x, z, h} \left((1 - s_1)x + \beta \cdot \hat{\phi}(x, z, h; s_1)^2 \right) \mathbb{E}_{F_0}[y] \\ \text{s.t.} \quad &h = s_1x - z - (s_1 - s_0^2) \geq 0, \quad x, z \geq 0, \end{aligned} \tag{A.12}$$

where

$$\hat{\phi}(x, z, h; s_1) \equiv \max \left\{ \sqrt{(1 - s_1)x}, \sqrt{1 - s_1} - \sqrt{h}, 1 - s_0, \sqrt{x} - \sqrt{z} \right\}. \tag{A.13}$$

Note that $(x, z, h) = (1, s_0^2, 0)$ is feasible in program (A.12) and leads to objective value

$$\left((1 - s_1) + \beta \cdot \max \left\{ \sqrt{1 - s_1}, 1 - s_0 \right\}^2 \right) \mathbb{E}_{F_0}[y].$$

If $x \geq 1 + \beta$, then

$$\begin{aligned} (1 - s_1)x + \beta \cdot \hat{\phi}(x, z, h; s_1)^2 &\geq (1 - s_1)(1 + \beta) + \beta(1 - s_0)^2 \\ &= (1 - s_1) + \beta(1 - s_1) + \beta(1 - s_0)^2 \\ &\geq (1 - s_1) + \beta \cdot \max \left\{ \sqrt{1 - s_1}, 1 - s_0 \right\}^2. \end{aligned}$$

Therefore, restricting $x \in [0, 1 + \beta]$ will not change the infimum of program (A.12). Moreover,

$$\max \{z, h\} \leq z + h = s_1x - (s_1 - s_0^2) \leq s_1x \leq x,$$

so restricting $(x, z, h) \in [0, 1 + \beta]^3$ will not change the infimum of program (A.12).

Consider the following program

$$\begin{aligned} \hat{\Psi}^*(s_1) &\equiv \sup_{x,z,h} \hat{\Psi}(x,z,h;s_1) \equiv -\left((1-s_1)x + \beta \cdot \hat{\phi}(x,z,h;s_1)^2\right) \\ \text{s.t.} \quad &(x,z,h) \in \hat{\Gamma}(s_1), \end{aligned} \tag{A.14}$$

where $\hat{\phi}$ is defined by equation (A.13), and $\hat{\Gamma}$ is defined as follows:

$$\hat{\Gamma}(s_1) \equiv \{(x,z,h) \in [0, 1+\beta]^3 : h = s_1x - z - (s_1 - s_0^2)\}.$$

By definition, $\hat{\Psi} : [0, 1+\beta]^3 \times [s_0^2, 1] \rightarrow \mathbb{R}$ is a continuous function, and $\hat{\Gamma} : [s_0^2, 1] \rightrightarrows [0, 1+\beta]^3$ is a compact-valued and nonempty-valued correspondence. Moreover, the infimum of program (A.12), $\hat{U}(s_1)$, is given by $(-\hat{\Psi}^*(s_1)) \cdot \mathbb{E}_{F_0}[y]$.

Note that for each s_1 , $\hat{\Gamma}(s_1)$ defines a plane intersecting a cube, and that the plane shifts linearly in s_1 . Thus, $\hat{\Gamma}$ is both upper and lower hemicontinuous. It then follows from Berge's maximum theorem that $\hat{\Psi}^*$ is continuous, and

$$\hat{\Gamma}^*(s_1) \equiv \{(x,z,h) \in \hat{\Gamma}(s_1) : \hat{\Psi}(x,z,h;s_1) = \hat{\Psi}^*(s_1)\}$$

is upper hemicontinuous with nonempty and compact values. As a consequence, a solution to program (A.14) exists for all s_1 , and the supremum can be replaced by maximum.

It follows that the infimum in program (A.12) and therefore the original program (A.11) can both be replaced by minimum, and the resulting minimum value $\hat{U}(s_1) = (-\hat{\Psi}^*(s_1)) \cdot \mathbb{E}_{F_0}[y]$ is continuous in s_1 . Hence, $\hat{U}(s_1)$ achieves a maximum over $[s_0^2, 1]$. This maximum is also the optimal guarantee over all linear contracts. $\square$

Proof of Theorem 2. According to Lemma A.5, among all linear first-period contracts, there exists an optimal one, call it w_1^* . If w_1 is any other (nonlinear) first-period contract that outperforms w_1^* , then by Lemma A.4, there is a linear contract that in turn does at least as well as w_1 . But this contradicts the fact that w_1^* is an optimal linear contract. Therefore, w_1^* is optimal among all first-period contracts. $\square$

Appendix B Constant Technology: General Set of Known Actions

In this appendix, we analyze the situation where the principal knows a *set of actions* A_0 available to the agents in the case of constant technology. The first main result is Lemma 3', which characterizes the principal's optimal second-period payoff guarantee $\hat{V}_2^*(w_1, a_1)$ in closed form and identifies the contract that attains it in various cases, analogous to Lemma 3 in the main text. Furthermore, we identify a sufficient condition on the set of known actions, *lower bound on marginal cost* (Definition B.1), which ensures that linear contracts still outperform nonlinear ones. This leads to the second main result, Theorem 2', which generalizes the optimality of linear contracts to richer environments.

In the first period, the principal believes that the true technology A could be any technology such that $A \supseteq A_0$. After the principal offers contract w_1 and observes the action a_1 chosen by agent 1, we adapt the rule of updating, *compatibility*, as follows:

Definition 1' (Compatible). *Given w_1 and $a_1 = (F_1, c_1)$, a technology A is compatible with (w_1, a_1) if*

1. $A \supseteq A_0 \cup \{a_1\}$.
2. $\mathbb{E}_F [w_1(y)] - c \leq \mathbb{E}_{F_1} [w_1(y)] - c_1$ for all $(F, c) \in A$.

B.1 Second Period Analysis

We first consider the second period of the dynamic relationship, where the principal has offered some first-period contract w_1 and observed agent 1's chosen action $a_1 = (F_1, c_1)$. She learns that the true technology A is compatible with (w_1, a_1) : it contains A_0 and a_1 , and does not contain any action strictly better than a_1 for agent 1 under w_1 .

Again, if the principal offers the same contract $w_2 = w_1$, agent 2 will choose a_1 since the two agents have the same technology. This exactly repeats the first-period payoff $\mathbb{E}_{F_1} [y - w_1(y)]$ in the second period. Moreover, if some initially known action $(F_0, c_0) \in A_0$ leads to a higher payoff for the principal, i.e., $\mathbb{E}_{F_0} [y - w_1(y)] > \mathbb{E}_{F_1} [y - w_1(y)]$, it might be tempting for the principal to try to obtain the payoff $\mathbb{E}_{F_0} [y - w_1(y)]$ instead. However, we have already seen that achieving this payoff would violate agent 2's incentive constraint, and agent 2 needs to be compensated for not choosing a_1 . The amount of compensation increases with the *incentive gap*, which may now vary for different actions.

Definition 2' (Incentive gap). *Given w_1 and $a_1 = (F_1, c_1)$, the incentive gap with respect to an action a , $g(a|w_1, a_1)$, denotes the difference in agent 1's payoff between choosing a_1 and a . Formally,*

$$g(a|w_1, a_1) \equiv (\mathbb{E}_{F_1} [w_1(y)] - c_1) - (\mathbb{E}_{F_a} [w_1(y)] - c_a).$$

Analogous to Lemma 3, part 1 of Lemma 3' shows that if $\mathbb{E}_{F_0} [y - w_1(y)] > g(a_0|w_1, a_1)$, the principal can offer a modified version of w_1 with compensation in order to guarantee that her payoff in the second period is at least $(\sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} - \sqrt{g(a_0|w_1, a_1)})^2$. Let

$$\Theta(w_1, a_1) \equiv \max_{a \in A_0 \cup \{a_1\}} \left\{ \sqrt{\mathbb{E}_{F_a} [y - w_1(y)]} - \sqrt{g(a|w_1, a_1)} \right\}, \quad (\text{B.1})$$

where we treat $w_2 = w_1$ as a special case of a modified version of w_1 (with no modification).¹⁷ The proof of Lemma 3' further shows that $\Theta(w_1, a_1)^2$ is the principal's optimal guarantee using a modified version of w_1 .

Note that the optimal static contract in Carroll (2015) is still available to the principal. By offering this contract following the procedure in Carroll (2015), the principal can guarantee that her payoff in the

¹⁷By definition, $g(a_1|w_1, a_1) = 0$. Moreover, it follows from agent 1's rationality that $g(a_0|w_1, a_1) \geq 0$ for all $a_0 \in A_0$.

second period is at least $\Phi(a_1)^2$, where Φ is defined by equation (2). Part 2 of Lemma 3' shows that when $\Phi(a_1) > \Theta(w_1, a_1)$, it is optimal for the principal to offer this optimal static contract in the second period, and doing so exactly attains payoff guarantee $\Phi(a_1)^2$.

We are now ready to present the main result of this subsection, Lemma 3', which characterizes the principal's optimal second-period payoff guarantee $\hat{V}_2^*(w_1, a_1)$, and establishes the optimality of the aforementioned contracts. It is optimal for the principal to offer either a modified version of w_1 with compensation, or a linear contract.

Lemma 3'. *Suppose the principal offers first-period contract w_1 , and agent 1 chooses a_1 in response. The principal's optimal second-period payoff guarantee is*

$$\hat{V}_2^*(w_1, a_1) = (\max\{\Theta(w_1, a_1), \Phi(a_1)\})^2. \quad (\text{B.2})$$

Specifically,

1. If $\Theta(w_1, a_1) \geq \Phi(a_1)$ and $a^* \in A_0 \cup \{a_1\}$ attains the maximum in equation (B.1), then the principal's optimal second-period payoff guarantee is achieved by a modified version of w_1 :

$$w_2(y) = w_1(y) + m \cdot (y - w_1(y)) \quad \text{with} \quad m = \sqrt{\frac{g(a^*|w_1, a_1)}{\mathbb{E}_{F_{a^*}}[y - w_1(y)]}} \in [0, 1]. \quad (\text{B.3})$$

2. If $\Theta(w_1, a_1) < \Phi(a_1)$ and $a^* \in A_0 \cup \{a_1\}$ attains the maximum in equation (2), then the principal's optimal second-period payoff guarantee is achieved by a linear contract:

$$w_2(y) = s_2 y \quad \text{with} \quad s_2 = \sqrt{\frac{c_{a^*}}{\mathbb{E}_{F_{a^*}}[y]}}. \quad (\text{B.4})$$

Proof. All proofs for this Appendix are in Appendix B.3. □

B.2 First Period Analysis

So far, we have focused on principal's problem in the second period and fully characterized her optimal second-period payoff guarantee. Now we analyze the principal's first-period problem of choosing a first-period contract w_1 to maximize her overall payoff guarantee $\hat{U}(w_1)$.

The following condition, *lower bound on marginal cost*, is sufficient to ensure that the principal's optimal overall payoff guarantee is achieved by a linear first-period contract.

Definition B.1 (Lower bound on marginal cost). *The known technology A_0 satisfies lower bound on marginal cost if, for any pair of actions $(F, c), (F', c') \in A_0$ with $0 < \mathbb{E}_F[y] < \mathbb{E}_{F'}[y]$, it holds that*

$$c' - c \geq \mathbb{E}_{F'}[y] - \mathbb{E}_F[y].$$

This condition provides linkage between different actions in the known technology A_0 . Moreover, it contains the economic meaning that, between known actions, the change in costs cannot be too small compared with the change in expected output. Thus, this condition sets a lower bound on the marginal cost of the known technology in discrete form.

The main result of the first period analysis is Theorem 2'.

Theorem 2'. *Suppose the known technology A_0 satisfies lower bound on marginal cost. In the case of constant technology, there exists a linear first-period contract w_1 that maximizes the principal's overall payoff guarantee $\hat{U}(w_1)$.*

Analogous to Theorems 1 and 2, the proof of Theorem 2' takes two steps: (1) Lemma A.4' improves any nonlinear first-period contract into a linear one; (2) Lemma A.5' shows that the maximum of the principal's first-period problem exists within the class of linear first-period contracts. We remark that the additional condition, lower bound on marginal cost, comes into play only in the first step of the proof (i.e., Lemma A.4').

We start from any arbitrary first-period contract w_1 , and construct another linear contract $\hat{w}_1$ that provides the principal with a weakly higher overall payoff guarantee. Let $a_0 = (F_0, c_0)$ be the action agent 1 will choose if the true technology $A = A_0$. The procedure of constructing the linear $\hat{w}_1$ is exactly the same as in the proof of Lemma A.4, given by equation (3). When the known technology satisfies lower bound on marginal cost, Lemma A.4' below shows that the principal's overall payoff guarantee is at least as high under $\hat{w}_1$ as it is under w_1 .

Lemma A.4'. *Suppose the known technology A_0 satisfies lower bound on marginal cost. Let w_1 be any first-period contract, and let $(F_0, c_0) \in A_0$ be agent 1's best response when the true technology A is just A_0 . The linear contract $\hat{w}_1$ defined by equation (3) satisfies $\hat{U}(\hat{w}_1) \geq \hat{U}(w_1)$.*

Similar to the proof of Lemma A.4, for any action that may be taken by agent 1 under $\hat{w}_1$ and some technology $A \supseteq A_0$, the proof of Lemma A.4' explicitly constructs an alternative action a'_1 that may be taken by agent 1 under w_1 and some other technology. The difference between this general case and the singleton case is that the principal's optimal second-period payoff guarantee $\hat{V}_2^*$ is given by a more general expression (B.2), and in particular maximum in Θ or Φ may be attained by $a^* \in A_0 \setminus \{a_0\}$. The condition *lower bound on marginal cost* disciplines the relationship between a_0 and a^* , which makes the proof method of Lemma A.4 generalizable. In subsequent research, we hope to examine whether this (or any such) restriction is necessary, in the sense that there exists a counterexample when it is violated.

By establishing Lemma A.4', we have shown that any nonlinear first-period contract can be improved by a linear one. To finalize the proof of Theorem 2', it suffices to show that, within the class of linear contracts, the maximum of $U(w_1)$ exists.

Lemma A.5'. *Within the class of linear first-period contracts, there exists an optimal one for the principal.*

The proof of Lemma A.5' requires to characterize the overall payoff guarantee of an arbitrary linear first-period contract. Assume the principal offers a linear first-period contract $w_1(y) = s_1 y$ with $s_1 \in [0, 1]$, and agent 1 chooses $a_1 = (F_1, c_1)$ in response. As is shown in Lemma 3', the principal's optimal second-period payoff guarantee $\hat{V}_2^*(w_1, a_1) = (\max\{\Theta(w_1, a_1), \Phi(a_1)\})^2$. Thus, her interim payoff guarantee is

$$\hat{U}(w_1|a_1) = \mathbb{E}_{F_1}[y - w_1(y)] + \beta \cdot \hat{V}_2^*(w_1, a_1) = (1 - s_1)\mathbb{E}_{F_1}[y] + \beta \cdot \hat{V}_2^*(w_1, a_1).$$

The worst-case overall payoff guarantee minimizes the above expression over all a_1 that agent 1 may choose under some technology. Note that agent 1 prefers action a_1 over all known actions $a \in A_0$ if and only if

$$(\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_a}[w_1(y)] - c_a) = (s_1\mathbb{E}_{F_1}[y] - c_1) - (s_1\mathbb{E}_{F_a}[y] - c_a) \geq 0, \quad \forall a \in A_0.$$

Moreover, agent 1 obtains at least his reservation payoff of zero, which can also be viewed as his payoff from the null action $(\delta_0, 0)$. Hence, the following program yields a lower bound on the principal's overall payoff guarantee

$$\begin{aligned} \inf_{F_1, c_1} \quad & (1 - s_1)\mathbb{E}_{F_1}[y] + \beta \cdot \hat{V}_2^*(w_1, (F_1, c_1)) \\ \text{s.t.} \quad & (s_1\mathbb{E}_{F_1}[y] - c_1) - (s_1\mathbb{E}_{F_a}[y] - c_a) \geq 0, \quad \forall a \in A_0 \cup \{(\delta_0, 0)\}, \end{aligned} \tag{B.5}$$

because the principal's interim payoff guarantee can never be strictly lower than the infimum given by program (B.5).

Conversely, for any feasible $a_1 = (F_1, c_1)$ in program (B.5), agent 1 would take action a_1 in response to w_1 when his technology $A_1 = A_0 \cup \{a_1\}$. The worst case over all such technologies leaves the principal with exactly her interim payoff guarantee, $\hat{U}(w_1|a_1) = (1 - s_1) \mathbb{E}_{F_1}[y] + \beta \cdot \hat{V}_2^*(w_1, a_1)$. Thus, if a solution to program (B.5) exists (i.e., if infimum may be replaced by minimum), then the principal's payoff guarantee cannot be strictly higher than its minimum value.

Therefore, the worst-case overall payoff guarantee of any linear first-period contract $w_1(y) = s_1 y$ is exactly characterized by program (B.5). In the proof of Lemma A.5' in Appendix B.3, we formally show the existence of minimum in this program, and its continuity in the first-period share s_1 using Berge's maximum theorem. Since the overall payoff guarantee is continuous in the first-period share s_1 , it achieves a maximum. This maximum is also the optimal guarantee over all linear contracts.

Combining Lemmas A.4' and A.5', we prove the main result of this section, Theorem 2', which establishes the optimality of a linear first-period contract.

B.3 Proofs for Appendix B

To prove Lemma 3', we start by establishing two lemmas, Lemmas B.1 and B.2, to show that the principal's payoff guarantees in the second period from offering the two contracts, (i) $w_2(y) = w_1(y) + m \cdot (y - w_1(y))$ with m defined by equation (B.3), and (ii) $w_2(y) = s_2 y$ with s_2 defined by equation (B.4), are exactly as claimed in the statement of Lemma 3'.

Lemma B.1. *If $\Theta(w_1, a_1) \geq \Phi(a_1)$ and $a^* = (F^*, c^*) \in A_0 \cup \{a_1\}$ attains the maximum in equation (B.1), and the principal offers $w_2(y) = w_1(y) + m \cdot (y - w_1(y))$ with m defined by equation (B.3), then her payoff guarantee in the second period is exactly*

$$\Theta(w_1, a_1)^2 = \left(\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g(a^*|w_1, a_1)} \right)^2.$$

Proof of Lemma B.1. Let $g^* \equiv g(a^*|w_1, a_1) = (\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F^*}[w_1(y)] - c^*) \geq 0$. We have $\Theta(w_1, a_1) = \sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g^*}$. From $\Theta(w_1, a_1) \geq \Phi(a_1) > 0$, it holds that

$$m = \sqrt{\frac{g^*}{\mathbb{E}_{F^*}[y - w_1(y)]}} \in [0, 1].$$

Suppose the principal offers $w_2(y) = w_1(y) + m \cdot (y - w_1(y))$ with m defined by equation (B.3). We first show that this guarantees her at least $\left(\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g^*} \right)^2$.

Let (F_2, c_2) be the action chosen by agent 2. By agent 1's rationality, we have

$$\mathbb{E}_{F_1}[w_1(y)] - c_1 \geq \mathbb{E}_{F_2}[w_1(y)] - c_2.$$

By agent 2's rationality, we have

$$\mathbb{E}_{F_2}[w_2(y)] - c_2 \geq \mathbb{E}_{F^*}[w_2(y)] - c^*.$$

Summing up the two inequalities, we obtain

$$\begin{aligned} m \cdot \mathbb{E}_{F_2}[y - w_1(y)] &= \mathbb{E}_{F_2}[w_2(y) - w_1(y)] \geq (\mathbb{E}_{F^*}[w_2(y)] - c^*) - (\mathbb{E}_{F_1}[w_1(y)] - c_1) \\ &= m \cdot \mathbb{E}_{F^*}[y - w_1(y)] - g^*, \end{aligned}$$

implying that

$$\mathbb{E}_{F_2}[y - w_1(y)] \geq \mathbb{E}_{F^*}[y - w_1(y)] - g^*/m.$$

Therefore, the principal's payoff in the second period is

$$\begin{aligned}\mathbb{E}_{F_2} [y - w_2(y)] &= \mathbb{E}_{F_2} [y - w_1(y)] - m \cdot \mathbb{E}_{F_2} [y - w_1(y)] = (1 - m) \mathbb{E}_{F_2} [y - w_1(y)] \\ &\geq (1 - m) (\mathbb{E}_{F^*} [y - w_1(y)] - g^*/m) = \left(\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} - \sqrt{g^*} \right)^2,\end{aligned}$$

as desired.

Next we show that her payoff guarantee from $w_2(y) = w_1(y) + m \cdot (y - w_1(y))$ cannot be strictly higher than $\left(\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} - \sqrt{g^*} \right)^2$, since this is exactly her payoff when the technology is $A = A_0 \cup \{a_1, (F', c')\}$, with $F' = (1 - m) F^* + m \cdot \delta_0$ and $c' = c^* - (m \cdot \mathbb{E}_{F^*} [w_1(y)] + g^*)$.

The proof takes three steps.

Step 1 $c^* \geq m \cdot \mathbb{E}_{F^*} [w_1(y)] + g^*$, so c' is indeed nonnegative.

From $\Theta(w_1, a_1) \geq \Phi(a_1)$, we obtain

$$\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} - \sqrt{g^*} = \Theta(w_1, a_1) \geq \Phi(a_1) \geq \sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{c^*},$$

which implies that

$$c^* \geq \left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} + \sqrt{g^*} \right)^2$$

It suffices to show

$$\begin{aligned}&\left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} + \sqrt{g^*} \right)^2 \geq m \cdot \mathbb{E}_{F^*} [w_1(y)] + g^* \\ \Leftrightarrow &\left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right)^2 \geq m \cdot \mathbb{E}_{F^*} [w_1(y)] - 2\sqrt{g^*} \cdot \left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right) \\ \Leftrightarrow &\left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right)^2 \geq m \cdot \left(\mathbb{E}_{F^*} [w_1(y)] - 2\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \cdot \left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right) \right).\end{aligned}\tag{B.6}$$

Note that

$$\begin{aligned}&\mathbb{E}_{F^*} [w_1(y)] - 2\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \cdot \left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right) \\ &= \mathbb{E}_{F^*} [w_1(y)] - 2\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \cdot \frac{\mathbb{E}_{F^*} [w_1(y)]}{\sqrt{\mathbb{E}_{F^*} [y]} + \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]}} \\ &= \frac{\mathbb{E}_{F^*} [w_1(y)]}{\sqrt{\mathbb{E}_{F^*} [y]} + \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]}} \cdot \left(\sqrt{\mathbb{E}_{F^*} [y]} + \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} - 2\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right) \\ &= \left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right) \cdot \left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right) = \left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right)^2.\end{aligned}$$

Therefore, inequality (B.6) is equivalent to

$$\left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right)^2 \geq m \cdot \left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \right)^2,$$

which is implied by the assumption that $\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} \geq \sqrt{g^*}$ (or equivalently, $m \leq 1$).

Step 2 $A = A_0 \cup \{a_1, (F', c')\}$ is compatible with (w_1, a_1) . That is, agent 1 chooses a_1 in response to w_1 .

Agent 1's payoff from (F', c') is

$$\begin{aligned}\mathbb{E}_{F'} [w_1(y)] - c' &= (1 - m) \mathbb{E}_{F^*} [w_1(y)] - c^* + (m \cdot \mathbb{E}_{F^*} [w_1(y)] + g^*) \\ &= (\mathbb{E}_{F^*} [w_1(y)] - c^*) + g^* = \mathbb{E}_{F_1} [w_1(y)] - c_1,\end{aligned}$$

so he would choose $a_1 = (F_1, c_1)$ in response to w_1 .

Note that agent 1 is actually indifferent between (F_1, c_1) and (F', c') , and we will show below that agent 2 is indifferent between (F^*, c^*) and (F', c') . Technically to ensure that agent 1 chooses (F_1, c_1) and agent 2 chooses (F', c') we can set $F' = (1 - m + \varepsilon) F^* + (m - \varepsilon) \delta_0$ and $c' = c^* - (m \cdot \mathbb{E}_{F^*} [w_1(y)] + g^*) + \varepsilon \cdot \mathbb{E}_{F^*} [w_1(y) + (m/2) \cdot (y - w_1(y))]$, and then let $\varepsilon \downarrow 0$. Many of the following cases of potential indifference shall be treated similarly, and we omit them for brevity.

Step 3 If $A = A_0 \cup \{a_1, (F', c')\}$, then agent 2 chooses (F', c') in response to w_2 , leading to a payoff of $\left(\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} - \sqrt{g^*}\right)^2$ for the principal.

Agent 2's payoff from (F', c') is

$$\begin{aligned}\mathbb{E}_{F'} [w_2(y)] - c' &= (1 - m) \mathbb{E}_{F^*} [w_1(y) + m \cdot (y - w_1(y))] - c^* + (m \cdot \mathbb{E}_{F^*} [w_1(y)] + g^*) \\ &= \mathbb{E}_{F^*} [w_1(y)] + m \cdot \mathbb{E}_{F^*} [y - w_1(y)] - m^2 \cdot \mathbb{E}_{F^*} [y - w_1(y)] - c^* + g^* \\ &= \mathbb{E}_{F^*} [w_2(y)] - g^* - c^* + g^* = \mathbb{E}_{F^*} [w_2(y)] - c^*.\end{aligned}$$

For any action $a_0 = (F_0, c_0) \in A_0 \cup \{a_1\}$, let $g_0 \equiv g(a_0 | w_1, a_1) = (\mathbb{E}_{F_1} [w_1(y)] - c_1) - (\mathbb{E}_{F_0} [w_1(y)] - c_0) \geq 0$. Agent 2's payoff from a_0 is

$$\mathbb{E}_{F_0} [w_2(y)] - c_0 = \mathbb{E}_{F_0} [w_1(y) + m \cdot (y - w_1(y))] - c_0 = m \cdot \mathbb{E}_{F_0} [y - w_1(y)] + (\mathbb{E}_{F_0} [w_1(y)] - c_0).$$

Note that

$$\begin{aligned}\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} - \sqrt{g^*} &= \Theta(w_1, a_1) \geq \sqrt{\mathbb{E}_{F_0} [y - w_1(y)]} - \sqrt{g_0} \\ \Rightarrow \mathbb{E}_{F_0} [y - w_1(y)] &\leq \left(\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} - \sqrt{g^*} + \sqrt{g_0}\right)^2.\end{aligned}$$

Moreover,

$$\mathbb{E}_{F_0} [w_1(y)] - c_0 = (\mathbb{E}_{F_1} [w_1(y)] - c_1) - g_0 = (\mathbb{E}_{F^*} [w_1(y)] - c^*) + g^* - g_0.$$

Thus, agent 2's payoff from a_0 ,

$$\begin{aligned}\mathbb{E}_{F_0} [w_2(y)] - c_0 &= m \cdot \mathbb{E}_{F_0} [y - w_1(y)] + (\mathbb{E}_{F_0} [w_1(y)] - c_0) \\ &\leq m \cdot \left(\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} - \sqrt{g^*} + \sqrt{g_0}\right)^2 + (\mathbb{E}_{F^*} [w_1(y)] - c^*) + g^* - g_0 \\ &\leq m \cdot \mathbb{E}_{F^*} [y - w_1(y)] + (\mathbb{E}_{F^*} [w_1(y)] - c^*) \\ &= \mathbb{E}_{F^*} [w_2(y)] - c^* = \mathbb{E}_{F'} [w_2(y)] - c',\end{aligned}\tag{B.7}$$

so he would choose (F', c') in response to w_2 . Recall $m = \sqrt{g^* / \mathbb{E}_{F^*} [y - w_1(y)]}$, so the last inequality (B.7)

is equivalent to

$$\begin{aligned} & m \cdot \left(\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} - \sqrt{g^*} + \sqrt{g_0} \right)^2 + g^* - g_0 \leq m \cdot \mathbb{E}_{F^*} [y - w_1(y)] \\ \Leftrightarrow & \left(1 - \sqrt{\frac{g^*}{\mathbb{E}_{F^*} [y - w_1(y)]}} \right) (\sqrt{g_0} - \sqrt{g^*}) \geq 0, \end{aligned}$$

which always holds.

This leaves the principal with a payoff of

$$\begin{aligned} \mathbb{E}_{F'} [y - w_2(y)] &= \mathbb{E}_{F'} [y - w_1(y)] - m \cdot \mathbb{E}_{F'} [y - w_1(y)] = (1 - m) \mathbb{E}_{F'} [y - w_1(y)] \\ &= (1 - m)^2 \mathbb{E}_{F^*} [y - w_1(y)] = \left(\sqrt{\mathbb{E}_{F^*} [y - w_1(y)]} - \sqrt{g^*} \right)^2, \end{aligned}$$

as desired.

This completes the proof. $\square$

Lemma B.2. *If $\Theta(w_1, a_1) < \Phi(a_1)$ and $(F^*, c^*) \in A_0 \cup \{a_1\}$ attains the maximum in equation (2), and the principal offers the linear contract $w_2(y) = s_2 y$ with s_2 defined by equation (B.4), then her payoff guarantee in the second period is exactly*

$$\Phi(a_1)^2 = \left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{c^*} \right)^2.$$

Proof of Lemma B.2. Suppose the principal offers the linear contract $w_2(y) = s_2 y$ with s_2 defined by equation (B.4). We first show that this guarantees her at least $\left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{c^*} \right)^2$.

Let (F_2, c_2) be the action chosen by agent 2. By agent 2's rationality, we have

$$\mathbb{E}_{F_2} [w_2(y)] - c_2 \geq \mathbb{E}_{F^*} [w_2(y)] - c^*,$$

which further implies that

$$s_2 \mathbb{E}_{F_2} [y] = \mathbb{E}_{F_2} [w_2(y)] \geq \mathbb{E}_{F_2} [w_2(y)] - c_2 \geq \mathbb{E}_{F^*} [w_2(y)] - c^* = s_2 \mathbb{E}_{F^*} [y] - c^*,$$

and hence

$$\mathbb{E}_{F_2} [y] \geq \mathbb{E}_{F^*} [y] - c^*/s_2.$$

Therefore, the principal's payoff in the second period is

$$\mathbb{E}_{F_2} [y - w_2(y)] = \mathbb{E}_{F_2} [(1 - s_2) y] \geq (1 - s_2) (\mathbb{E}_{F^*} [y] - c^*/s_2) = \left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{c^*} \right)^2,$$

as desired.

Next we show that her payoff guarantee from this linear contract cannot be strictly higher, since $\left(\sqrt{\mathbb{E}_{F^*} [y]} - \sqrt{c^*} \right)^2$ is exactly her payoff when the technology is $A = A_0 \cup \{a_1, (F', 0)\}$, with $F' = \lambda F^* + (1 - \lambda)\delta_0$ where $\lambda = 1 - \sqrt{c^*/\mathbb{E}_{F^*} [y]} \in [0, 1]$.

The proof takes two steps. Let $g^* \equiv g(a^*|w_1, a_1) = (\mathbb{E}_{F_1} [w_1(y)] - c_1) - (\mathbb{E}_{F^*} [w_1(y)] - c^*) \geq 0$.

Step 1 $A = A_0 \cup \{a_1, (F', 0)\}$ is compatible with (w_1, a_1) . That is, agent 1 chooses a_1 in response to w_1 .

Agent 1's payoff from $(F', 0)$ is $\mathbb{E}_{F'} [w_1(y)] = \lambda \mathbb{E}_{F^*} [w_1(y)] = \left(1 - \sqrt{c^*/\mathbb{E}_{F^*} [y]} \right) \mathbb{E}_{F^*} [w_1(y)]$, and we

have

$$\begin{aligned} \left(1 - \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}}\right) \mathbb{E}_{F^*}[w_1(y)] \leq \mathbb{E}_{F_1}[w_1(y)] - c_1 &\Leftrightarrow \left(1 - \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}}\right) \mathbb{E}_{F^*}[w_1(y)] \leq (\mathbb{E}_{F^*}[w_1(y)] - c^*) + g^* \\ &\Leftrightarrow \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \mathbb{E}_{F^*}[w_1(y)] - c^* + g^* \geq 0. \end{aligned}$$

From

$$\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*} = \Phi(a_1) > \Theta(w_1, a_1) \geq \sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g^*},$$

we obtain

$$\mathbb{E}_{F^*}[w_1(y)] \geq \mathbb{E}_{F^*}[y] - \left(\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*} + \sqrt{g^*}\right)^2,$$

and thus

$$\begin{aligned} \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \mathbb{E}_{F^*}[w_1(y)] - c^* + g^* &\geq \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \cdot \left(\mathbb{E}_{F^*}[y] - \left(\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*} + \sqrt{g^*}\right)^2\right) - c^* + g^* \\ &= \left(1 - \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}}\right) \left(\sqrt{c^*} - \sqrt{g^*}\right)^2 \geq 0, \end{aligned}$$

as desired. So we indeed have $\mathbb{E}_{F'}[w_1(y)] \leq \mathbb{E}_{F_1}[w_1(y)] - c_1$, implying that agent 1 would choose $a_1 = (F_1, c_1)$ in response to w_1 .

Step 2 If $A = A_0 \cup \{a_1, (F', 0)\}$, then agent 2 chooses $(F', 0)$ in response to w_2 , leading to a payoff of $\left(\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*}\right)^2$ for the principal.

Agent 2's payoff from $(F', 0)$ is

$$\begin{aligned} \mathbb{E}_{F'}[w_2(y)] &= \lambda \mathbb{E}_{F^*}[s_2 y] = \left(1 - \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}}\right) \cdot \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \cdot \mathbb{E}_{F^*}[y] \\ &= \left(\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*}\right) \sqrt{c^*} = \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \cdot \mathbb{E}_{F^*}[y] - c^* \\ &= s_2 \mathbb{E}_{F^*}[y] - c^* = \mathbb{E}_{F^*}[w_2(y)] - c^*. \end{aligned}$$

For any action $a_0 = (F_0, c_0) \in A_0 \cup \{a_1\}$, agent 2's payoff from a_0 is

$$\mathbb{E}_{F_0}[w_2(y)] - c_0 = \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \cdot \mathbb{E}_{F_0}[y] - c_0,$$

and we have

$$\sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \cdot \mathbb{E}_{F_0}[y] - c_0 \leq \mathbb{E}_{F^*}[w_2(y)] - c^* \Leftrightarrow \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \cdot \mathbb{E}_{F_0}[y] - c_0 \leq \left(\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*}\right) \sqrt{c^*}.$$

From $\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*} = \Phi(a_1) \geq \sqrt{\mathbb{E}_{F_0}[y]} - \sqrt{c_0}$, we obtain $\mathbb{E}_{F_0}[y] \leq (\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*} + \sqrt{c_0})^2$, and thus

$$\begin{aligned} & (\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*}) \sqrt{c^*} - \left(\sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \cdot \mathbb{E}_{F_0}[y] - c_0 \right) \\ & \geq (\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*}) \sqrt{c^*} - \left(\sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \cdot (\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*} + \sqrt{c_0})^2 - c_0 \right) \\ & = \left(1 - \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \right) (\sqrt{c^*} - \sqrt{c_0})^2 \geq 0, \end{aligned}$$

as desired. So we indeed have $\mathbb{E}_{F_0}[w_2(y)] - c_0 \leq \mathbb{E}_{F^*}[w_2(y)] - c^* = \mathbb{E}_{F'}[w_2(y)]$, implying that agent 2 would choose $(F', 0)$ in response to w_2 .

This leaves the principal with a payoff of

$$\begin{aligned} \mathbb{E}_{F'}[y - w_2(y)] &= \lambda \mathbb{E}_{F^*}[(1 - s_2)y] = \left(1 - \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \right) \left(1 - \sqrt{\frac{c^*}{\mathbb{E}_{F^*}[y]}} \right) \cdot \mathbb{E}_{F^*}[y] \\ &= \left(\sqrt{\mathbb{E}_{F^*}[y]} - \sqrt{c^*} \right)^2, \end{aligned}$$

as desired.

This completes the proof. $\square$

We are now ready to prove Lemma 3'.

Proof of Lemma 3'. Combining Lemmas B.1 and B.2, we have shown that by offering the best of the two contracts: (i) $w_2(y) = w_1(y) + m \cdot (y - w_1(y))$ with m defined by equation (B.3), and (ii) $w_2(y) = s_2 y$ with s_2 defined by equation (B.4), the principal's payoff guarantee in the second period is exactly given by $(\max\{\Theta(w_1, a_1), \Phi(a_1)\})^2$. The principal's optimal second-period payoff guarantee, $V_2^*(w_1, a_1)$, is thus at least $(\max\{\Theta(w_1, a_1), \Phi(a_1)\})^2$.

Now consider an arbitrary second-period contract w_2 . It suffices to show that the principal's payoff guarantee is not strictly higher than $(\max\{\Theta(w_1, a_1), \Phi(a_1)\})^2$ under w_2 .

Let $a_0 = (F_0, c_0)$ be the action agent 2 will choose if the true technology is exactly $A_0 \cup \{a_1\}$. Consider the following three cases.

Case 1. $\mathbb{E}_{F_0}[w_2(y)] < c_0$.

Consider the second-period contract w_2 when $A = A_0 \cup \{a_1, (\delta_0, 0)\}$, which is compatible with (w_1, a_1) . Agent 2's payoff from $(\delta_0, 0)$ is

$$w_2(0) \geq 0 > \mathbb{E}_{F_0}[w_2(y)] - c_0,$$

so he would prefer to take action $(\delta_0, 0)$. This leaves the principal with a payoff of

$$-w_2(0) \leq 0 \leq \Phi(a_1)^2,$$

as desired.

Case 2. $\mathbb{E}_{F_0} [w_2(y)] \geq c_0$, and it holds that

$$\begin{aligned} \text{either (i)} \quad & \mathbb{E}_{F_0} [w_1(y)] \leq \mathbb{E}_{F_1} [w_1(y)] - c_1, \\ \text{or (ii)} \quad & \mathbb{E}_{F_0} [w_2(y)] < \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_1} [w_1(y)] - c_1)} c_0. \end{aligned} \quad (\text{B.8})$$

Let $\lambda = 1 - c_0/\mathbb{E}_{F_0} [w_2(y)] \in [0, 1]$ and let F' be the mixture $\lambda F_0 + (1 - \lambda)\delta_0$. Consider the technology $A = A_0 \cup \{a_1, (F', 0)\}$. We proceed with two steps.

Step 1 A is compatible with (w_1, a_1) . That is, agent 1 chooses a_1 in response to w_1 .
Agent 1's payoff from $(F', 0)$ is

$$\begin{aligned} \mathbb{E}_{F'} [w_1(y)] &= \lambda \mathbb{E}_{F_0} [w_1(y)] = \mathbb{E}_{F_0} [w_1(y)] - \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_2(y)]} c_0 \\ &< \mathbb{E}_{F_1} [w_1(y)] - c_1. \end{aligned} \quad (\text{B.9})$$

Note that inequality (B.9) holds exactly due to the assumptions in (B.8). So agent 1 would prefer to take action $a_1 = (F_1, c_1)$ when $A = A_0 \cup \{a_1, (F', 0)\}$.

Step 2 Agent 2 chooses $(F', 0)$ in response to w_2 , resulting in the principal's payoff no more than $\Phi(a_1)^2$.
Agent 2's payoff from $(F', 0)$ is

$$\begin{aligned} \mathbb{E}_{F'} [w_2(y)] &= \lambda \mathbb{E}_{F_0} [w_2(y)] + (1 - \lambda)w_2(0) \\ &\geq \lambda \mathbb{E}_{F_0} [w_2(y)] = \mathbb{E}_{F_0} [w_2(y)] - c_0. \end{aligned}$$

So he would prefer to take action $(F', 0)$ when $A = A_0 \cup \{a_1, (F', 0)\}$.

This leaves the principal with a payoff of

$$\begin{aligned} \mathbb{E}_{F'} [y - w_2(y)] &= \lambda \mathbb{E}_{F_0} [y - w_2(y)] + (1 - \lambda)(0 - w_2(0)) \\ &\leq \lambda \mathbb{E}_{F_0} [y - w_2(y)] = \left(1 - \frac{c_0}{\mathbb{E}_{F_0} [w_2(y)]}\right) (\mathbb{E}_{F_0} [y] - \mathbb{E}_{F_0} [w_2(y)]) \\ &\leq \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0}\right)^2, \end{aligned} \quad (\text{B.10})$$

which is no more than $\Phi(a_1)^2$, as desired. The last inequality (B.10),

$$\begin{aligned} \left(1 - \frac{c_0}{\mathbb{E}_{F_0} [w_2(y)]}\right) (\mathbb{E}_{F_0} [y] - \mathbb{E}_{F_0} [w_2(y)]) &\leq \left(\sqrt{\mathbb{E}_{F_0} [y]} - \sqrt{c_0}\right)^2 \\ \Leftrightarrow \left(\sqrt{\mathbb{E}_{F_0} [w_2(y)]} - \sqrt{\frac{c_0 \mathbb{E}_{F_0} [y]}{\mathbb{E}_{F_0} [w_2(y)]}}\right)^2 &\geq 0, \end{aligned}$$

which always holds.

Case 3. Both inequalities in (B.8) are reversed, i.e.,

$$\mathbb{E}_{F_0} [w_1(y)] > \mathbb{E}_{F_1} [w_1(y)] - c_1 \quad \text{and} \quad \mathbb{E}_{F_0} [w_2(y)] \geq \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_1} [w_1(y)] - c_1)} c_0.$$

Let

$$\lambda = \frac{(\mathbb{E}_{F_0} [w_2(y)] - c_0) - (\mathbb{E}_{F_1} [w_1(y)] - c_1)}{\mathbb{E}_{F_0} [w_2(y)] - \mathbb{E}_{F_0} [w_1(y)]},$$

$$c' = \frac{\mathbb{E}_{F_0} [w_1(y)] (\mathbb{E}_{F_0} [w_2(y)] - c_0) - \mathbb{E}_{F_0} [w_2(y)] (\mathbb{E}_{F_1} [w_1(y)] - c_1)}{\mathbb{E}_{F_0} [w_2(y)] - \mathbb{E}_{F_0} [w_1(y)]},$$

and let F' be the mixture $\lambda F_0 + (1 - \lambda)\delta_0$. Consider the technology $A = A_0 \cup \{a_1, (F', c')\}$. We proceed with three steps.

Step 1 $\lambda \in [0, 1]$ and $c' \geq 0$, so (F', c') is a valid action.

Note that

$$\mathbb{E}_{F_0} [w_2(y)] \geq \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_1} [w_1(y)] - c_1)} c_0 \geq \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_0} [w_1(y)] - c_0)} c_0 = \mathbb{E}_{F_0} [w_1(y)],$$

so the denominator of λ and c' is positive.

Moreover,

$$\begin{aligned} \mathbb{E}_{F_0} [w_2(y)] - c_0 &\geq \frac{\mathbb{E}_{F_1} [w_1(y)] - c_1}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_1} [w_1(y)] - c_1)} c_0 \\ &\geq \frac{\mathbb{E}_{F_1} [w_1(y)] - c_1}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_0} [w_1(y)] - c_0)} c_0 = \mathbb{E}_{F_1} [w_1(y)] - c_1, \end{aligned}$$

so the numerator of λ is positive.

The numerator of c' is positive because

$$\begin{aligned} \mathbb{E}_{F_0} [w_1(y)] (\mathbb{E}_{F_0} [w_2(y)] - c_0) &\geq \mathbb{E}_{F_0} [w_2(y)] (\mathbb{E}_{F_1} [w_1(y)] - c_1) \\ \Leftrightarrow \mathbb{E}_{F_0} [w_2(y)] &\geq \frac{\mathbb{E}_{F_0} [w_1(y)]}{\mathbb{E}_{F_0} [w_1(y)] - (\mathbb{E}_{F_1} [w_1(y)] - c_1)} c_0. \end{aligned}$$

Finally,

$$\begin{aligned} (\mathbb{E}_{F_0} [w_2(y)] - c_0) - (\mathbb{E}_{F_1} [w_1(y)] - c_1) &\leq \mathbb{E}_{F_0} [w_2(y)] - \mathbb{E}_{F_0} [w_1(y)] \\ \Leftrightarrow \mathbb{E}_{F_0} [w_1(y)] - c_0 &\leq \mathbb{E}_{F_1} [w_1(y)] - c_1, \end{aligned}$$

so λ is indeed smaller than 1.

Step 2 A is compatible with (w_1, a_1) . That is, agent 1 chooses a_1 in response to w_1 .

Agent 1's payoff from (F', c') is

$$\mathbb{E}_{F'} [w_1(y)] - c' = \lambda \mathbb{E}_{F_0} [w_1(y)] - c' = \mathbb{E}_{F_1} [w_1(y)] - c_1,$$

so he would prefer to take action $a_1 = (F_1, c_1)$ when $A = A_0 \cup \{a_1, (F', c')\}$.

Step 3 Agent 2 chooses (F', c') in response to w_2 , resulting in the principal's payoff no more than $\Theta(w_1, a_1)^2$.

Agent 2's payoff from (F', c') is

$$\begin{aligned}\mathbb{E}_{F'}[w_2(y)] - c' &= \lambda \mathbb{E}_{F_0}[w_2(y)] + (1 - \lambda)w_2(0) - c' \\ &\geq \lambda \mathbb{E}_{F_0}[w_2(y)] - c' = \mathbb{E}_{F_0}[w_2(y)] - c_0.\end{aligned}$$

So he would prefer to take action (F', c') when $A = A_0 \cup \{a_1, (F', c')\}$.

This leaves the principal with a payoff of

$$\begin{aligned}\mathbb{E}_{F'}[y - w_2(y)] &= \lambda \mathbb{E}_{F_0}[y - w_2(y)] + (1 - \lambda)(0 - w_2(0)) \\ &\leq \lambda \mathbb{E}_{F_0}[y - w_2(y)] = \frac{(\mathbb{E}_{F_0}[w_2(y)] - c_0) - (\mathbb{E}_{F_1}[w_1(y)] - c_1)}{\mathbb{E}_{F_0}[w_2(y)] - \mathbb{E}_{F_0}[w_1(y)]} (\mathbb{E}_{F_0}[y] - \mathbb{E}_{F_0}[w_2(y)]) \\ &\leq \left(\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g(a_0|w_1, a_1)} \right)^2,\end{aligned}\tag{B.11}$$

which is no more than $\Phi(w_1, a_1)^2$, as desired. The last inequality (B.11),

$$\begin{aligned}&\frac{(\mathbb{E}_{F_0}[w_2(y)] - c_0) - (\mathbb{E}_{F_1}[w_1(y)] - c_1)}{\mathbb{E}_{F_0}[w_2(y)] - \mathbb{E}_{F_0}[w_1(y)]} (\mathbb{E}_{F_0}[y] - \mathbb{E}_{F_0}[w_2(y)]) \leq \left(\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g(a_0|w_1, a_1)} \right)^2 \\ \Leftrightarrow &\frac{(\mathbb{E}_{F_0}[w_2(y)] - \mathbb{E}_{F_0}[w_1(y)] - \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} \cdot \sqrt{g(a_0|w_1, a_1)})^2}{\mathbb{E}_{F_0}[w_2(y)] - \mathbb{E}_{F_0}[w_1(y)]} \geq 0,\end{aligned}$$

which always holds. (Recall that $g(a_0|w_1, a_1) = (\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_0}[w_1(y)] - c_0) \geq 0$.)

Summing up the above three cases, we prove that the principal's payoff guarantee is not strictly higher than $(\max\{\Theta(w_1, a_1), \Phi(a_1)\})^2$ under any second-period contract w_2 .

This completes the proof. $\square$

B.3.1 Proofs for Subsection B.2

To prove Lemma A.4', we start by establishing the following Lemma B.3.

Lemma B.3. *Suppose the known technology A_0 satisfies lower bound on marginal cost. If $\Theta(w_1, a_1) \geq \Phi(a_1)$ and $a^* = (F^*, c^*) \in A_0$ attains the maximum in equation (B.1), then (i) $c^* \leq c_0$, (ii) $\mathbb{E}_{F^*}[y] \leq \mathbb{E}_{F_0}[y]$, and (iii) $\mathbb{E}_{F^*}[w_1(y)] \leq \mathbb{E}_{F^*}[\hat{w}_1(y)] = s_1 \mathbb{E}_{F^*}[y]$, where $\hat{w}_1$ is defined by equation (3).*

Proof of Lemma B.3. Let $g_0 \equiv g(a_0|w_1, a_1) = (\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_0}[w_1(y)] - c_0) \geq 0$, and $g^* \equiv g(a^*|w_1, a_1) = (\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F^*}[w_1(y)] - c^*) \geq 0$. By assumption, we have

$$\mathbb{E}_{F_0}[w_1(y)] - c_0 \geq \mathbb{E}_{F^*}[w_1(y)] - c^* \quad \Rightarrow \quad g^* \geq g_0.$$

Note that

$$\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g^*} = \Theta(w_1, a_1) \geq \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} - \sqrt{g_0}.\tag{B.12}$$

We first argue that $c^* \leq c_0$ must hold, otherwise there will be a contradiction to the assumption that A_0 satisfies lower bound on marginal cost.

Suppose not, i.e., $c^* > c_0$. Consider the following two cases.

Case 1. $\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} \geq \sqrt{g_0}$.

From equation (B.12) we obtain

$$\begin{aligned} \frac{(\mathbb{E}_{F^*}[y - w_1(y)]) - (\mathbb{E}_{F_0}[y - w_1(y)])}{\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} + \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]}} &= \frac{\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]}}{\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} + \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]}} \\ &\geq \sqrt{g^*} - \sqrt{g_0} = \frac{(\mathbb{E}_{F_0}[w_1(y)] - c_0) - (\mathbb{E}_{F^*}[w_1(y)] - c^*)}{\sqrt{g^*} + \sqrt{g_0}}. \end{aligned}$$

Since $\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} > \sqrt{g^*}$ and $\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} \geq \sqrt{g_0}$, the above expression implies that

$$\begin{aligned} &(\mathbb{E}_{F^*}[y - w_1(y)]) - (\mathbb{E}_{F_0}[y - w_1(y)]) > (\mathbb{E}_{F_0}[w_1(y)] - c_0) - (\mathbb{E}_{F^*}[w_1(y)] - c^*) \\ \Rightarrow &\mathbb{E}_{F^*}[y] - \mathbb{E}_{F_0}[y] > c^* - c_0 > 0, \end{aligned}$$

a contradiction to the assumption that A_0 satisfies lower bound on marginal cost!

Case 2. $\sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} < \sqrt{g_0}$.

We have

$$\mathbb{E}_{F_0}[y - w_1(y)] < g_0 = (\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_0}[w_1(y)] - c_0) \Rightarrow \mathbb{E}_{F_0}[y] - c_0 < \mathbb{E}_{F_1}[w_1(y)] - c_1.$$

Similarly, from $\Theta(w_1, a_1) \geq \Phi(a_1) > 0$, we have $\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g^*} > 0$, and thus

$$\mathbb{E}_{F^*}[y - w_1(y)] > g^* = (\mathbb{E}_{F_1}[w_1(y)] - c_1) - (\mathbb{E}_{F^*}[w_1(y)] - c^*) \Rightarrow \mathbb{E}_{F^*}[y] - c^* > \mathbb{E}_{F_1}[w_1(y)] - c_1.$$

It follows that

$$\mathbb{E}_{F^*}[y] - c^* > \mathbb{E}_{F_0}[y] - c_0 \Rightarrow \mathbb{E}_{F^*}[y] - \mathbb{E}_{F_0}[y] > c^* - c_0 > 0,$$

another contradiction to the assumption that A_0 satisfies lower bound on marginal cost!

Summing up the above two cases, we show that $c^* \leq c_0$. It follows from lower bound on marginal cost that $\mathbb{E}_{F^*}[y] \leq \mathbb{E}_{F_0}[y]$.

Moreover, $\mathbb{E}_{F_0}[w_1(y)] - c_0 \geq \mathbb{E}_{F^*}[w_1(y)] - c^*$ implies that

$$\mathbb{E}_{F_0}[w_1(y)] - \mathbb{E}_{F^*}[w_1(y)] \geq c_0 - c^* \geq 0 \Rightarrow \mathbb{E}_{F_0}[w_1(y)] \geq \mathbb{E}_{F^*}[w_1(y)].$$

Equation (B.12) implies that

$$\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{\mathbb{E}_{F_0}[y - w_1(y)]} \geq \sqrt{g^*} - \sqrt{g_0} \geq 0 \Rightarrow \mathbb{E}_{F^*}[y - w_1(y)] \geq \mathbb{E}_{F_0}[y - w_1(y)].$$

Combining the above two inequalities, we have

$$\begin{aligned} &\frac{\mathbb{E}_{F^*}[y - w_1(y)]}{\mathbb{E}_{F^*}[w_1(y)]} \geq \frac{\mathbb{E}_{F_0}[y - w_1(y)]}{\mathbb{E}_{F_0}[w_1(y)]} \\ \Rightarrow &\frac{\mathbb{E}_{F^*}[y]}{\mathbb{E}_{F^*}[w_1(y)]} \geq \frac{\mathbb{E}_{F_0}[y]}{\mathbb{E}_{F_0}[w_1(y)]} = \frac{1}{s_1} \\ \Rightarrow &\mathbb{E}_{F^*}[w_1(y)] \leq s_1 \mathbb{E}_{F^*}[y], \end{aligned} \tag{B.13}$$

as desired. The equality in (B.13) follows from the definition in (3). $\square$

Proof of Lemma A.4'. Consider an arbitrary action $a_1 = (F_1, c_1)$ agent 1 would take under contract $\hat{w}_1$. We

need to show that the principal's interim payoff guarantee, $U(\hat{w}_1|a_1)$, is at least $U(w_1)$. Lemma 3' shows that the principal's optimal second-period payoff guarantee is

$$\hat{V}_2^*(\hat{w}_1, a_1) = (\max\{\Theta(\hat{w}_1, a_1), \Phi(a_1)\})^2,$$

where

$$\begin{aligned}\Theta(\hat{w}_1, a_1) &= \max_{a \in A_0 \cup \{a_1\}} \left\{ \sqrt{\mathbb{E}_{F_a}[y - \hat{w}_1(y)]} - \sqrt{g(a|\hat{w}_1, a_1)} \right\}, \\ \Phi(a_1) &= \max_{a \in A_0 \cup \{a_1\}} \left\{ \sqrt{\mathbb{E}_{F_a}[y]} - \sqrt{c_a} \right\},\end{aligned}$$

and her interim payoff guarantee is

$$\hat{U}(\hat{w}_1|a_1) = \mathbb{E}_{F_1}[y - \hat{w}_1(y)] + \beta \cdot \hat{V}_2^*(\hat{w}_1, a_1).$$

It suffices to construct another action a'_1 , which may be taken by agent 1 under w_1 and some other technology, such that $\hat{U}(w_1|a'_1) \leq \hat{U}(\hat{w}_1|a_1)$. By assumption, a_0 is agent 1's best response if $A = A_0$, so an action a'_1 may be taken by agent 1 under w_1 if and only if the incentive gap with respect to a_0 is nonnegative, i.e., $g(a_0|w_1, a'_1) \geq 0$. Consider the following two cases.

Case 1. $\mathbb{E}_{F_1}[y] \geq \mathbb{E}_{F_0}[y]$.

Let $a'_1 = a_0$. When agent 1 takes action a_0 in response to w_1 , the principal's resulting payoff in the first period is

$$\mathbb{E}_{F_0}[y - w_1(y)] = (1 - s_1) \mathbb{E}_{F_0}[y] \leq (1 - s_1) \mathbb{E}_{F_1}[y] = \mathbb{E}_{F_1}[y - \hat{w}_1(y)],$$

so her payoff in the first period under $(w_1|a_0)$ is weakly lower than under $(\hat{w}_1|a_1)$.

Moreover, it follows from Lemma 3' that the principal's optimal second-period payoff guarantee is

$$\hat{V}_2^*(w_1, a_0) = (\max\{\Theta(w_1, a_0), \Phi(a_0)\})^2.$$

We now show that $\hat{V}_2^*(w_1, a_0) \leq \hat{V}_2^*(\hat{w}_1, a_1)$, which is equivalent to

$$\max\{\Theta(w_1, a_0), \Phi(a_0)\} \leq \max\{\Theta(\hat{w}_1, a_1), \Phi(a_1)\}.$$

Note that

$$\begin{aligned}\Theta(w_1, a_0) &= \max_{a \in A_0} \left\{ \sqrt{\mathbb{E}_{F_a}[y - w_1(y)]} - \sqrt{g(a|w_1, a_0)} \right\}, \\ \Phi(a_0) &= \max_{a \in A_0} \left\{ \sqrt{\mathbb{E}_{F_a}[y]} - \sqrt{c_a} \right\}.\end{aligned}$$

By definition we have $0 < \Phi(a_0) \leq \Phi(a_1)$. Thus, it suffices to show that whenever $\Theta(w_1, a_0) > \Phi(a_0)$, it holds that $\Theta(w_1, a_0) \leq \Theta(\hat{w}_1, a_1)$.

Let $a^* = (F^*, c^*) \in A_0$ attains the maximum in $\Theta(w_1, a_0)$. It follows from Lemma B.3 that $\mathbb{E}_{F^*}[y] \leq \mathbb{E}_{F_0}[y] \leq \mathbb{E}_{F_1}[y]$ and $\mathbb{E}_{F^*}[w_1(y)] \leq s_1 \mathbb{E}_{F^*}[y]$.

We claim that

$$\Theta(w_1, a_0) = \sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g(a^*|w_1, a_0)} \leq \sqrt{\mathbb{E}_{F_1}[y - \hat{w}_1(y)]} \leq \Theta(\hat{w}_1, a_1).$$

must hold. Suppose not, then

$$\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g(a^*|w_1, a_0)} > \sqrt{\mathbb{E}_{F_1}[y - \hat{w}_1(y)]},$$

which implies that

$$\sqrt{(1 - s_1) \mathbb{E}_{F^*}[y]} \geq \sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g(a^*|w_1, a_0)} > \sqrt{\mathbb{E}_{F_1}[y - \hat{w}_1(y)]} = \sqrt{(1 - s_1) \mathbb{E}_{F_1}[y]},$$

a contradiction to $\mathbb{E}_{F^*}[y] \leq \mathbb{E}_{F_1}[y]$!

Therefore, whenever $\Theta(w_1, a_0) > \Phi(a_0)$, it holds that $\Theta(w_1, a_0) \leq \Theta(\hat{w}_1, a_1)$, which implies $\hat{V}_2^*(w_1, a_0) \leq \hat{V}_2^*(\hat{w}_1, a_1)$. The principal's interim payoff guarantee is

$$\begin{aligned} \hat{U}(w_1|a_0) &= \mathbb{E}_{F_0}[y - w_1(y)] + \beta \cdot \hat{V}_2^*(w_1, a_0) \\ &\leq \mathbb{E}_{F_1}[y - \hat{w}_1(y)] + \beta \cdot \hat{V}_2^*(\hat{w}_1, a_1) = \hat{U}(\hat{w}_1|a_1), \end{aligned}$$

as desired.

Case 2. $\mathbb{E}_{F_1}[y] < \mathbb{E}_{F_0}[y]$.

Let $\lambda = \mathbb{E}_{F_1}[y]/\mathbb{E}_{F_0}[y] \in [0, 1]$ and let F'_1 be the mixture $\lambda F_0 + (1 - \lambda) \delta_0$. Note that $\mathbb{E}_{F'_1}[y] = \mathbb{E}_{F_1}[y]$. Consider $a'_1 = (F'_1, c_1)$. For any action a , the corresponding incentive gap with respect to a is

$$g(a|w_1, a'_1) = (\mathbb{E}_{F'_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_a}[w_1(y)] - c_a).$$

Note that

$$\mathbb{E}_{F'_1}[w_1(y)] - c_1 = \lambda \mathbb{E}_{F_0}[w_1(y)] - c_1 = \lambda s_1 \mathbb{E}_{F_0}[y] - c_1 = s_1 \mathbb{E}_{F_1}[y] - c_1 = \mathbb{E}_{F_1}[\hat{w}_1(y)] - c_1,$$

and

$$\mathbb{E}_{F_0}[w_1(y)] - c_0 = s_1 \mathbb{E}_{F_0}[y] - c_0 = \mathbb{E}_{F_0}[\hat{w}_1(y)] - c_0.$$

Thus,

$$\begin{aligned} g(a_0|w_1, a'_1) &= (\mathbb{E}_{F'_1}[w_1(y)] - c_1) - (\mathbb{E}_{F_0}[w_1(y)] - c_0) \\ &= (\mathbb{E}_{F_1}[\hat{w}_1(y)] - c_1) - (\mathbb{E}_{F_0}[\hat{w}_1(y)] - c_0) \\ &= g(a_0|\hat{w}_1, a_1) \geq 0, \end{aligned}$$

implying that a'_1 may be chosen by agent 1 in response to w_1 under some technology.

When agent 1 chooses action a'_1 in response, the principal's resulting payoff in the first period is

$$\mathbb{E}_{F'_1}[y - w_1(y)] = \lambda \mathbb{E}_{F_0}[y - w_1(y)] = \lambda (1 - s_1) \mathbb{E}_{F_0}[y] = (1 - s_1) \mathbb{E}_{F_1}[y] = \mathbb{E}_{F_1}[y - \hat{w}_1(y)],$$

so her payoff in the first period under $(w_1|a'_1)$ and under $(\hat{w}_1|a_1)$ are exactly equal.

Moreover, it follows from Lemma 3' that the principal's optimal second-period payoff guarantee under $(w_1|a'_1)$ is

$$\hat{V}_2^*(w_1, a'_1) = \left(\max \{ \Theta(w_1, a'_1), \Phi(a'_1) \} \right)^2.$$

We now show that $\hat{V}_2^*(w_1, a'_1) \leq \hat{V}_2^*(\hat{w}_1, a'_1)$, which is equivalent to

$$\max \left\{ \Theta(w_1, a'_1), \Phi(a'_1) \right\} \leq \max \left\{ \Theta(\hat{w}_1, a_1), \Phi(a_1) \right\}.$$

Note that

$$\begin{aligned} \Theta(w_1, a'_1) &= \max_{a \in A_0 \cup \{a'_1\}} \left\{ \sqrt{\mathbb{E}_{F_a}[y - w_1(y)]} - \sqrt{g(a|w_1, a'_1)} \right\}, \\ \Phi(a'_1) &= \max_{a \in A_0 \cup \{a'_1\}} \left\{ \sqrt{\mathbb{E}_{F_a}[y]} - \sqrt{c_a} \right\}. \end{aligned}$$

From $\mathbb{E}_{F'_1}[y] = \mathbb{E}_{F_1}[y]$, it follows that $\Phi(a'_1) = \Phi(a_1) > 0$. Thus, it suffices to show that whenever $\Theta(w_1, a'_1) > \Phi(a'_1)$, it holds that $\Theta(w_1, a'_1) \leq \Theta(\hat{w}_1, a_1)$.

Let $a^* = (F^*, c^*) \in A_0 \cup \{a'_1\}$ attains the maximum in $\Theta(w_1, a'_1)$.

1. If $a^* = a'_1$, then

$$\begin{aligned} \Theta(w_1, a'_1) &= \sqrt{\mathbb{E}_{F'_1}[y - w_1(y)]} - \sqrt{g(a'_1|w_1, a'_1)} \\ &= \sqrt{\mathbb{E}_{F_1}[y - \hat{w}_1(y)]} - \sqrt{g(a_1|\hat{w}_1, a_1)} \leq \Theta(\hat{w}_1, a_1), \end{aligned}$$

as desired.

2. If $a^* \in A_0$, then it follows from Lemma B.3 that $\mathbb{E}_{F^*}[w_1(y)] \leq \mathbb{E}_{F^*}[\hat{w}_1(y)]$.

From $\Theta(w_1, a'_1) > \Phi(a'_1) > 0$, we have $\Theta(w_1, a'_1) = \sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g(a^*|w_1, a'_1)} > 0$, and thus

$$\begin{aligned} \mathbb{E}_{F^*}[y - w_1(y)] &> g(a^*|w_1, a'_1) = (\mathbb{E}_{F'_1}[w_1(y)] - c_1) - (\mathbb{E}_{F^*}[w_1(y)] - c^*) \\ \Rightarrow \mathbb{E}_{F^*}[y] - c^* &> \mathbb{E}_{F'_1}[w_1(y)] - c_1 = \mathbb{E}_{F_1}[\hat{w}_1(y)] - c_1 \\ \Rightarrow \mathbb{E}_{F^*}[y - \hat{w}_1(y)] &> (\mathbb{E}_{F_1}[\hat{w}_1(y)] - c_1) - (\mathbb{E}_{F^*}[\hat{w}_1(y)] - c^*) = g(a^*|\hat{w}_1, a_1). \end{aligned}$$

We claim that

$$\Theta(w_1, a'_1) = \sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g(a^*|w_1, a'_1)} \leq \sqrt{\mathbb{E}_{F^*}[y - \hat{w}_1(y)]} - \sqrt{g(a^*|\hat{w}_1, a_1)} \leq \Theta(\hat{w}_1, a_1).$$

must hold. Suppose not, then

$$\begin{aligned} &\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{g(a^*|w_1, a'_1)} \leq \sqrt{\mathbb{E}_{F^*}[y - \hat{w}_1(y)]} - \sqrt{g(a^*|\hat{w}_1, a_1)} \\ \Leftrightarrow &\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} - \sqrt{\mathbb{E}_{F^*}[y - \hat{w}_1(y)]} \leq \sqrt{g(a^*|w_1, a'_1)} - \sqrt{g(a^*|\hat{w}_1, a_1)} \\ \Leftrightarrow &\frac{\mathbb{E}_{F^*}[y - w_1(y)] - \mathbb{E}_{F^*}[y - \hat{w}_1(y)]}{\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} + \sqrt{\mathbb{E}_{F^*}[y - \hat{w}_1(y)]}} \leq \frac{g(a^*|w_1, a'_1) - g(a^*|\hat{w}_1, a_1)}{\sqrt{g(a^*|w_1, a'_1)} + \sqrt{g(a^*|\hat{w}_1, a_1)}}. \end{aligned} \tag{B.14}$$

Note that

$$\mathbb{E}_{F^*}[y - w_1(y)] - \mathbb{E}_{F^*}[y - \hat{w}_1(y)] = \mathbb{E}_{F^*}[\hat{w}_1(y)] - \mathbb{E}_{F^*}[w_1(y)] \geq 0,$$

and that

$$\begin{aligned}
& g(a^*|w_1, a'_1) - g(a^*|\hat{w}_1, a_1) \\
&= ((\mathbb{E}_{F'_1}[w_1(y)] - c_1) - (\mathbb{E}_{F^*}[w_1(y)] - c^*)) - ((\mathbb{E}_{F_1}[\hat{w}_1(y)] - c_1) - (\mathbb{E}_{F^*}[\hat{w}_1(y)] - c^*)) \\
&= \mathbb{E}_{F^*}[\hat{w}_1(y)] - \mathbb{E}_{F^*}[w_1(y)] \geq 0.
\end{aligned}$$

Therefore, inequality (B.14) is equivalent to

$$\frac{\mathbb{E}_{F^*}[\hat{w}_1(y)] - \mathbb{E}_{F^*}[w_1(y)]}{\sqrt{\mathbb{E}_{F^*}[y - w_1(y)]} + \sqrt{\mathbb{E}_{F^*}[y - \hat{w}_1(y)]}} \leq \frac{\mathbb{E}_{F^*}[\hat{w}_1(y)] - \mathbb{E}_{F^*}[w_1(y)]}{\sqrt{g(a^*|w_1, a'_1)} + \sqrt{g(a^*|\hat{w}_1, a_1)}},$$

which is implied by $\mathbb{E}_{F^*}[y - w_1(y)] > g(a^*|w_1, a'_1)$ and $\mathbb{E}_{F^*}[y - \hat{w}_1(y)] > g(a^*|\hat{w}_1, a_1)$.

Therefore, whenever $\Theta(w_1, a'_1) > \Phi(a'_1)$, it holds that $\Theta(w_1, a'_1) \leq \Theta(\hat{w}_1, a'_1)$, which implies $\hat{V}_2^*(w_1, a'_1) \leq \hat{V}_2^*(\hat{w}_1, a_1)$. The principal's interim payoff guarantee is

$$\begin{aligned}
\hat{U}(w_1|a'_1) &= \mathbb{E}_{F'_1}[y - w_1(y)] + \beta \cdot \hat{V}_2^*(w_1, a'_1) \\
&\leq \mathbb{E}_{F_1}[y - \hat{w}_1(y)] + \beta \cdot \hat{V}_2^*(\hat{w}_1, a_1) = \hat{U}(\hat{w}_1|a_1),
\end{aligned}$$

as desired.

This completes the proof. $\square$

Proof of Lemma A.5'. We first reformulate program (B.5) as an equivalent maximization problem with continuous objective function and compact feasible region. Slightly abusing notation, we use $\hat{U}(s_1)$ instead of $\hat{U}(w_1)$ to denote the infimum value of program (B.5).

Plug $w_1(y) = s_1 y$ into equation (B.1). We may rewrite $\Theta(w_1, a_1)$ as

$$\Theta(w_1, a_1) = \max_{a \in A_0 \cup \{a_1\}} \left\{ \sqrt{(1 - s_1) \mathbb{E}_{F_a}[y]} - \sqrt{g(a|w_1, a_1)} \right\}.$$

Similarly, for $a \in A_0 \cup \{a_1\}$,

$$g(a|w_1, a_1) = (s_1 \mathbb{E}_{F_1}[y] - c_1) - (s_1 \mathbb{E}_{F_a}[y] - c_a) \geq 0.$$

Note that both the objective and the constraints of program (B.5) depend on the choice variables (F_1, c_1) only through the value of $(\mathbb{E}_{F_1}[y], c_1)$. Rewrite $\mathbb{E}_{F_1}[y] = x$ and $c_1 = z$ with $x, z \geq 0$. Plugging into the original program (B.5), we obtain an equivalent program

$$\begin{aligned}
\hat{U}(s_1) &= \inf_{x, z} (1 - s_1)x + \beta \cdot \max \{ \theta(x, z; s_1), \phi(x, z) \}^2 \\
\text{s.t.} \quad & s_1 x - z \geq \max_{a \in A_0 \cup \{(\delta_0, 0)\}} \{ s_1 \mathbb{E}_{F_a}[y] - c_a \}, \quad x, z \geq 0,
\end{aligned} \tag{B.15}$$

where

$$\theta(x, z; s_1) \equiv \max \left\{ \sqrt{(1 - s_1)x}, \max_{a \in A_0} \left\{ \sqrt{(1 - s_1) \mathbb{E}_{F_a}[y]} - \sqrt{(s_1 x - z) - (s_1 \mathbb{E}_{F_a}[y] - c_a)} \right\} \right\}, \tag{B.16}$$

and ϕ is defined by equation (A.2).

Let $\bar{x} \equiv \max_{a \in A_0} \mathbb{E}_{F_a}[y] > 0$, and $\bar{v} \equiv \max_{a \in A_0} \{ \sqrt{\mathbb{E}_{F_a}[y]} - \sqrt{c_a} \} > 0$. Suppose

$$(F_0, c_0) \in \arg \max_{a \in A_0 \cup \{(\delta_0, 0)\}} \{s_1 \mathbb{E}_{F_a}[y] - c_a\}.$$

Note that $(x_0, z_0) = (\mathbb{E}_{F_0}[y], c_0)$ is feasible in program (B.15) and leads to objective value

$$(1 - s_1)x_0 + \beta \cdot \max \{ \theta(x_0, z_0; s_1), \phi(x_0, z_0) \}^2 \leq (1 - s_1)\bar{x} + \beta \cdot \max \{ \sqrt{(1 - s_1)\bar{x}}, \bar{v} \}^2.$$

If $x \geq (1 + \beta)\bar{x}$, then

$$\begin{aligned} (1 - s_1)x + \beta \cdot \max \{ \theta(x, z; s_1), \phi(x, z) \}^2 &\geq (1 - s_1)(1 + \beta)\bar{x} + \beta \cdot \bar{v}^2 \\ &= (1 - s_1)\bar{x} + \beta(1 - s_1)\bar{x} + \beta \cdot \bar{v}^2 \\ &\geq (1 - s_1) + \beta \cdot \max \{ \sqrt{(1 - s_1)\bar{x}}, \bar{v} \}^2. \end{aligned}$$

Therefore, restricting $x \in [0, (1 + \beta)\bar{x}]$ will not change the infimum of program (B.15). Moreover,

$$s_1x - z \geq 0 \quad \Rightarrow \quad z \leq s_1x \leq x,$$

so restricting $(x, z) \in [0, (1 + \beta)\bar{x}]^2$ will not change the infimum of program (B.15).

Consider the following program

$$\begin{aligned} \hat{\Psi}^*(s_1) &\equiv \sup_{x, z} \hat{\Psi}(x, z; s_1) \equiv - \left((1 - s_1)x + \beta \cdot \max \{ \theta(x, z; s_1), \phi(x, z) \}^2 \right) \\ \text{s.t.} \quad &(x, z) \in \hat{\Gamma}(s_1), \end{aligned} \tag{B.17}$$

where θ is defined by equation (B.16), ϕ is defined by equation (A.2), and $\hat{\Gamma}$ is defined as follows:

$$\hat{\Gamma}(s_1) \equiv \left\{ (x, z) \in [0, (1 + \beta)\bar{x}]^2 : s_1x - z \geq \max_{a \in A_0 \cup \{(\delta_0, 0)\}} \{s_1 \mathbb{E}_{F_a}[y] - c_a\} \right\}.$$

By definition, $\hat{\Psi} : [0, (1 + \beta)\bar{x}]^2 \times [0, 1] \rightarrow \mathbb{R}$ is a continuous function, and $\hat{\Gamma} : [0, 1] \rightrightarrows [0, (1 + \beta)\bar{x}]^2$ is a compact-valued and nonempty-valued correspondence. Moreover, the infimum of program (B.15), $\hat{U}(s_1)$, is given by $-\hat{\Psi}^*(s_1)$.

Note that for each s_1 , $\hat{\Gamma}(s_1)$ defines a half plane intersecting a square, and that the half plane shifts linearly in s_1 . Thus, $\hat{\Gamma}$ is both upper and lower hemicontinuous. It then follows from Berge's maximum theorem that $\hat{\Psi}^*$ is continuous, and

$$\hat{\Gamma}^*(s_1) \equiv \left\{ (x, z) \in \hat{\Gamma}(s_1) : \hat{\Psi}(x, z; s_1) = \hat{\Psi}^*(s_1) \right\}$$

is upper hemicontinuous with nonempty and compact values. As a consequence, a solution to program (B.17) exists for all s_1 , and the supremum can be replaced by maximum.

It follows that the infimum in program (B.15) and therefore the original program (B.5) can both be replaced by minimum, and the resulting minimum value $\hat{U}(s_1) = -\hat{\Psi}^*(s_1)$ is continuous in s_1 . Hence, $\hat{U}(s_1)$ achieves a maximum over $[0, 1]$. This maximum is also the optimal guarantee over all linear contracts. $\square$

Proof of Theorem 2'. According to Lemma A.5', among all linear first-period contracts, there exists an optimal one, call it w_1^* . If w_1 is any other (nonlinear) first-period contract that outperforms w_1^* , then by Lemma A.4', there is a linear contract that in turn does at least as well as w_1 . But this contradicts the fact that w_1^* is an optimal linear contract. Therefore, w_1^* is optimal among all first-period contracts. $\square$

Appendix C Optimal First-period Contract

In this appendix, we examine the structure of the optimal linear first-period contract in our dynamic model, and compare it with the optimal static contract identified by [Carroll \(2015\)](#). This requires an exact calculation of the overall payoff guarantee from an arbitrary linear first-period contract, which becomes complicated when the principal knows a general set A_0 of available actions. In particular, in response to a linear first-period contract $w(y) = s_1 y$, the optimal payoff that agent 1 can obtain from *known* actions, $\max_{a \in A_0} \{s_1 \mathbb{E}_{F_a}[y] - c_a\}$, changes with respect to s_1 in an intractable manner. This payoff, however, is a key component of the constraint in the programs that characterize the principal's overall payoff guarantee. For this reason, we focus on the case where the principal knows only one action $a_0 = (F_0, c_0)$ available.

We demonstrate that the principal's second-period payoff guarantee takes a simpler form in the case of advancing technology (equation (2)). It turns out that the principal's overall payoff guarantee is also easier to characterize in this situation. In the proof of Theorem 1, we set up a program (5) that characterizes the principal's overall payoff guarantee from any linear first-period contract. We explicitly solve the program (5) for any first-period share s_1 , and the resulting overall payoff guarantee U is depicted in Figure 2. From this calculation, we can show that the optimal first-period share s_1^* exists and is unique. Moreover, in Figure 2, the optimal first-period share is greater than $s_0 \equiv \sqrt{c_0/\mathbb{E}_{F_0}[y]}$, the optimal static share in [Carroll \(2015\)](#).

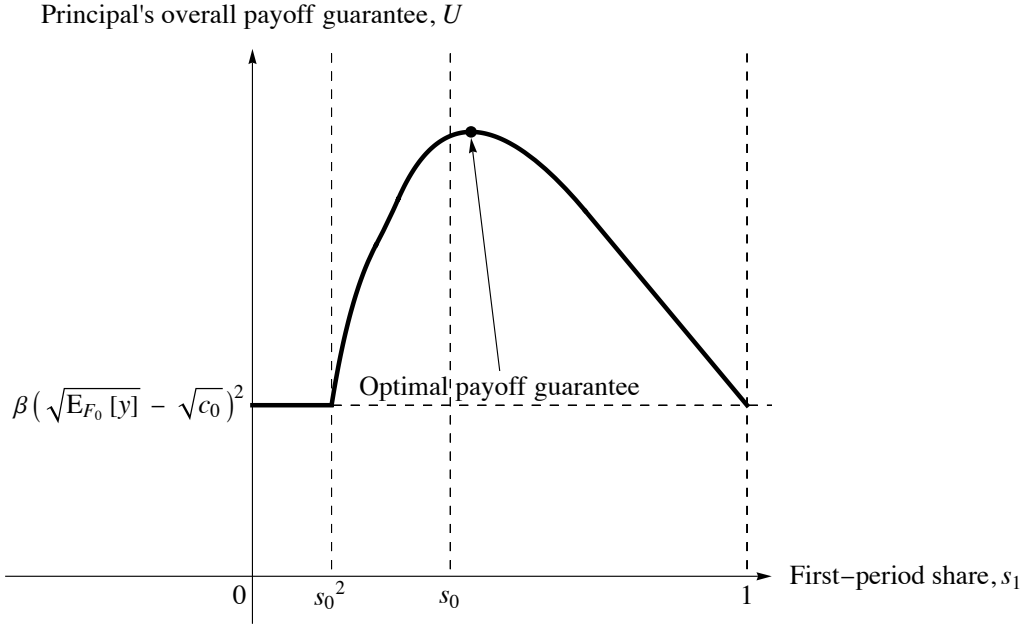


Figure 2: Overall payoff guarantee in the case of advancing technology ($s_0 = 0.4$, $\beta = 0.8$).

Proposition C.1 formally establishes this observation and exactly characterizes the optimal first-period share. It reveals an *exploration effect* where the optimal first-period share offered to agent 1 is always larger than the optimal static share s_0 . Moreover, the exploration effect increases as the principal becomes more patient (β increases), provided that $\beta < 1$. When $\beta > 1$, it starts to decrease, and vanishes as $\beta \rightarrow \infty$.

Proposition C.1. *Suppose the principal knows only one available action $a_0 = (F_0, c_0)$, and let $s_0 \equiv \sqrt{c_0/\mathbb{E}_{F_0}[y]}$ denote the optimal static share. In the case of advancing technology, the optimal first-period share s_1^* is unique, and satisfies the following properties:*

1. For all $\beta \in (0, \infty)$, the optimal first-period share is larger than the optimal static share, i.e., $s_1^* > s_0$.
2. In both limiting cases $\beta \rightarrow 0$ and $\beta \rightarrow \infty$, s_1^* approaches s_0 .

3. s_1^* is strictly increasing in β if $\beta < 1$, and is strictly decreasing if $\beta > 1$.

Proof. All proofs for this Appendix are in Appendix C.1. □

The pattern identified by Proposition C.1 is illustrated in Figure 3. It is straightforward to understand

Optimal first-period share, s_1^*

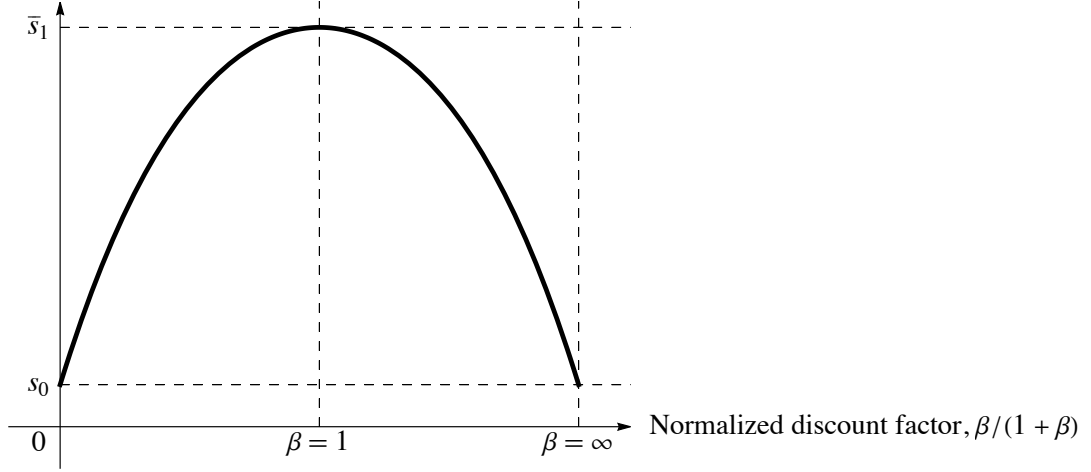


Figure 3: The optimal first-period share s_1^* in the case of advancing technology ($s_0 = 0.4$).

the result that the dynamic model converges to the static model as the discount factor β approaches 0. To get intuition behind the opposite case, that is, when β approaches infinity, the optimal first-period share s_1^* approaches the optimal static share s_0 again, note that unlike in standard models where patience automatically leads to the option value of exploration, here the principal is concerned with the worst-case discovery. In the limiting case $\beta \rightarrow \infty$ where only the second period matters, there is no incentive for her to raise the first-period share s_1 from s_0 , precisely because the worst-case technology always leaves the principal without any valuable discovery. The principal is thus essentially indifferent among any first-period contract in this limiting case, making the opportunity to explore in the first period completely useless to her.

In the case of constant technology, the principal adopts a more complex rule of updating (i.e., compatibility). Under all possible parameters choices, we aim to compute the exact solution to the analogous program (A.11), which characterizes the overall payoff guarantee of any linear first-period contract $w_1(y) = s_1 y$. Current results show that, for a range of parameter values (specifically, β not too large), the resulting worst-case payoff guarantee $\hat{U}$ is a bell-shaped curve as depicted in Figure 4. From this figure, the optimal first-period share appears to be unique, and smaller than the optimal static share s_0 .

Now we explain why the principal chooses to lower the share offered to agent 1 compared to the optimal static share in Carroll (2015). Note that this result is different from the previous case of advancing technology due to the distinct rule of updating, thus resulting in a different optimal second-period payoff guarantee (equation (6)). Within the parameter values we tried, the true worst-case technology A is such that, after offering first-period contract w_1 and observing agent 1's selected action a_1 , the principal optimally selects the second response among the four candidates of optimal second-period contracts, namely, a modified w_1 with compensation to agent 2. Based on this observation, it won't be worst-case optimal for the principal to offer a strictly higher share compared to the optimal static share in the first period, in anticipation of an even higher share in the subsequent period. Instead, the principal benefits from reducing the share in the first period to hedge against the risk of increasing the share in the second period.

We hope to finish the subsequent calculations to formally confirm this observation, in order to better understand the exploration effect in the case of constant technology. In particular, we are interested in

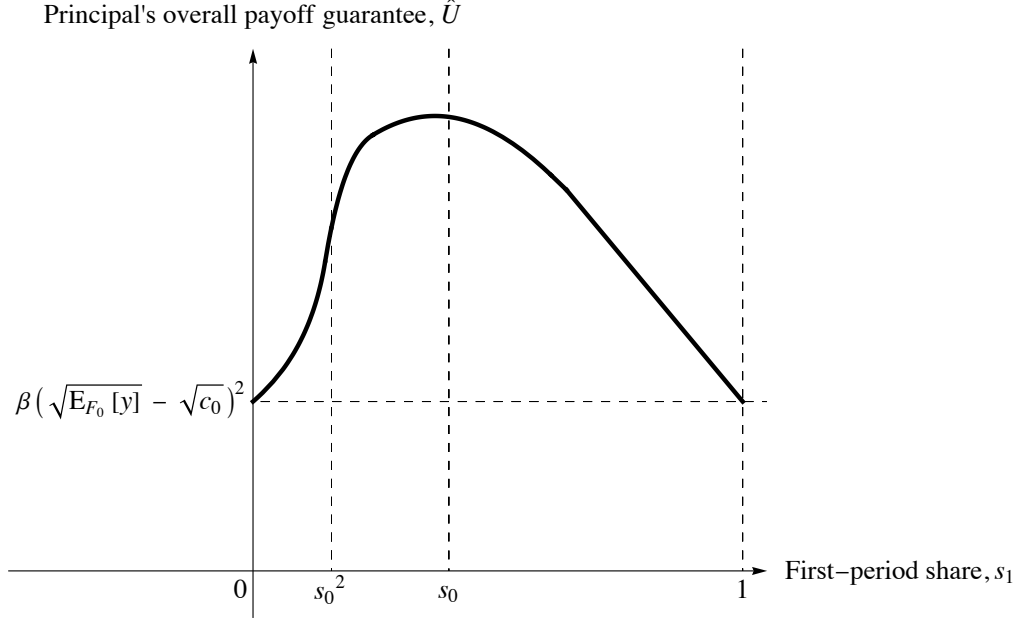


Figure 4: Overall payoff guarantee in the case of constant technology ($s_0 = 0.4, \beta = 0.8$).

whether the optimal first-period share s_1^* approaches the optimal static share s_0 again as the discount factor β approaches infinity.

C.1 Proofs for Appendix C

Proof of Proposition C.1. Suppose that the principal knows only one available action $a_0 = (F_0, c_0)$, and let $s_0 \equiv \sqrt{c_0/\mathbb{E}_{F_0}[y]}$ denote the optimal static share. We first solve program (5) for any first-period share s_1 . Program (5) simplifies to (write $c_0 = s_0^2 \mathbb{E}_{F_0}[y]$):

$$\begin{aligned} \inf_{F_1, c_1} \quad & (1 - s_1) \mathbb{E}_{F_1}[y] + \beta \cdot \max \left\{ \sqrt{\mathbb{E}_{F_1}[y]} - \sqrt{c_1}, (1 - s_0) \sqrt{\mathbb{E}_{F_0}[y]} \right\}^2 \\ \text{s.t.} \quad & (s_1 \mathbb{E}_{F_1}[y] - c_1) \geq \max \left\{ (s_1 - s_0^2) \mathbb{E}_{F_0}[y], 0 \right\}. \end{aligned} \quad (\text{C.1})$$

Note that both the objective and the constraints of program (C.1) depend on the choice variables (F_1, c_1) only through the value of $(\mathbb{E}_{F_1}[y], c_1)$. Rewrite $\mathbb{E}_{F_1}[y] = x \mathbb{E}_{F_0}[y]$ and $c_1 = z \mathbb{E}_{F_0}[y]$ with $x, z \geq 0$. Plugging into program (C.1) and cancelling out $\mathbb{E}_{F_0}[y]$ from both sides of the constraints, we obtain an equivalent program

$$\begin{aligned} \inf_{x, z} \quad & \left((1 - s_1) x + \beta \cdot \max \left\{ \sqrt{x} - \sqrt{z}, 1 - s_0 \right\}^2 \right) \mathbb{E}_{F_0}[y] \\ \text{s.t.} \quad & s_1 x - z \geq \max \left\{ s_1 - s_0^2, 0 \right\}, \quad x, z \geq 0. \end{aligned} \quad (\text{C.2})$$

If $s_1 < s_0^2$, then clearly the objective in program (C.2) attains minimum when $(x, z) = (0, 0)$, and it follows that $U(s_1) = \beta(1 - s_0)^2 \mathbb{E}_{F_0}[y]$.

Henceforth, assume $s_1 \in [s_0^2, 1]$. Moreover, the constraint in program (C.2) must bind; otherwise we could reduce x to lower the objective. Thus, we can write $z = s_1 x - (s_1 - s_0^2)$ and further simplify the

program:

$$\begin{aligned} \inf_x \quad & \left((1-s_1)x + \beta \cdot \max \left\{ \sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)}, 1 - s_0 \right\}^2 \right) \mathbb{E}_{F_0}[y] \\ \text{s.t.} \quad & x \geq 1 - s_0^2/s_1. \end{aligned} \quad (\text{C.3})$$

Let

$$\underline{s} \equiv \frac{-(2+\beta) + 6s_0 - (2-\beta)s_0^2 + (1-s_0)^{3/2} \sqrt{(2+\beta)^2 - (2-\beta)^2 s_0}}{2((2-\beta) - (1-\beta)s_0)} \quad \text{and} \quad \bar{s} \equiv \frac{(1+\beta) - \beta s_0}{\beta + (1-\beta)s_0} s_0. \quad (\text{C.4})$$

We have $\frac{s_0}{2-s_0} < \underline{s} < s_0 < \bar{s} < 1$.

$$1. \quad s_0^2 \leq s_1 \leq \frac{s_0}{2-s_0}.$$

It follows that $\sqrt{1 - s_0^2/s_1} \leq 1 - s_0$. So we have

$$(1-s_1)x + \beta \cdot \max \left\{ \sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)}, 1 - s_0 \right\}^2 \geq (1-s_1) \left(1 - \frac{s_0^2}{s_1} \right) + \beta(1-s_0)^2,$$

which is attained at $x = 1 - s_0^2/s_1$. This shows that within this range

$$U(s_1) = \left((1-s_1) \left(1 - \frac{s_0^2}{s_1} \right) + \beta(1-s_0)^2 \right) \mathbb{E}_{F_0}[y].$$

$$2. \quad \frac{s_0}{2-s_0} < s_1 \leq \underline{s}.$$

We argue that

$$(1-s_1)x + \beta \cdot \max \left\{ \sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)}, 1 - s_0 \right\}^2 \geq \frac{(1+s_1-2s_0)^2}{1-s_1} + \beta(1-s_0)^2,$$

which is attained at $x = \left(\frac{1+s_1-2s_0}{1-s_1} \right)^2$.

Note that if $x > \left(\frac{1+s_1-2s_0}{1-s_1} \right)^2$, we have

$$(1-s_1)x + \beta \cdot \max \left\{ \sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)}, 1 - s_0 \right\}^2 > \frac{(1+s_1-2s_0)^2}{1-s_1} + \beta(1-s_0)^2,$$

as desired. Thus, it suffices to focus on $x \in \left[1 - \frac{s_0^2}{s_1}, \left(\frac{1+s_1-2s_0}{1-s_1} \right)^2 \right]$, and in this range

$$\sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)} \geq 1 - s_0.$$

So the objective in program (C.3) becomes

$$\left((1-s_1)x + \beta \cdot \left(\sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)} \right)^2 \right) \mathbb{E}_{F_0}[y] \equiv h_1(x) \mathbb{E}_{F_0}[y]. \quad (\text{C.5})$$

We have

$$h_1'(x) \rightarrow -\infty \text{ as } x \rightarrow \left(1 - \frac{s_0^2}{s_1}\right)^+ \quad \text{and} \quad h_1''(x) = \frac{\beta(s_1 - s_0^2)^2}{2x^{3/2}(s_1x - (s_1 - s_0^2))^{3/2}} > 0.$$

Therefore, $h_1(x) \geq h_1\left(\left(\frac{1+s_1-2s_0}{1-s_1}\right)^2\right)$ for all $x \in \left[1 - \frac{s_0^2}{s_1}, \left(\frac{1+s_1-2s_0}{1-s_1}\right)^2\right]$ if and only if

$$h_1'\left(\left(\frac{1+s_1-2s_0}{1-s_1}\right)^2\right) \leq 0 \quad \Leftrightarrow \quad s_1 \leq \underline{s},$$

which holds by the assumption ($\underline{s}$ is defined in equation (C.4)). This shows that within this range

$$U(s_1) = \left(\frac{(1+s_1-2s_0)^2}{1-s_1} + \beta(1-s_0)^2\right) \mathbb{E}_{F_0}[y].$$

3. $\underline{s} < s_1 < \bar{s}$. We argue that

$$(1-s_1)x + \beta \cdot \max\left\{\sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)}, 1-s_0\right\}^2 \geq h_1(x^*),$$

where $h_1(\cdot)$ is defined by equation (C.5), and x^* is the unique interior minimizer of $h_1(\cdot)$.

Specifically, the closed-form expression of x^* is given by

$$x^* = \frac{(s_1 - s_0^2)\left((1+\beta) - (1-\beta)s_1 + \sqrt{((1+\beta) - (1-\beta)s_1)^2 - 4\beta^2s_1}\right)}{2s_1\sqrt{((1+\beta) - (1-\beta)s_1)^2 - 4\beta^2s_1}}. \quad (\text{C.6})$$

From the assumption that $\underline{s} < s_1 < \bar{s}$, we have $x^* < \min\left\{\left(\frac{1+s_1-2s_0}{1-s_1}\right)^2, 1\right\}$, and hence

$$\sqrt{x^*} - \sqrt{s_1x^* - (s_1 - s_0^2)} > 1 - s_0.$$

This confirms that the objective in program (C.3) becomes $h_1(x^*) \mathbb{E}_{F_0}[y]$ at $x = x^*$.

Moreover, for all $x \in \left[1 - \frac{s_0^2}{s_1}, \min\left\{\left(\frac{1+s_1-2s_0}{1-s_1}\right)^2, 1\right\}\right]$, we have $h_1(x) \geq h_1(x^*)$; for all $x > \min\left\{\left(\frac{1+s_1-2s_0}{1-s_1}\right)^2, 1\right\}$, we have

$$\begin{aligned} & (1-s_1)x + \beta \cdot \max\left\{\sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)}, 1-s_0\right\}^2 \\ & > (1-s_1) \min\left\{\left(\frac{1+s_1-2s_0}{1-s_1}\right)^2, 1\right\} + \beta(1-s_0)^2 = h_1\left(\min\left\{\left(\frac{1+s_1-2s_0}{1-s_1}\right)^2, 1\right\}\right) > h_1(x^*). \end{aligned}$$

This shows that within this range

$$U(s_1) = h_1(x^*) \mathbb{E}_{F_0}[y].$$

4. $s_1 \geq \bar{s}$.

We argue that

$$(1 - s_1)x + \beta \cdot \max \left\{ \sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)}, 1 - s_0 \right\}^2 \geq (1 - s_1) + \beta(1 - s_0)^2,$$

which is attained at $x = 1$.

Note that if $x > 1$, we have

$$(1 - s_1)x + \beta \cdot \max \left\{ \sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)}, 1 - s_0 \right\}^2 > (1 - s_1) + \beta(1 - s_0)^2,$$

as desired. Thus, it suffices to focus on $x \in \left[1 - \frac{s_0^2}{s_1}, 1\right]$. Since $s_1 \geq \bar{s} > s_0$, in this range we have

$$\sqrt{x} - \sqrt{s_1x - (s_1 - s_0^2)} \geq 1 - s_0.$$

Again, the objective in program (C.3) becomes $h_1(x) \mathbb{E}_{F_0}[y]$ with $h_1(\cdot)$ defined by equation (C.5).

Similarly, $h_1(x) \geq h_1(1)$ for all $x \in \left[1 - \frac{s_0^2}{s_1}, 1\right]$ if and only if

$$h'_1(1) \leq 0 \quad \Leftrightarrow \quad s_1 \geq \bar{s},$$

which holds by the assumption ($\bar{s}$ is defined in equation (C.4)). This shows that within this range

$$U(s_1) = \left((1 - s_1) + \beta(1 - s_0)^2\right) \mathbb{E}_{F_0}[y].$$

So far we have solved program (5) for any first-period share s_1 :

$$U(s_1) = \begin{cases} \beta(1 - s_0)^2 \mathbb{E}_{F_0}[y], & s_1 < s_0^2, \\ \left((1 - s_1) \left(1 - \frac{s_0^2}{s_1}\right) + \beta(1 - s_0)^2\right) \mathbb{E}_{F_0}[y], & s_0^2 \leq s_1 \leq \frac{s_0}{2 - s_0}, \\ \left(\frac{(1 + s_1 - 2s_0)^2}{1 - s_1} + \beta(1 - s_0)^2\right) \mathbb{E}_{F_0}[y], & \frac{s_0}{2 - s_0} < s_1 \leq \underline{s}, \\ h_1(x^*) \mathbb{E}_{F_0}[y], & \underline{s} < s_1 < \bar{s}, \\ \left((1 - s_1) + \beta(1 - s_0)^2\right) \mathbb{E}_{F_0}[y], & s_1 \geq \bar{s}, \end{cases}$$

where $\underline{s}$ and $\bar{s}$ are defined by equation (C.4), $h_1(\cdot)$ is defined by equation (C.5), and x^* is defined by equation (C.6).

Following this calculation, we first show that the optimal first-period share s_1^* is unique. Note that $U(\cdot)$ is constant for $s_1 < s_0^2$, strictly increasing for $s_0^2 \leq s_1 \leq \underline{s}$, and strictly decreasing for $s_1 \geq \bar{s}$. It remains to show that $U(\cdot)$ has a unique interior maximizer within the range $\underline{s} < s_1 < \bar{s}$. We can further simplify the expression:

$$U(s_1) = \frac{(s_1 - s_0^2) \left((1 + \beta) - (1 + \beta)s_1 + \sqrt{((1 + \beta) - (1 - \beta)s_1)^2 - 4\beta^2 s_1} \right)}{2s_1} \mathbb{E}_{F_0}[y] \equiv h_2(s_1) \mathbb{E}_{F_0}[y].$$

Since $h_2'' < 0$, it suffices to show that $h_2'(\underline{s}) > 0$ and $h_2'(\bar{s}) < 0$. In fact, we have

$$h_2'(s_0) = \frac{\beta(1-s_0)}{\sqrt{((1+\beta)-(1-\beta)s_1)^2 - 4\beta^2 s_1}} > 0 \quad \text{and} \quad h_2'(\bar{s}) = -1 < 0,$$

so there exists a unique number $s_1^* \in (s_0, \bar{s}) \subset (\underline{s}, \bar{s})$ such that $h_2'(s_1^*) = 0$. Therefore, the optimal first-period share is unique, and satisfies $s_1^* > s_0$ for all $\beta \in (0, \infty)$.

Now we consider the comparative statics of s_1^* in β . From $h_2'(s_1^*) = 0$, we can write s_1^* as the unique solution $s_1 \in (s_0, \bar{s})$ to the following quartic equation:

$$\beta s_0^4 - s_1^2 \left((1-\beta)^2 s_1^2 - 2(1+\beta^2) s_1 + (1+\beta+\beta^2) \right) + s_0^2 \left((1-\beta)^2 s_1^2 - 2(1+\beta+\beta^2) s_1 + (1+\beta)^2 \right) = 0.$$

Dividing both sides by β and write $\nu = \beta + 1/\beta - 2 \geq 0$, it follows that s_1^* is the unique solution $s_1 \in (s_0, \bar{s})$ to

$$h_3(s_1, \nu) \equiv s_0^4 - s_1^2 \left(\nu s_1^2 - 2(2+\nu) s_1 + (3+\nu) \right) + s_0^2 \left(\nu s_1^2 - 2(3+\nu) s_1 + (4+\nu) \right) = 0.$$

In both limiting cases $\beta \rightarrow 0$ and $\beta \rightarrow \infty$, we have $\nu \rightarrow \infty$, and s_1^* approaches the solution to

$$-s_1^2 (s_1^2 - 2s_1 + 1) + s_0^2 (s_1^2 - 2s_1 + 1) = 0,$$

that is, $s_1^* \rightarrow s_0$. Moreover, to show s_1^* is strictly increasing in β if $\beta < 1$, and is strictly decreasing if $\beta > 1$, it is equivalent to show s_1^* is strictly decreasing in ν . From the implicit function theorem,

$$\frac{\partial s_1^*}{\partial \nu} \propto \frac{\partial h_3}{\partial \nu}(s_1^*, \nu) = -(s_1^* - s_0)(s_1^* + s_0)(1 - s_1^*)^2 < 0,$$

as desired.

This completes the proof. □